# Newtons laws of motion (NLM)

## Newton's laws:

### I- law : (law of inertia):

↳ importance → Definition of 1) Force 2) Inertia

force: It is the form of pull or push that acts on the body which changes or tries to change the state of body

rest ——— uniform motion

## Different types of forces:

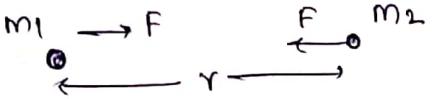
1) Field forces.
(Non - contact forces)

ex: Gravitational force
Electromagnetic force
Nuclear forces.

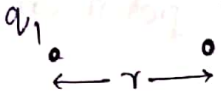
2) Contact forces.

ex: musular force
Normal reaction force
frictional force
Tension force in string /spring.

### Gravitational force.



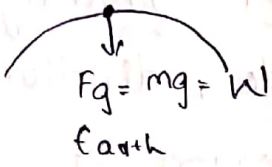

$$F = \frac{G m_1 m_2}{r^2}$$



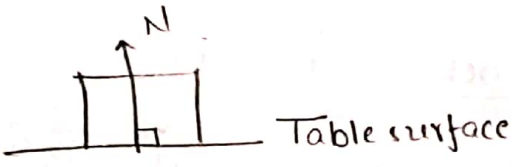

$$F_e = \frac{k q_1 q_2}{r^2}$$

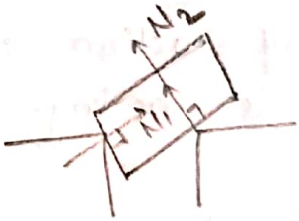


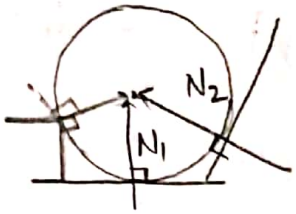

$F_g = mg = W.$

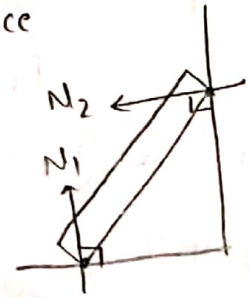


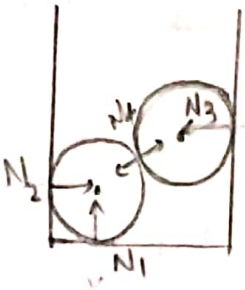

### Normal reaction force:



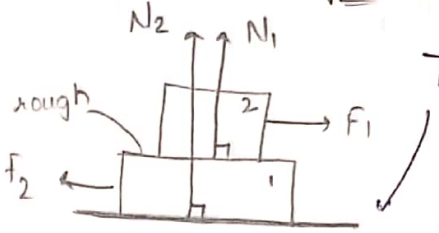



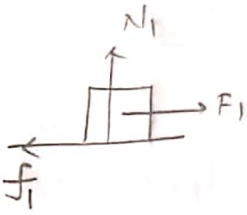




Scanned with CamScanner

## Frictional force:

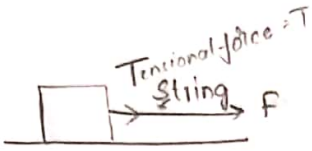


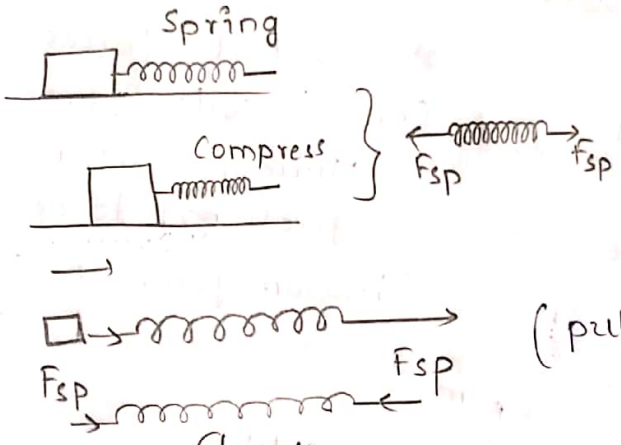




$f_1$ & $f_2$ opp to rel force




String force is always a pulling force

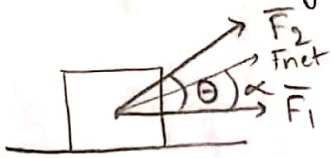



## Force is a vector quantity:

It has magnitude, direction

* point of application of force → the point on the body where the force is acting.
* line of action of force → the line along which force is acting.

★




$$\bar{F}_{net} = \bar{F}_2 + \bar{F}_1$$

$$= F_1 \hat{i} + F_2\cos\theta\,\hat{i} + F_2\sin\theta\,\hat{j}$$

$$\bar{F} = (F_1 + F_2\cos\theta)\,\hat{i} + (F_2\sin\theta)\,\hat{j}$$

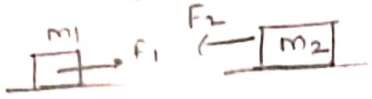
$$\tan\alpha = \frac{F_2\sin\theta}{F_1 + F_2\cos\theta} \Rightarrow \alpha = \tan^{-1}\left(\frac{F_2\sin\theta}{F_1 + F_2\cos\theta}\right)$$

with $x$-axis

Scanned with CamScanner

## Inertia:

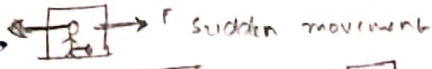
- the property of inability of the body to change its state (rest & uniform motion) by itself.

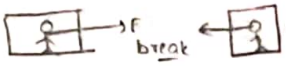



$(m_2 > m_1)$

* mass is a measure of inertia mass(↑) → Inertia(↑)

### Types of inertia

1) Inertia of rest: → 

2) Inertia of motion: 

3) Inertia of direction




**I - law:** Every body continues to be in a state of rest or of uniform motion along st. line unless net external force acts on it. (Equilibrium)

Equilibrium: ( $F_{net} = 0$ ) & $\underbrace{\tau_{net} = 0}_{\text{Torque}}$.

## Momentum of a body (P):

linear momentum : (P) = mass x velocity

$$\bar{p} = m\bar{v}$$

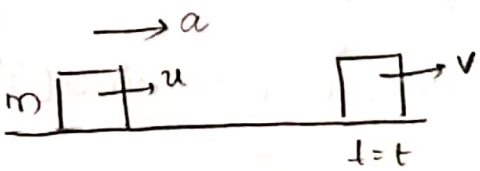
$$\boxed{\bar{p} = m\bar{v}}$$

It is a vector quantity & its direction is always along the direction of motion. (along $\bar{v}$)

SI unit: kgm/s

CGS unit: gm cm/s

Ex.




$\bar{v} = \bar{u} + \bar{a}t$

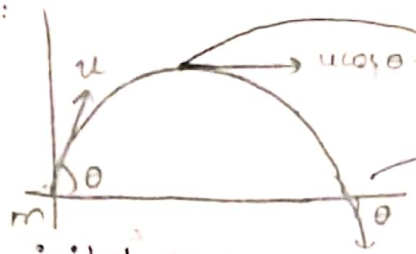
$\bar{v} = (u + at)\hat{i}$

∴ momentum at time t

$$\bar{p} = m\bar{v} = m(u + at)\hat{i},$$

Scanned with CamScanner

Eg:




$\bar{p} = mu\cos\theta\,\hat{i}$

$\bar{p} = mu\cos\theta\,\hat{i} - mu\sin\theta\,\hat{j}$

inital momentum

$\bar{u} = u\cos\theta\,\hat{i} + u\sin\theta\,\hat{j}$

$\therefore \bar{p} = mu\cos\theta\,\hat{i} + mu\sin\theta\,\hat{j}$

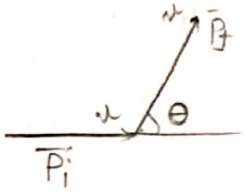
Change in momentum ($\Delta\bar{p}$):

$\Delta\bar{p} = \bar{p}_f - \bar{p}_i$

If m is constant

$\Delta\bar{p} = m\bar{v}_f - m\bar{v}_i$

$= m(\bar{v}_f - \bar{v}_i)$



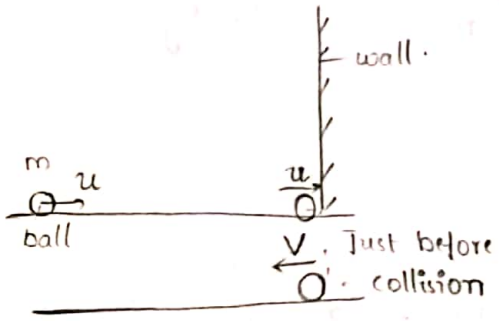

$\Delta\bar{p} = \bar{p}_f - \bar{p}_i$

$= mu\cos\theta\,\hat{i} + mu\sin\theta\,\hat{j} - mu\,\hat{i}$

$\Delta\bar{p} = mu(\cos\theta - 1)\,\hat{i} + mu\sin\theta\,\hat{j}$

$|\Delta\bar{p}| = 2mu\sin(\theta/2)$

Applications:




Change in momentum

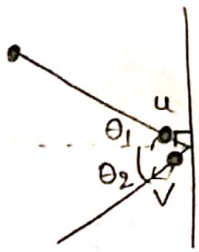
$\Delta\bar{p} = \bar{p}_f - \bar{p}_i$

$= mv(-\hat{i}) - mu\,\hat{i}$

$\Delta\bar{p} = m(v+u)(-\hat{i})$

$\Delta\bar{p} = m(v+u)$, away from the wall.

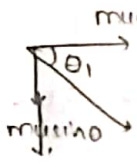
g neglected




$\Delta\bar{p} = \bar{p}_f - \bar{p}_i$

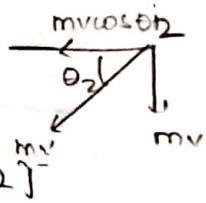
$= m\bar{v} - m\bar{u}$

$\bar{p}_i = m\bar{u}$




$\bar{p}_i = mu\cos\theta\,\hat{i} + mu\sin\theta(-\hat{j})$

$\bar{p}_f = m\bar{v}$




$= mv\cos\theta_2(-\hat{i}) + mv\sin\theta_2\,\hat{j}$

$\Delta\bar{p} = (mv\cos\theta_2 + mu\cos\theta_1)\,\hat{i} + (mv\sin\theta_2 - mu\sin\theta_1)(-\hat{j})$

Scanned with CamScanner

$\bar{\Delta p} = -m(u\cos\theta_1 + v\cos\theta_2)\hat{i} + m(v\sin\theta_2 - u\sin\theta_1)\hat{j}$

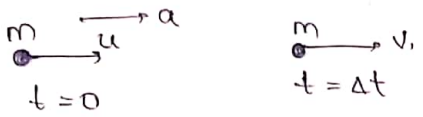
∴ for elastic collision

$$\bar{\Delta p} = -2mu\cos\theta_1\,(i)$$

Newtons II law:

The rate of change in momentum of a body is directly proportional to net acceleration external force acts on it & takes place in the direction of change in momentum.

$$\Sigma F_{ext} \propto \frac{\Delta \bar{p}}{\Delta t}$$




$\bar{p}_i = m\bar{u}$ Here $\bar{v} = \bar{u} + \bar{a}t$

$\bar{p}_f = m\bar{v}$

change in momentum.

$$\bar{\Delta p} = \bar{p}_f - \bar{p}_i = m(\bar{v} - \bar{u})$$

rate of change in momentum

$$= \frac{\bar{\Delta p}}{\Delta t} = \frac{m(\bar{v} - \bar{u})}{\Delta t} = \frac{m\bar{a}t}{\Delta t}$$

$$\frac{\Delta \bar{p}}{\Delta t} = m\bar{a}$$

From II - law,

$$\bar{F} \propto \frac{\Delta \bar{p}}{\Delta t}$$

$$\bar{F} \propto m\bar{a}$$

$$\bar{F} = km\bar{a}$$

$k$ = propotionality unit

Unit force:

$F = 1$ for $m = 1\,kg$ & $a = 1\,m/s^2$

$$1 = k(1)(1)$$

$$\Longrightarrow k = 1$$

$$\bar{F} = m\bar{a}$$

net force

Scanned with CamScanner

& $\bar{F} = k\frac{\Delta\bar{P}}{\Delta t}$

$\bar{F} = \frac{\Delta\bar{P}}{\Delta t}$

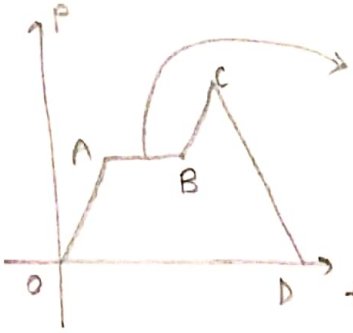
Average force & $\bar{F}_{net} = \lim_{\Delta t \to 0} \frac{\Delta\bar{P}}{\Delta t} = \frac{d\bar{P}}{dt}$

$\bar{F} = \frac{d\bar{P}}{dt}$ = rate of change of momentum

* Slope of p-t graph gives "force",

(i) Slope of secant line give Average force

(ii) Slope of tangent line give instantaneous force.




Slope = 0.
$\therefore F = 0$
$ma = 0$
$a = 0$
$\Rightarrow v = \text{const}$.

* $(F_x\hat{i} + F_y\hat{j} + F_z\hat{k}) = m(a_x\hat{i} + a_y\hat{j} + a_z\hat{k})$

$F_x = ma_x$, $F_y = ma_y$, $F_z = ma_z$

$\therefore$ net force

In general

$$\bar{F} = \frac{\Delta\bar{P}}{dt} = \frac{d}{dt}(m\bar{v})$$

$$\boxed{\bar{F} = m\frac{d\bar{v}}{dt} + \bar{v}\frac{dm}{dt}}$$

If $m = \text{const}$ then $\frac{dm}{dt} = 0$

$$\bar{F} = \frac{m d\bar{v}}{dt}$$

$$\boxed{\bar{F} = m\bar{a}}$$

If only $\bar{v} = \text{const}$ then $\frac{d\bar{v}}{dt} = 0$

$$\boxed{\bar{F} = \frac{\bar{v}\,dm}{dt}} \Rightarrow \text{variable mass system.}$$

Scanned with CamScanner

## Problems :

F = Force

**F = const**

$ma = F$

$a = \frac{F}{m}$ = const

1) $\bar{v} = \bar{u} + \bar{a}t$

2) $\bar{s} = \bar{u}t + \frac{1}{2}\bar{a}t^2$

3) $\bar{v}\cdot\bar{v} - \bar{u}\cdot\bar{u} = 2\bar{a}\cdot\bar{s}$

4) $S_n = u + a(2n-1)$

**F(x)**

$F = 3x$

$ma = 3x$

$a = \frac{3x}{m}$

$v\frac{dv}{dx} = \frac{3x}{m}$

$\int v\,dv = \int \frac{3x}{m}dx$

**F(t)**

$F = 5t$

$ma = 5t$

$a = \frac{5t}{m}$

$\frac{dv}{dt} = \frac{5t}{m}$

$\int dv = \int \frac{5t}{m}dt$

**F(v)**

$F = \frac{-b}{v^2}$

$ma = \frac{-b}{v^2}$

$a = \frac{-b}{mv^2}$

$\frac{dv}{dt} = \frac{-b}{mv^2}$

$\int v^2 dv = \int -\frac{b}{m}dt$

## Newtons III law:

to every action there is a equal & opp reaction

** action = −(Reaction)

action & reaction both doesn't act on, same body

$\bar{F}_{AB}$ = Force on "A" due to "B"

$\bar{F}_{BA}$ = Force on "B" due to "A"

$$\boxed{\bar{F}_{AB} = -\bar{F}_{BA}}$$

* $\bar{F}_{AB} = \frac{d\bar{P}_A}{dt} = \frac{\Delta\bar{P}_A}{\Delta t}$

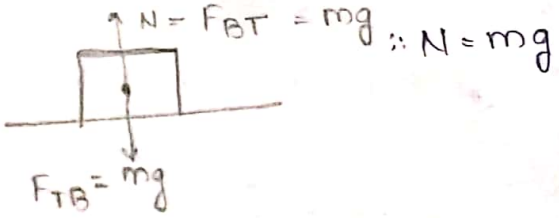
& $\bar{F}_{BA} = \frac{d\bar{P}_B}{dt} = \frac{\Delta\bar{P}_B}{\Delta t}$

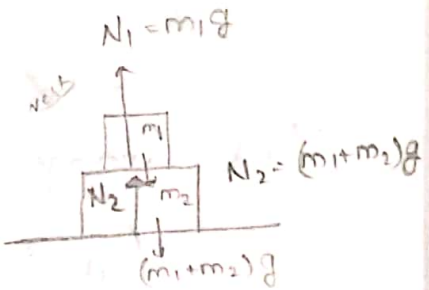
$\bar{F} = 0$

$m\bar{a} = 0$

$\bar{a} = 0$ → $v = 0$ / $v$ = const

## Applications of III law:

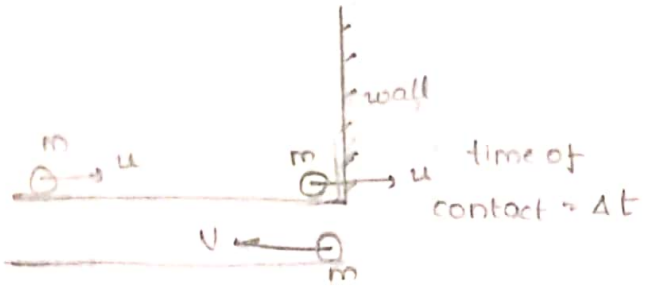



Scanned with CamScanner

Applications

1)




Change in momentum of ball.

$\Delta \bar{P} = \bar{P}_f - \bar{P}_i$

$= -mv\hat{i} - mu\hat{i}$

$= m(v+u)(-\hat{i})$

∴ Average force acting on the ball due to wall

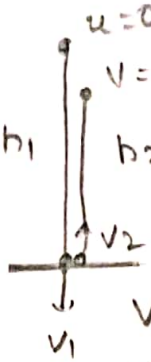
$$F_{b/w} = \frac{\Delta \bar{P}_b}{\Delta t} = \frac{m(v+u)(-i)}{\Delta t}$$

From III law

Ave. force acting on the wall due to ball

$$\bar{F}_{w/b} = \frac{\Delta \bar{P}_w}{\Delta t} = -\bar{F}_{b/w}$$

$$\bar{F}_{w/b} = \frac{m(v+u)\,\hat{i}}{\Delta t}$$




Time of contact = Δt.

Force on the floor due to ball,

$$F = \frac{m(V_1+V_2)}{\Delta t}, \text{ downward.}$$

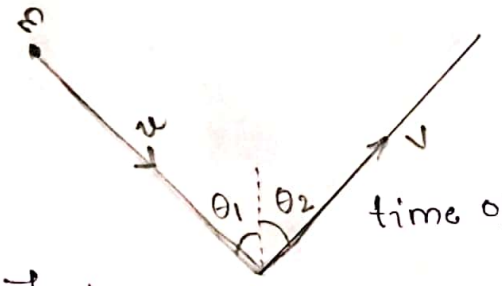
$V_1^2 - 0 = 2gh_1$

$V_1 = \sqrt{2gh_1}$

$V_2 = \sqrt{2gh_2}$

$$= \frac{m(\sqrt{2gh_1} + \sqrt{2gh_2})}{\Delta t}$$

$$= \frac{m\sqrt{2g}\,(\sqrt{h_1}+\sqrt{h_2})}{\Delta t}$$




time of contact = Δt

Find average force on the floor due to ball.

$$F_{f/b} = \frac{\Delta \bar{P}}{\Delta t}$$

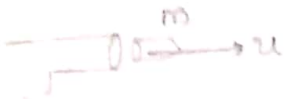
$$F_f = mv\sin\theta_2\,\hat{i} + mv\cos\theta_2\,\hat{j}$$

Scanned with CamScanner

$\bar{P}_i = mv\sin\theta_1\,\hat{i} - mv\cos\theta_1\,\hat{j}$

$F_{ave} = \frac{\Delta\bar{P}}{\Delta t} = \frac{\bar{P}_f - \bar{P}_i}{\Delta t}$

* **Gun - bullets problem:**




$m$ = mass of each bullet

$u$ = speed of each bullet

$\therefore N$ = number of bullets fired in $t$ sec.

$\therefore n = \frac{N}{t}$ = Number of bullets fired per second.

* change in momentum of each bullet

$\bar{P}_f - \bar{P}_i = mu - 0 = mu$

Total change in momentum of $N$ - bullets

$\Delta\bar{P} = Nmu$.

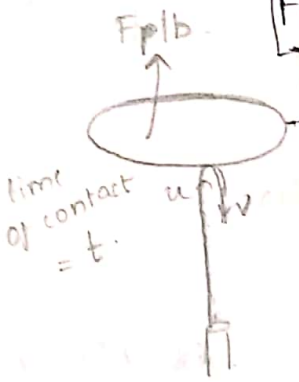
$\therefore$ Average force on bullets by gun,

$= \frac{\Delta\bar{P}}{t} = \frac{Nmu}{t} = mnu(\rightarrow)$

From III law

$\therefore$ force on gun by bullets

$\boxed{F_{G|b} = mnu\,(\leftarrow)}$




$dp = m(v+u)\downarrow$

$\Delta P = Nm(v+u)(\downarrow)$

$F_{avg} = \frac{\Delta P}{t} = \frac{Nm(v+u)}{t}$.

$F_{b|p} = mn(v+u)(\downarrow)$

III law

$\bar{F}_{P|b} = -\bar{F}_{b|p}$

$= mn(v+u)(\uparrow)$

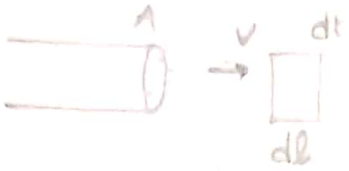
$\therefore$ If the plate remains at rest,

$F_{P|b} = Mg$ (weight of plate)

$\boxed{mn(v+u) = Mg.}$

Scanned with CamScanner

## Force on pipe due to ejection of liquid:




$V$ = speed of liquid w.r.to pipe

* Volume of liquid flowing out the pipe per second (= rate of flow of liquid)

$$\frac{d\bar{v}}{dt} = \frac{A\,dl}{dt} = AV$$

Volume rate $\propto V_{rel}$ ( $A$ = constant )

mass of the liquid flowing out the pipe per sec

$$\frac{dm}{dt} = \frac{(dv)\rho}{dt} = AV\rho \quad \therefore \rho = \text{density of liquid.}$$

* Force on the ejection liquid due to pipe,

$$F_{l/p} = V_{rel}\frac{dm}{dt}. \qquad V_{rel} = \text{Velocity of liquid w.r.to pipe.}$$

$$= V\frac{dm}{dt}$$

$$= V(AV\rho)$$

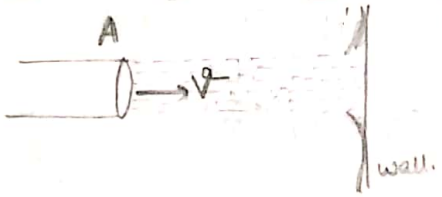
$$\boxed{F_{l/p} = AV^2\rho} \longrightarrow$$

III law

Force on pipe due to ejecting liquid

$$\boxed{F_{p/l} = AV^2\rho} \ (\longleftarrow)$$

## Force on wall due to jet of water:




$\frac{dm}{dt} = AV\rho$

rate of mass of liquid hitting the surface

After hitting, water jet dead drops

* $\bar{P}_i = (dm)\,v = (d\bar{v})\rho v = A(\Delta l)\rho v$

$\bar{P}_f = 0$

$\Delta\bar{P} = \bar{P}_f - \bar{P}_i = 0 - A\rho v(\Delta l)$

$$F_{j/w} = \frac{\Delta\bar{P}}{\Delta t} = -A\rho v\left(\frac{\Delta l}{\Delta t}\right) = -A\rho v^2$$

$$\boxed{F_{j/w} = A\rho v^2 \ (\longleftarrow)}$$

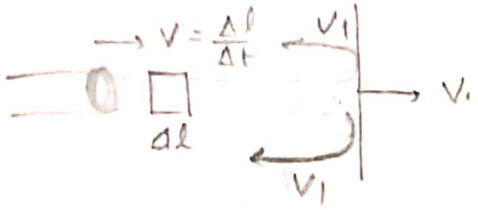
Scanned with CamScanner

$F_{w/j} = A\rho v^2 (\rightarrow)$

or $\langle F \rangle = V_{rel}\left(\frac{dm}{dt}\right)$

$= V\,(A v \rho) = A v^2 \rho\,(\rightarrow)$

Jet rebounces after hitting the wall:




$\bar{P}_i = (\Delta m)\bar{V} = A(\Delta l)\rho V.$

$\bar{P}_f = (\Delta m) V_1 = -A(\Delta l)\rho V_1$

$\frac{\Delta \bar{P}}{\Delta t} = -\frac{(A\rho V_1 \Delta l + A\rho V \Delta l)}{\Delta t}$ $\quad V = \frac{\Delta l}{\Delta t}$

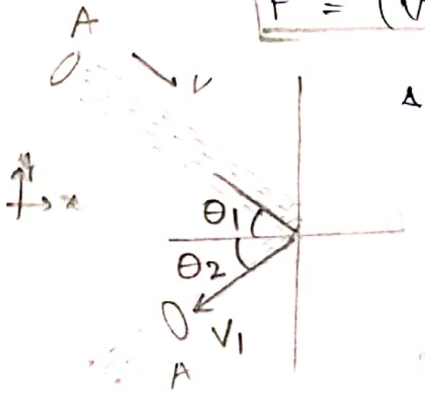
$$\boxed{F_{j/w} = -A v \rho (V + V_1)}$$

$$\boxed{F_{w/j} = A v \rho (V + V_1)}$$

II method

$$F = V_{rel}\left(\frac{dm}{dt}\right)$$

$$\boxed{F = (V + V_1) A v \rho}$$




$\Delta m = (\Delta V)\rho$

$= A \Delta l \rho$

$\bar{P}_i = (\Delta m)\bar{V}$ $\qquad \Delta P = \bar{P}_f - \bar{P}_i$

$\bar{P}_f = (\Delta m)\bar{V}_1$ $\qquad F = \frac{\Delta \bar{P}}{\Delta t}$

$= \Delta m\left(-V_1 \cos\theta_2 (-\hat{i}) + V_1 \sin\theta_2 (-\hat{j})\right)$

$\bar{P}_i = \Delta m\left(V \cos\theta_1 (-\hat{i}) + V \sin\theta_1 (-\hat{j})\right)$

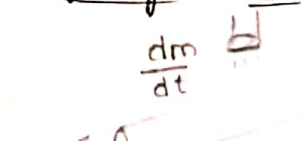
$\Delta \bar{P} = \bar{P}_f - \bar{P}_i = \Delta m\left(-V_1 \cos\theta_2 \hat{i} - V \cos\theta_1 \hat{i}\right)$

$+ \Delta m\left(V_1 \sin\theta_2 (-\hat{j}) + V \sin\theta_1 \hat{j}\right)$

$= \Delta m\left(V_1 \cos\theta_2 + V \cos\theta_1\right)(-\hat{i}) + \Delta m\left(V \sin\theta_1 - V_1 \sin\theta_2\right)\hat{j}$

$F = \frac{\Delta \bar{P}}{\Delta t}$ $\qquad \Delta m = A\rho\Delta l$

Conveyer belt:




$\frac{dm}{dt}$ = mass per unit time (added / removed) (kg/sec)

$V$ = const = speed of belt

Scanned with CamScanner

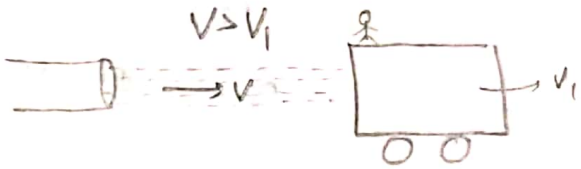
Extra force required to maintain const speed,

$$F = V_{rel} \frac{dm}{dt} \qquad V_{rel} = V_{s/b} = V_s - V_b$$

$$\boxed{F = V \frac{dm}{dt}}$$

Case i:

At pipe




$$\frac{dm}{dt} = AV\rho$$

rate of mass of water hitting the moving truck,

$$\frac{dm}{dt} = A(V-V_1)\rho \longrightarrow (1).$$

Velocity of jet w.r.to truck

$$V_{rel} = V - V_1$$

Force on moving surface

$$F = V_{rel} \frac{dm}{dt}$$

$$F = (V-V_1) A (V-V_1)\rho$$

$$\boxed{F = A(V-V_1)^2 \rho}$$

Case ii: If $V = V_1$ then $F = 0$.




$$\frac{dm}{dt} = A(V+V_1)\rho$$ At the moving surface

$$V_{rel} = V + V_1$$

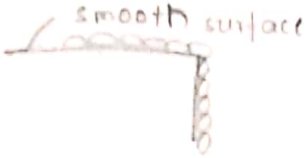
Force on moving surface

$$f = V_{rel} \frac{dm}{dt}$$

$$= (V_1+V) A (V+V_1) \rho$$

$$\boxed{F = A(V+V_1)^2 \rho}$$

Scanned with CamScanner




Find the speed of chain at the time of leaving the table

$n^{th}$ part of chain is lying on the table & then released



$$m \longrightarrow L$$
$$\frac{m}{L} \longleftarrow 1$$
$$\frac{m}{L}x \longleftarrow x$$

→ weight of hanging part of chain

$$F = \frac{m}{L} x g$$

$$Ma = \frac{m}{L} x g$$

$$a = \frac{gx}{L} \qquad L \to M$$

$$v\frac{dv}{dx} = \frac{g}{L}x$$

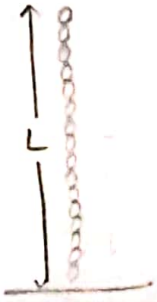
$$\int_0^v v\,dv = \int_{x=\frac{L}{n}}^{x=L} \frac{g}{L} x\,dx$$

$$\frac{v^2}{2} = \frac{g}{L} \times \left[\frac{x^2}{2}\right]_{\frac{L}{n}}^{L}$$

$$v^2 = \frac{g}{L}\left(L^2 - \frac{L^2}{n^2}\right)$$

$$= gL\left(\frac{n^2-1}{n^2}\right)$$

$$v = \sqrt{gL\left(1 - \frac{1}{n^2}\right)}$$




now chain released on to the table link by link. Find the force exerted by the chain on the table when a length $y$ is on to the table.

Scanned with CamScanner

Total force = Force due to weight of chain on the table + Force due to change in momentum of chain imparted to the table

$$F = F_1 + F_2$$

$$= \frac{M}{L} y g + \frac{\frac{m}{L}(dy)\sqrt{2gy}}{dt}$$

$$= \frac{M}{L} y g + \frac{m}{L}\left(\sqrt{2gy}\right)\left(\sqrt{2gy}\right)$$

$$= \frac{M}{L} y g + \frac{m}{L}(2gy).$$

$$\boxed{F = \frac{3myg}{L}} \star$$

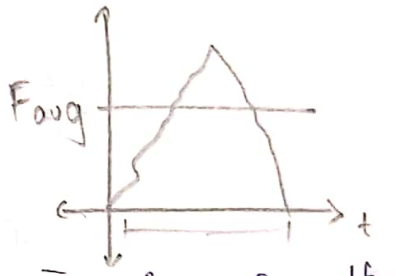
## Impulse (I):

### Impulsive force:

A very large force acting on a body in a short duration of time is called "Impulsive force"

Impulse $\bar{I}$ = force × time.

$$\star \quad \boxed{\bar{I} = \bar{F}t}$$




Impulse is a vector

Its direction is always along $\bar{F}$. i.e, in the direction of change in momentum.

we have $\bar{F} = \frac{d\bar{p}}{dt}$

$$d\bar{p} = \bar{F}dt$$

$$\int_{P_1}^{P_2} d\bar{p} = \int_{t_1}^{t_2} \bar{F}dt$$

$$\bar{P}_2 - \bar{P}_1 = \int_{t_1}^{t_2} \bar{F}dt \longrightarrow \begin{matrix} F = \text{const} \\ F = f(t) \end{matrix}$$

Scanned with CamScanner

$$\bar{I} = \int_{t_1}^{t_2} \bar{F}\,dt = \bar{P}_2 - \bar{P}_1$$

Impulse is always equal to change in momentum

we have,

$$\bar{P}_i \pm \bar{I} = \bar{P}_f \quad \text{i.e} \quad \bar{P}_i \pm \int_{t_1}^{t_2} \bar{F}\,dt = \bar{P}_f$$

* $\bar{P}_i + I = P_f$ (Impulse is given in direction of $P_i$)
* $P_i - I = P_f$ (Impulse is given opposite to the direction of $\bar{P}_i$)




$$-P_i + I = +P_f$$

$$I = P_f + P_i$$




$$P_i + I = P_f$$

$$I = P_f - P_i$$

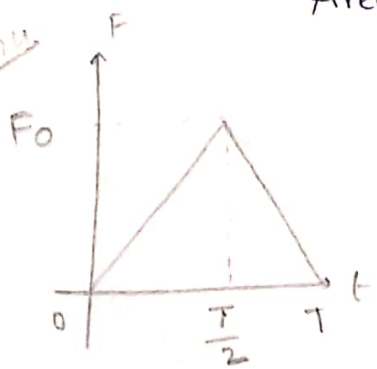
net impulse

$$\bar{F}_1 t_1 + \bar{F}_2 t_2 + \bar{F}_3 t_3 = \bar{P}_f - \bar{P}_i$$

$$I = \int_{t_1}^{t_2} F\,dt = \bar{P}_f - \bar{P}_i$$

Area of F-t graph gives "Impulse"

29/5/22







Area = Impulse

$$\frac{1}{2} T F_0 = mu - 0$$

$$\rightarrow F_0 = \frac{2um}{T}$$

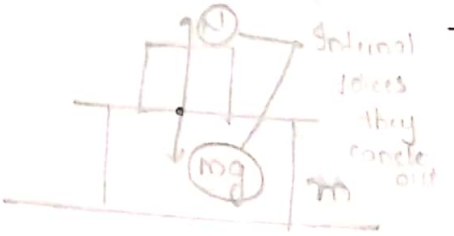
Average force,

$$\langle F \rangle = \frac{\int_{t_1}^{t_2} F\,dt}{\int_{t_1}^{t_2} dt} = \frac{\text{Impulse}}{\int_0^T dt} = \frac{\frac{1}{2} T F_0}{T} \Rightarrow \langle F \rangle = \frac{F_0}{2}$$

Scanned with CamScanner

# Free Body diagram (FBD):

System → collection of bodies / particles under observation



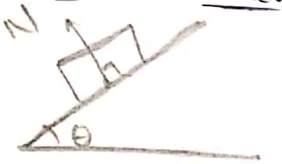

→ System is represented with point mass.
↳ System has common acceleration for all parts.

→ list out the external forces acting on the system
→ Chose x, y, z axis of coordinate system
→ Use $\boxed{\Sigma F = ma}$

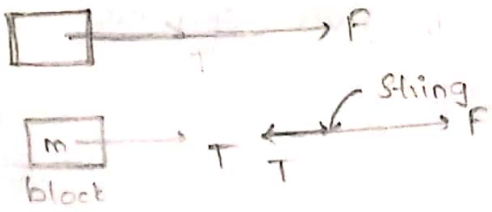
ie $\Sigma F_x = ma_x$ , $\Sigma F_y = ma_y$ , $\Sigma F_z = ma_z$

External forces
1) Applied force , gravitational forces
2) Normal reaction
3) Friction
4) Tension in string , spring force
5) Pseudo force.

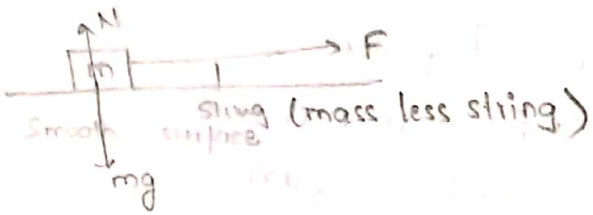
## Normal reaction:

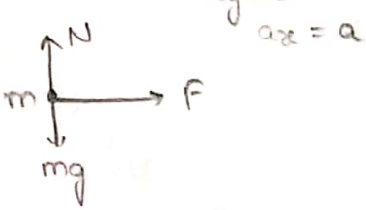



## Tension in a string:




★




$\Sigma F_y = ma_y = 0$

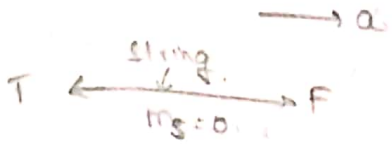
$\Sigma F_y = 0$ $\quad N - mg = 0.$ $\quad \boxed{N = mg}$

Scanned with CamScanner

$\Sigma F_x = ma_x$

$F = ma$

$a = \frac{F}{m}$




$F - T = m_s a$

$F - T = 0$

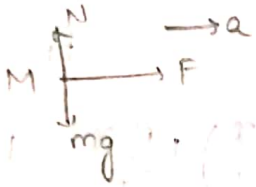
$F = T$

* Tension is same at all points in the mass less string.

* Tension in the rope | string:

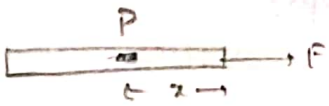



$\therefore N = mg.$

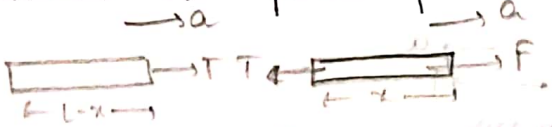


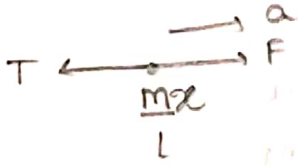

$F = Ma$

$a = \frac{F}{M}$




Tension at point P:




$L \rightarrow M$

$1 \rightarrow \frac{M}{L}$

$x \rightarrow \frac{M}{L}x$

$F - T = \frac{Mx}{L} a$

$F - T = \frac{Mx}{L} \frac{F}{M}$

$T = F - \frac{Fx}{L}$

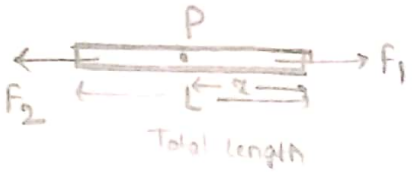
$= F\left(1 - \frac{x}{L}\right)$

Scanned with CamScanner

At $x = L$ (Free end)

$$T = F\left(1 - \frac{L}{L}\right) = 0 \text{ //.}$$

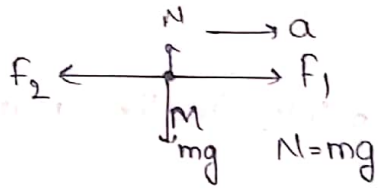
**




M total mass.

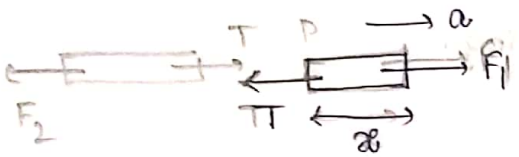
1) $F_1 > F_2$, $a \rightarrow$ to right
2) $F_2 > F_1$, $a \rightarrow$ to left
3) $F_1 = F_2$, $a = 0$.

let $F_1 > F_2$




$$F_1 - F_2 = Ma$$

$$a = \frac{F_1 - F_2}{M} \rightarrow (1)$$




$$F_1 - T = \frac{M}{L} x\, a$$

$$F_1 - T = \frac{M}{L} x \times \frac{F_1 - F_2}{M}$$

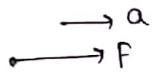
$$T = F_1\left(1 - \frac{x}{L}\right) + \frac{F_2 x}{L} \qquad T = F_1 - \left(\frac{F_1 - F_2}{L}\right) x.$$

Connecting bodies by strings / ropes:

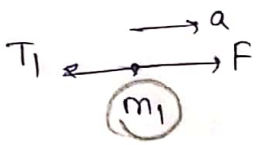



acceleration of system

$$a = \frac{F}{m_1 + m_2 + m_3}$$




$m_1 + m_2 + m_3$

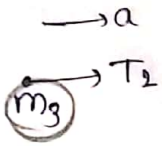



$$F - T_1 = m_1 a$$

$$F - T_1 = \frac{F m_1}{m_1 + m_2 + m_3}$$

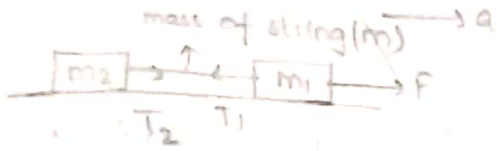
$$T_1 = F - \frac{F m_1}{m_1 + m_2 + m_3}$$

$$= \frac{(m_2 + m_3) F}{m_1 + m_2 + m_3}$$

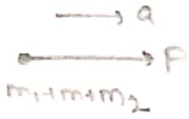



$$T_2 = m_3 a \qquad T_2 = \frac{m_3 F}{m_1 + m_2 + m_3}$$

Scanned with CamScanner




1) $a = \dfrac{F}{m_1 + m + m_2}$




2) 


$$F - T_1 = m_1 a$$

$$F - T_1 = \frac{m_1 F}{m_1 + m + m_2}$$

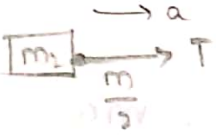
$$T_1 = \frac{(m + m_2) F}{m_1 + m_2 + m}$$

3) 


$$T_2 = m_2 a$$

$$T_2 = \frac{m_2 F}{m + m_1 + m_2}$$

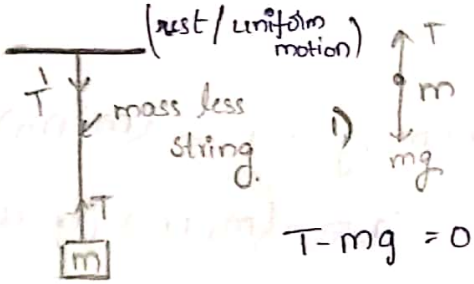
Tension at mid point of wire:



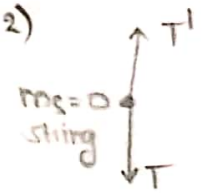

$$T = (m_2 + m/2)\, a$$

$$T = \frac{(m_2 + \frac{m}{2}) F}{m_1 + m_2 + m}$$

Objects suspended using strings / ropes:

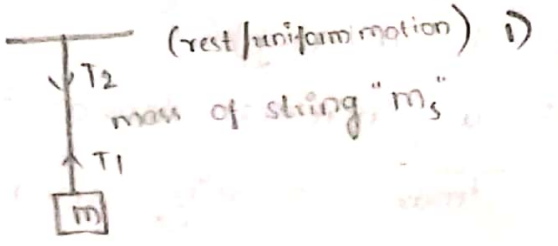



$$T - mg = 0 \implies T = mg$$

2) 


$$T^1 - T = 0$$

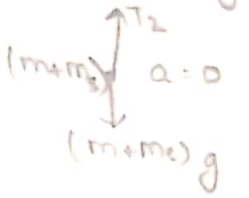
$$\therefore T^1 = T$$




$$T_1 = mg.$$

Scanned with CamScanner

At ceiling

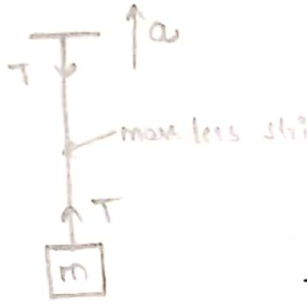


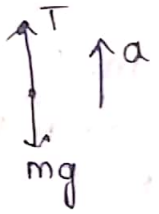

$T_2 = (m+m_3)g$

$T_2 > T_1$




$T = (m+m_3')g$

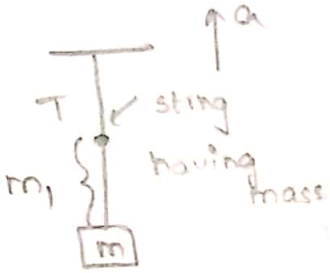


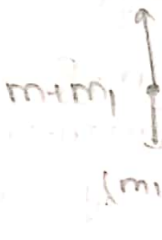




$T - mg = ma$

$T = m(a+g)$

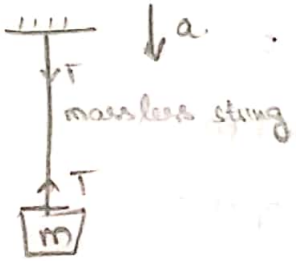
$T = T_{max}$ = breaking support.

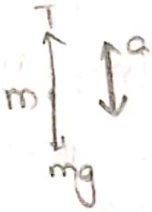



$T - (m_1+m)g = (m_1+m)a$

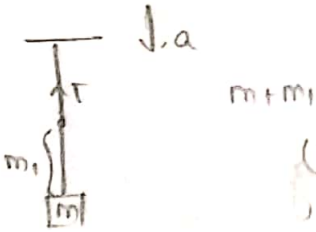
$T = (m_1+m)(a+g)$

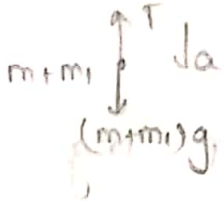



$mg - T = ma$

$T = m(g-a)$

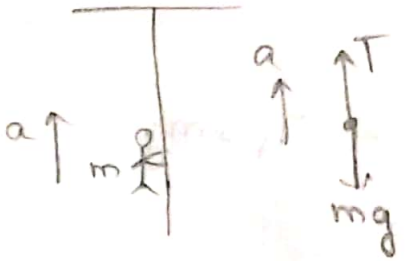



$(m+m_1)g - T = (m+m_1)a$

$T = (m_1+m)(g-a)$

→ A man climbing up the rope.




$T - mg = ma$

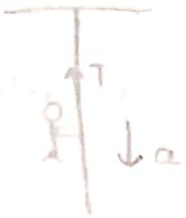
$\boxed{T = m(a+g)}$

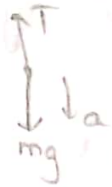
as a increases tension increases

$T_{max}$ = Breaking strength.

Scanned with CamScanner

## Climbing down the rope

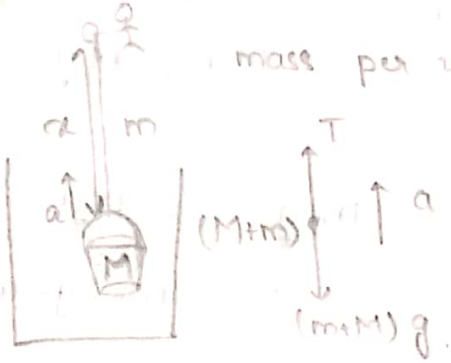



$mg - T = ma$

$T = m(g-a)$

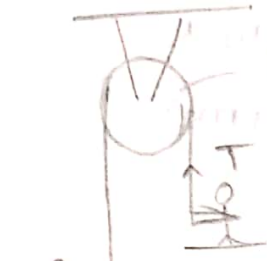
mass per unit length $= \lambda$




$T - (m+M)g = (M+m)a$

$T = (M+m)(g+a)$

$T = (M+\lambda x)(g+a)$

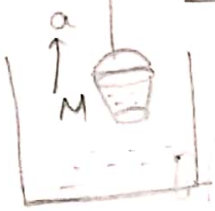


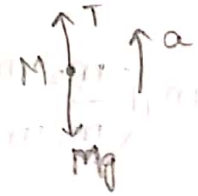




$T + N = mg$

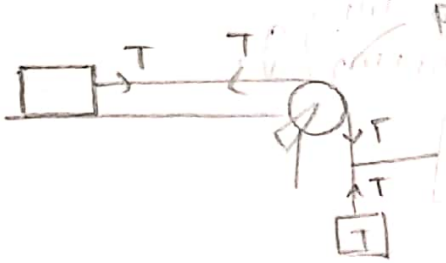
$N = mg - T$



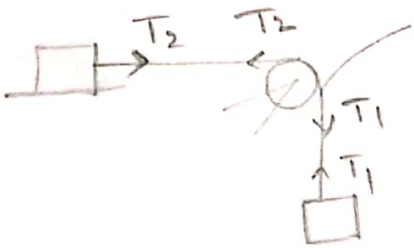




$T - Mg = Ma$

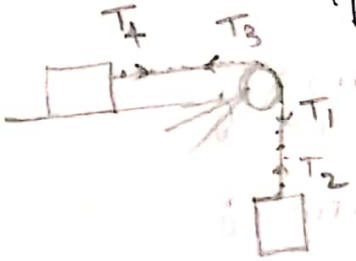
$T = M(g+a)$

* 


Pulley & string are, <u>mass less</u> & <u>friction less</u>



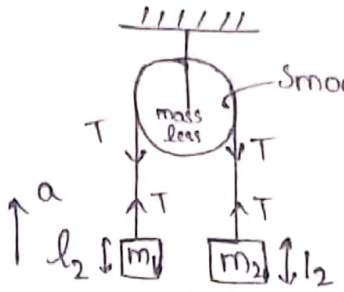

pulley is <u>rough</u> string is <u>massless</u>




Pulley is rough & string has mass

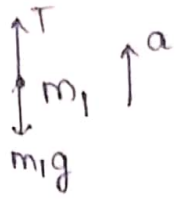
Scanned with CamScanner

# Atwood's machine:



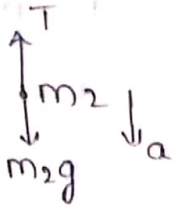

$(m_2 > m_1)$

when the system is released,




$$T - m_1 g = m_1 a \longrightarrow (1)$$

$$T = m_1(a+g)$$




$$m_2 g - T = m_2 a \longrightarrow (2)$$

$$T = m_2(g-a)$$

(1) + (2) $\quad m_2 g - m_1 g = m_1 a + m_2 a$

$$(m_2 - m_1) g = a(m_1 + m_2)$$

acceleration of blocks in atwood machine $\boxed{a = \dfrac{(m_2 - m_1) g}{m_1 + m_2}}$

Eq -(1) becomes $T - m_1 g = m_1 \left(\dfrac{m_2 - m_1}{m_1 + m_2}\right) g.$

$$T = m_1 g + m_1 \left(\frac{m_2 - m_1}{m_1 + m_2}\right) g.$$

$$= m_1 g \left[\frac{2m_2}{m_1 + m_2}\right]$$

$$\boxed{T = \frac{2 m_1 m_2 g}{m_1 + m_2}}$$

★ Points:

① Thrust on the pulley




$$T' = 2T$$

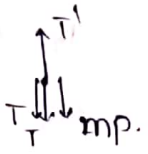
$$T' = 2\left(\frac{2 m_1 m_2}{m_1 + m_2}\right) g$$

$$T' = \frac{(4 m_1 m_2) g}{m_1 + m_2}$$

Scanned with CamScanner

If pulley have mass $m_p$

$$T' = 2T + m_p$$




$$T' = \frac{+m_1 m_2 g}{m_1 + m_2} + m_p$$

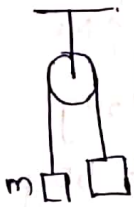
② Relative acceleration of blocks

$$a_{rel} = \bar{a}_{12} = a_1 - a_2$$
$$= a - (-a)$$
$$= 2a$$

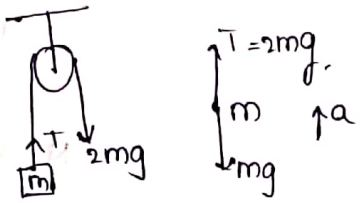
③ Time taken by the blocks to cross each other

$$S_{rel} = \cancel{u_{rel}}^{0}\, t + \frac{1}{2} a_{rel}\, t^2$$
$$l_1 + l_2 = \frac{1}{2}(2a)\, t^2$$
$$t = \sqrt{\frac{l_1 + l_2}{a}}$$

Ex:

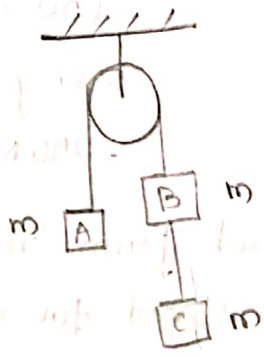



$$a = \frac{(2m - m)g}{2m + m} = \frac{g}{3}.$$



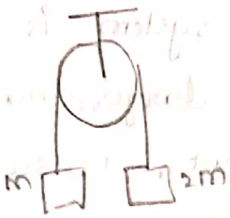

$$a = \frac{2mg - mg}{m} = g.$$

Ex:

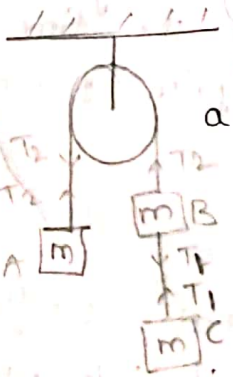



Calculate the tension in the string connecting A to B & B to C ?

Sol:

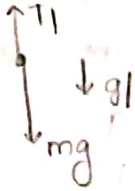



$$a = \frac{(2m - m)g}{2m + m} = \frac{g}{3}.$$




$a = g/3$

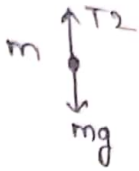
For C




$$mg - T_1 = \frac{mg}{3}$$

$$\therefore T_1 = \frac{2mg}{3}. \quad \text{B \& C}$$

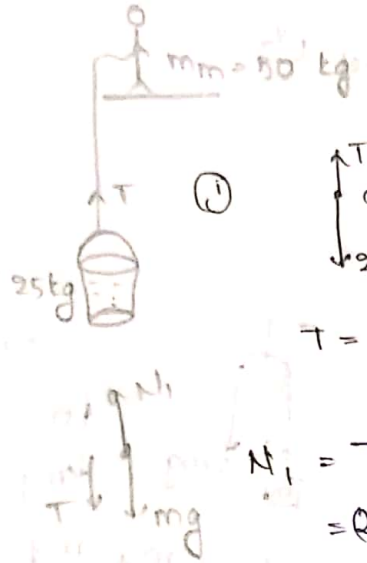
Scanned with CamScanner




For A $T_2 - mg = \frac{mg}{3}$

$T_2 = \frac{4mg}{3}$ (A & B)

A bucket of mass 25 kg is raised by A man of mass 50 kg in 2 ways as shown, what is the action of the floor by the man in 2 cases, if the floor heals to a normal force of 700 N, which mode should the man adopt to lift the bucket without the floor healding?



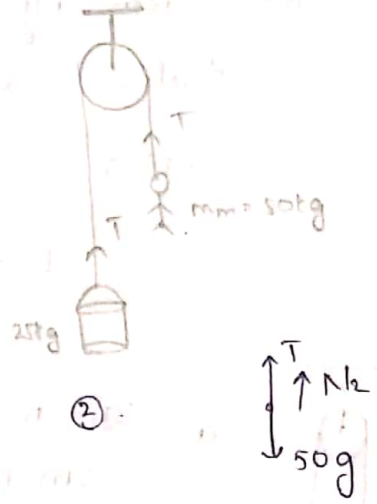

$T = 25g$

$N_1 = T + 50g$

$= (25+50)g$

$= 75g$

$= 750$ N.




$T + N_2 = 50g$

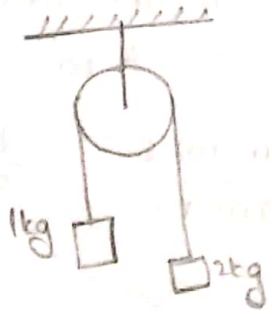
$N_2 = 50g - T$

$= (50-25)g$

$= 25g.$

$= 250$ N.

Ex:




The system is released from rest the larger mass is stoped for a moment, 1.0 sec after the system is set into motion. Find the time elapsed before the string is tight again ($g = 10 m/s$)

Sol: $a = \frac{(2-1)g}{2+1} = \frac{g}{3} = \frac{10}{3} m/s^2$

Velocity of 1kg block at $t = 1$ sec $V = u + at = 0 + \frac{10}{3}(1)$

$V = 10/3$ m/s

Scanned with CamScanner

Now larger block is stoped & then released string becomes tight when

$$S_{1kg} = S_{2kg}$$

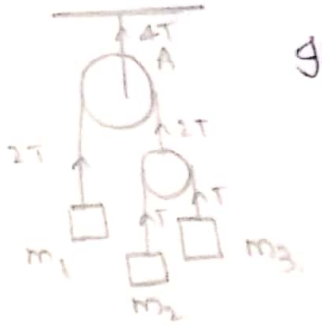
$$V_1 t - \frac{1}{2} g t^2 = \frac{1}{2} g t^2$$

$$g t^2 = V_1 t$$

$$t = \frac{V_1}{g} = \frac{10}{3 \times 10}$$

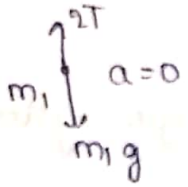
$$\underline{t = \frac{1}{3} \text{ sec}}$$

Ex:




If $m_1$ is at rest, find relation b/w $m_1$ $m_2$ & $m_3$.

Here $m_1$ is at rest → pulley B is at rest.




$$2T = m_1 g \rightarrow (1)$$

Again we have.

$$T = \frac{2 m_2 m_3 g}{m_2 + m_2}$$

$$\therefore 2T = m_1 g.$$

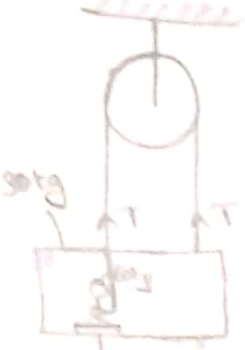
$$\frac{4 m_2 m_3 g}{m_2 + m_3} = m_1 g$$

$$\frac{4}{m_1} = \frac{m_2 + m_3}{m_2 m_3}$$

$$\frac{4}{m_1} = \frac{1}{m_2} + \frac{1}{m_3}$$

A man of mass 60kg is standing on a weighing machine kept in a box of mass 30 kg If the man manages to keep the box stationary find the reading of weighing machine.

Scanned with CamScanner



weighing machine reading = Normal reaction

F.B.D for man



$N + T = m_1 g$

$a = 0$

$N + T = 60g \longrightarrow (1)$

FBD for box.



$T - N = 30g$

$T + N = 60g \longrightarrow (2)$ & $\longrightarrow (1)$

$2T = 90g$

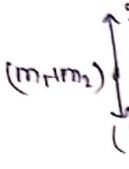
$T = 45g$

$= 450\ N$

$\therefore N = 15g = 150\ N$

m-II for finding tension

& man & box is taken as single system.



$2T = (m_1 + m_2)g = 90g$

$\Rightarrow T = 45g$

In the above problem what force should the man exert on the rope to get his correct weight on the machine. $\therefore N = mg$.

Here system moves with some acceleration 'a' (man + box)

F.B.D of man.



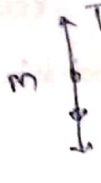
$T + N - Mg = Ma$ [~~$N = mg$~~]

$N = Mg$

$T = Ma \longrightarrow (1)$

$= 60a$

F.B.D of box



$T - mg - N = ma$

$T - (M+m)g = ma$

$60a - (90)g = 30a$

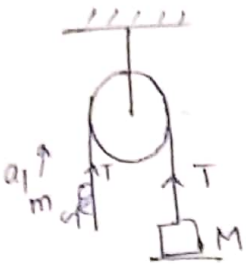
Scanned with CamScanner

$30a = 90g$

$a = \frac{900}{30}$

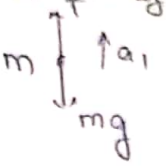
$a = 30\ m/s^2$

$T = 60a = 60 \times 30 = 1800\ N$



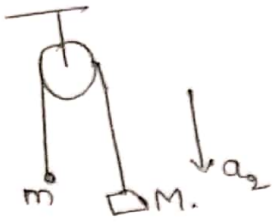

$T = Mg \rightarrow (1)$

for monkey.




$T - mg = ma_1$

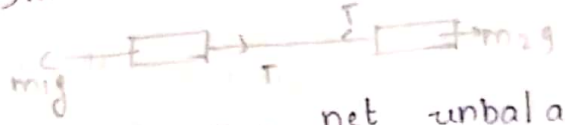
$a_1 = \left(\frac{M-m}{m}\right)g \rightarrow (1)$

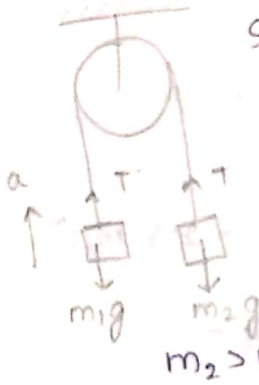



$a_2 = \left(\frac{M-m}{M+m}\right)g \rightarrow (2).$

$\Delta a = |a_1 - a_2|.$

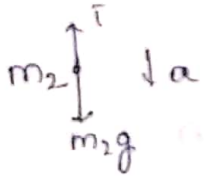
Short cut



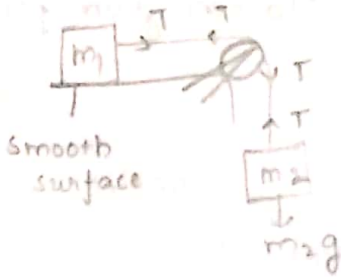

$a = \frac{\text{net unbalance of force}}{\text{Total mass}}$

$= \frac{m_2 g - m_1 g}{m_1 + m_2} = \frac{(m_2 - m_1)g}{m_1 + m_2}$

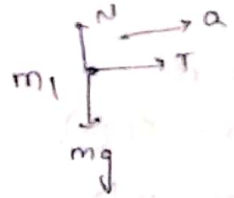



$m_2 g - T = m_2 a$

$T = m_2(g - a).$



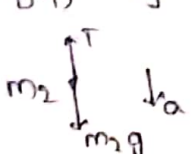

F·B·D of $m_1$




$N = m_1 g$

$T = m_1 a \rightarrow (1).$

F·B·D of $m_2$




$m_2 g - T = m_2 a \rightarrow (2)$

T

Eq (1) + (2).

$m_2 g = (m_2 + m_1)a$

$a = \frac{m_2 g}{m_1 + m_2}$

Scanned with CamScanner

$$T = \frac{m_1 m_2 g}{(m_1 + m_2)}$$

Thrust on pulley

$$T' = \sqrt{2}\, T$$



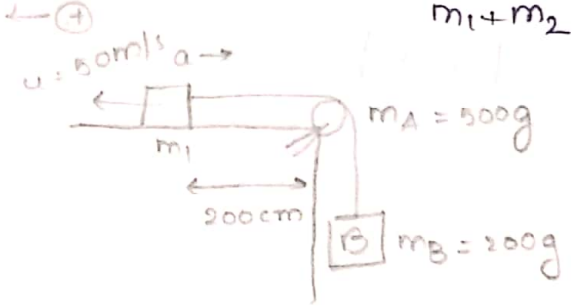

Short cut




$$a = \frac{m_2 g}{m_1 + m_2}$$

& $$T = m_1 a = \frac{m_1 m_2 g}{m_1 + m_2}$$

Ex:




Find the position of block A at $t = 1$ sec. & velocity of block A at $t = 1$ sec

Sol: $$a = \frac{m_B g}{m_A + m_B} = \frac{200\, g}{700} = \frac{2}{7} \times 9.8 = 2.8 \text{ m/s}^2$$

$$= 280 \text{ cm/s}^2$$

* $$S = ut + \frac{1}{2} a t^2$$

$$= 50(1) - \frac{1}{2}(280)(1)^2$$

$$= 50 - 140 = -90 \text{ cm}.$$

Position of block "A" is at 110 cm from pulley,

* Velocity

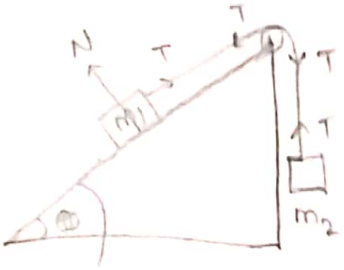
$$v = u + at$$

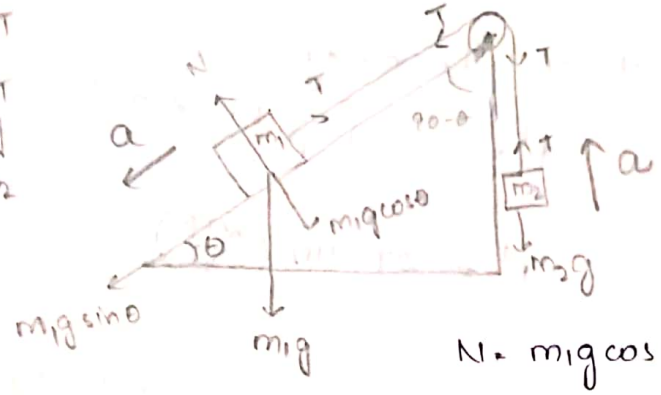
$$= 50 - 280(1)$$

$$= -230 \text{ cm/s}$$

i.e $v = 230$ cm/s to right.

Scanned with CamScanner



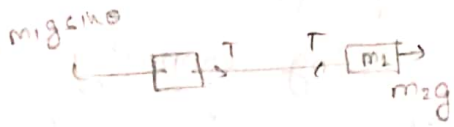




$N = m_1 g\cos\theta$

① $m_1 g\sin\theta = m_2 g$

$\Rightarrow m_1\sin\theta = m_2$ then $a = 0$

② If $m_1 g\sin\theta > m_2 g$




$\Rightarrow m_1\sin\theta > m_2$

$$a = \frac{m_1\sin\theta g - m_2 g}{m_1\sin\theta + m_2}$$

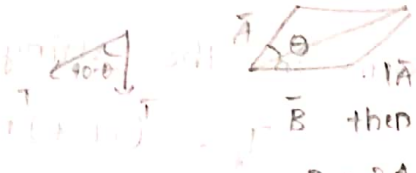
$$a = \frac{(m_1\sin\theta - m_2)g}{m_1+m_2}$$

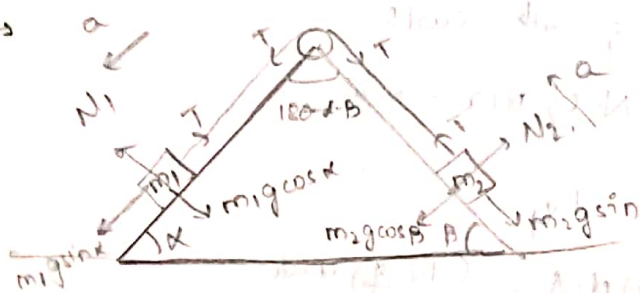
& $T - m_2 g = m_2 a$.

$$T = m_2(g+a)$$

Thrust on the pulley

$$T' = 2T\cos\left(\frac{90-\theta}{2}\right)$$




→ If $m_1 g\sin\alpha = m_2 g\sin\beta$

$m_1\sin\alpha = m_2\sin\beta$

$\Rightarrow a = 0$.

★ → $m_1\sin\alpha > m_2\sin\beta$

then $a = \frac{m_1 g\sin\alpha - m_2 g\sin\beta}{m_1+m_2}$

& $m_1 g\sin\alpha - T = m_1 a$

From this T can be found.

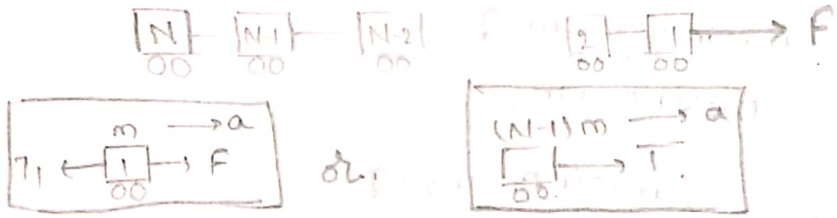
Scanned with CamScanner

* Thrust on the pulley

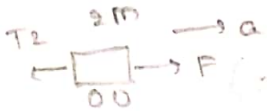
$$T' = 2T \cos\left(\frac{180-(\alpha+\beta)}{2}\right)$$

$$T' = 2T \sin\left(\frac{\alpha+\beta}{2}\right)$$

Ex: N identical carts connected by massless string. If the tension in the string connecting 4th & 5th carts is twice the tension in the string connecting 8th & 9th carts. Find the (i) tension in last string (ii) & N.




$F - T_1 = ma$ $\qquad$ $T = (N-1)ma$.




$F - T_2 = 2ma$ $\qquad$ $T_2 = (N-2)ma$.

Tension in the string b/w 4th & 5th carts

$$T_4 = (N-4)ma \longrightarrow (1).$$

T b/w 8th & 9th carts

$$T_8 = (N-8)ma \longrightarrow (2)$$

Given $2T_8 = T_4$

$$2(N-8)m\cancel{a} = (N-4)m\cancel{a}$$

$$2N - 8 = N - 4$$

$$N =$$

$$2(N-8)m\cancel{a} = (N-4)m\cancel{a}$$

$$2N - 16 = N - 4$$

$$\boxed{N = 12}$$

Scanned with CamScanner

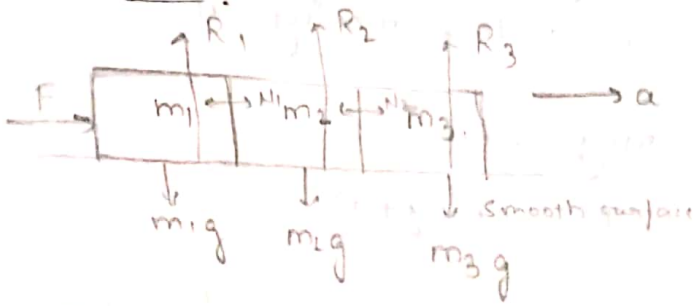
Tension in last string

$$T_N = ma$$

$$= m \times \frac{F}{12m}$$

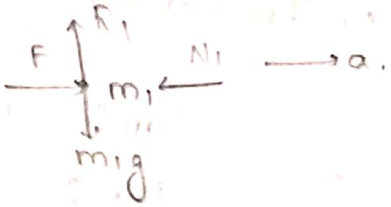
$$T_N = \frac{F}{12}$$

Contact bodies:




$$R_1 = m_1 g.$$

& $F - N_1 = m_1 a \longrightarrow (1)$




$$R_2 = m_2 g$$

$$N_1 - N_2 = m_2 a \longrightarrow (2).$$

$$R_3 = m_3 g.$$

$$N_2 = m_3 a \longrightarrow (3)$$

adding (1) & (2) & (3)

$$F = (m_1 + m_2 + m_3)\, a$$

$$\Rightarrow a = \frac{F}{(m_1 + m_2 + m_3)}$$

$$N_2 = m_3 a$$

$$= \frac{m_3 F}{(m_1 + m_2 + m_3)}$$, contact force b/w $m_2$ & $m_3$

& $F - N_1 = m_1 a$

$$N_1 = F - m_1 a = F - \frac{m_1 F}{m_1 + m_2 + m_3}$$

$$N_1 = \frac{(m_3 + m_2) F}{m_1 + m_2 + m_3}$$ Contact force b/w $m_1$ & $m_2$

Scanned with CamScanner

Ex: 10 coins each of mass m are placed one on other then Reaction force exerted on 6th coin by the 5th coin from the bottom?

Sol: R = weight of all coins above 5th coin

$$R = 5mg$$

* If the coin system moving up with acceleration 'a'



then

$$R - 5mg = 5ma$$

$$R = 5m(g+a)$$

Equilibrium:

↳ Translational equilibrium → V=0 ← rest; V = const.

$\Sigma F = 0$

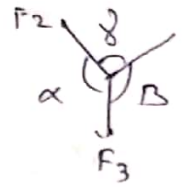
$\Sigma F_x = 0$

$\Sigma F_y = 0$

$\Sigma F_z = 0$

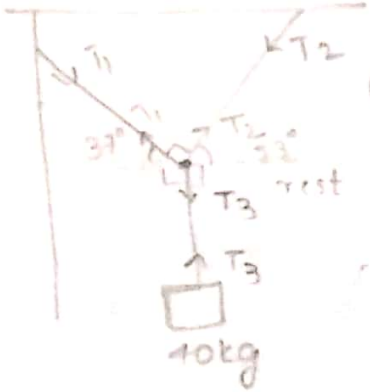
or

lami's theorem (coplanar forces)




$$\frac{F_1}{\alpha} = \frac{F_2}{\beta} = \frac{F_3}{\gamma}$$

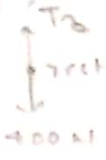
Ex:




Find tension in all strings.

m-I:

$T_3 = 400\ N \rightarrow (1)$




& lami's theorem:

$$\frac{T_1}{\sin(90+53)} = \frac{T_2}{\sin(90+37)} = \frac{T_3}{\sin 90}$$

$$T_1 = T_3 \cos 53$$

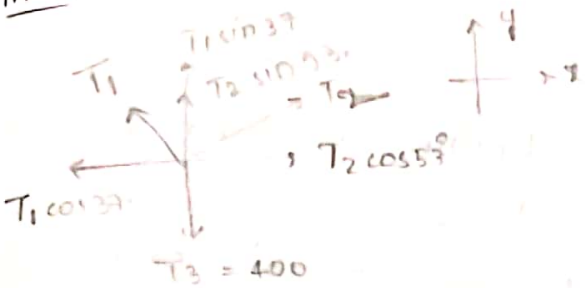
$$= 400 \times \frac{3}{5} = 240\ N$$

& $T_2 = T_3 \cos 37$

$$= 400 \times \frac{4}{5} = 320\ N.$$

Scanned with CamScanner

m-II :




$\Sigma F_x = 0$.

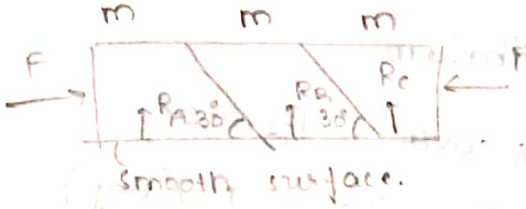
$T_2 \cos 53 = T_1 \cos 37 \rightarrow (1)$

$\Sigma F_y = 0$.

$T_1 \sin 37 + T_2 \cos 37 = T_3 \rightarrow (2)$

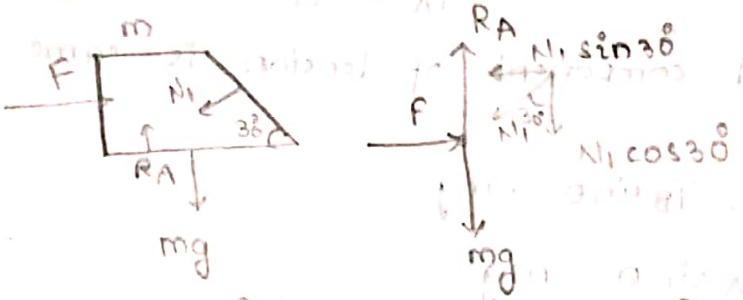
Solving (1) & (2) we get

$T_1$ & $T_2$



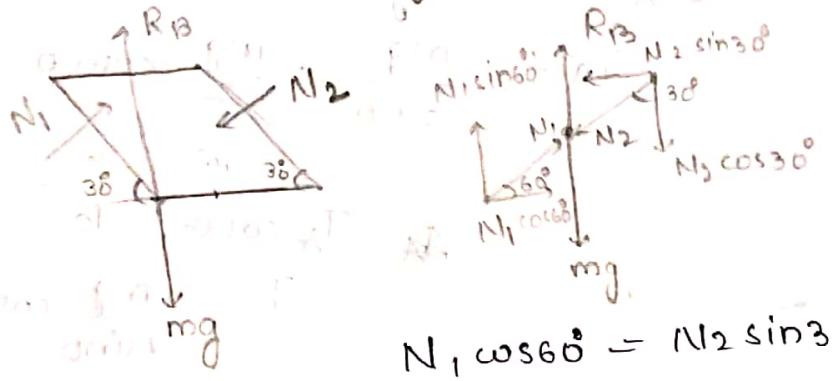

Find $R_A : R_B : R_C$.

∴ System is in equilibrium



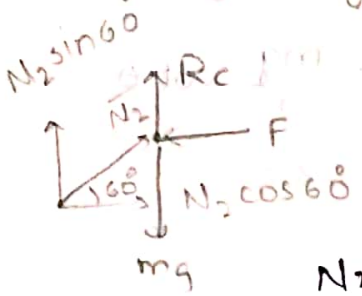

$F = N_1 \sin 30° \rightarrow (1)$

$R_A = mg + N_1 \cos 30° \rightarrow (2)$




$N_1 \cos 60° = N_2 \sin 30° \rightarrow (3)$

$R_B + N_1 \sin 60° = N_2 \cos 30° + mg \rightarrow (4)$




$N_2 \cos 60° = F \rightarrow (5)$

$N_2 \sin 60° + R_C = mg \rightarrow (6)$

Ans $R_A : R_B : R_C = 3 : 1 : 2$.

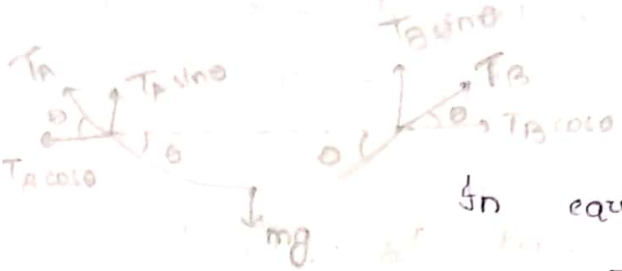
Scanned with CamScanner

## Uniform chain:




Find ① Tension in chain at A & B

② Tension at mid point of chain




In equilibrium

$$T_A \cos\theta = T_B \cos\theta$$

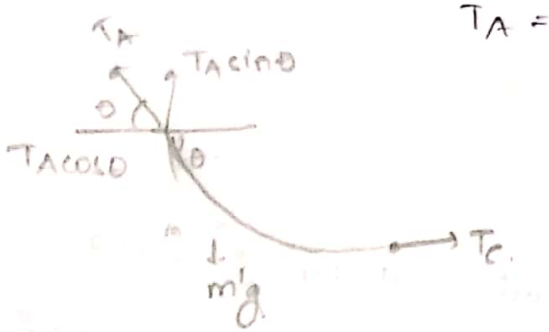
$$T_A = T_B = T \text{ (say)}$$

Clearly Horizontal component of tension is same every where

$$T_A \sin\theta + T_B \sin\theta = mg$$

$$2T_A \sin\theta = mg$$

$$T_A = \frac{mg}{2\sin\theta} = \frac{mg}{2} \operatorname{cosec}\theta$$




$$T_A \cos\theta = T_C$$

$$T_C = \frac{mg}{2\sin\theta} \cos\theta$$

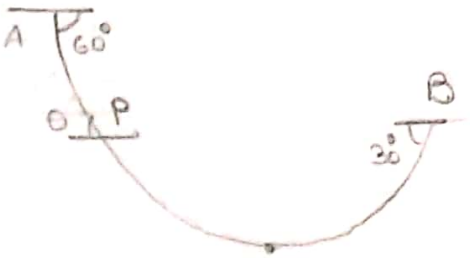
$$\boxed{T_C = \frac{mg}{2} \cot\theta}$$

mass of AC part of chain:

$$T_A \sin\theta = m'g$$

$$\frac{mg}{2\sin\theta} \sin\theta = m'g$$

$$\boxed{m' = m/2}$$



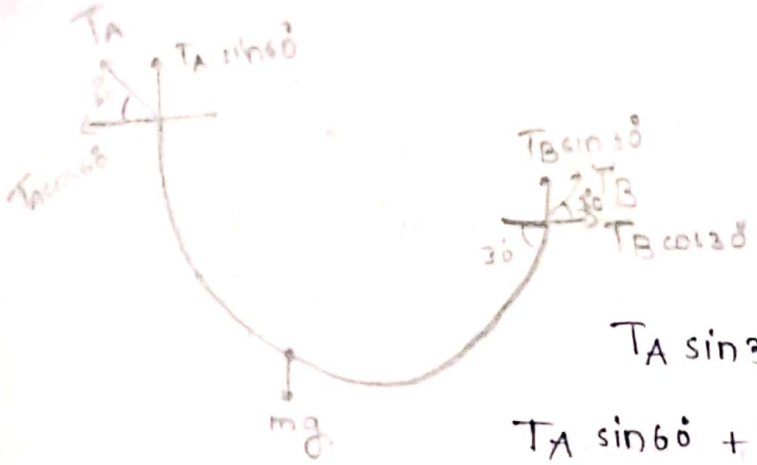

$\theta = 15°$
$\cos 15 = 0.96$
$\sin 15 = 0.25$
} Given

Find ① Tension in string at A & B

② Tension in the range at 'P'

③ mass of chain of part AP.

Scanned with CamScanner




$$T_A \sin 30° = T_B \cos 30° \rightarrow (1)$$

$$T_A \sin 60° + T_B \sin 30° = mg \rightarrow (2)$$

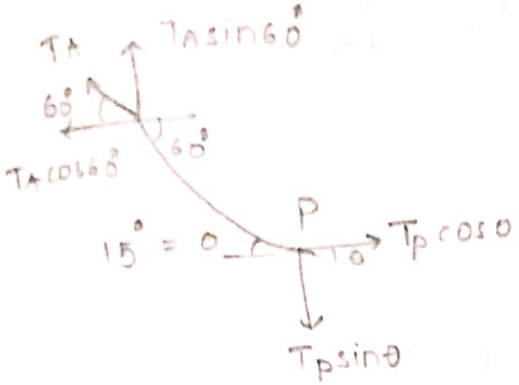
$$T_A = \frac{T_B \cos 30°}{\sin 30°} = T_B \sqrt{3}$$

$$\frac{T_A \sqrt{3}}{2} + \frac{T_B}{2} = mg.$$

$$\frac{3T_B}{2} + \frac{T_B}{2} = mg.$$

$$2T_B = mg$$

$$T_B = \frac{mg}{2} \qquad T_A = \frac{mg\sqrt{3}}{2}$$




$$T_A \cos 60 = T_{BP} \cos 15°$$

$$T_A \sin 60 = m'g + T_A \sin 15°$$

$$\frac{mg\sqrt{3}}{2} \times \frac{1}{2} = T_P\, 0.96$$

$$T_P = \frac{mg\sqrt{3}}{4 \times 0.96} = \frac{mg(1.7)}{3.84}$$

$$T_A \sin 60° = m'g + T_P \sin 15°$$

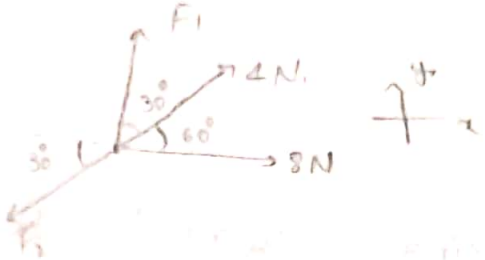
$$m'g = \frac{3(mg)}{4} - \frac{T_A \cos 60°}{\cos 15} \times \sin 15°$$

$$= \frac{3mg}{4} - \frac{mg(1.7)}{3.84} \times 0.25.$$

$$= \frac{mg}{4}\left[3 - \frac{1.7}{3.84}\right].$$

Scanned with CamScanner

Ex:




Four concurrent forces keep the object in equilibrium Find magnitude of $F_1$ & $F_2$

$F_1 = \frac{4}{\sqrt{3}} N \qquad F_2 = \frac{20}{\sqrt{3}} N.$

$F_2 \cos 30^\circ$ ⟷ $8N + 4\cos 60^\circ$.

$F_1 + 4\cos 30^\circ$ ↕ $F_2 \sin 30^\circ$

$$\frac{F_2\sqrt{3}}{2} = 8 + \frac{4}{2}$$

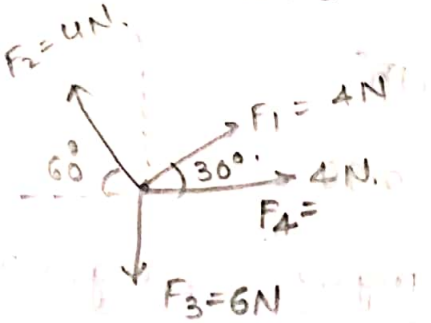
$$\frac{F_2\sqrt{3}}{2} = 10$$

$$F_2 = \frac{20}{\sqrt{3}}.$$

$$\frac{F_2}{2} = F_1 + \frac{4\sqrt{3}}{2}$$

$$\frac{10\times 2}{\sqrt{3}\times 2} - \frac{4\sqrt{3}\times\sqrt{3}}{2\times\sqrt{3}} = F_1.$$

$$\frac{20-12}{2\sqrt{3}} = \frac{8}{2\sqrt{3}} = \frac{4}{\sqrt{3}} = F_1$$




Find the resultant of all these forces

$\bar{F}_{net} = 2\left[(\sqrt{3}+1)\hat{i} + (1+\sqrt{3})\hat{j}\right]$

$$F_y = F_2\sin 60^\circ \hat{j} + F_1 \sin 30^\circ \hat{j} - 6\hat{j}$$

$$= 2\sqrt{3}\,\hat{j} + 2\hat{j} - 6\hat{j}$$

$$= 2\sqrt{3}\,\hat{j} - 4\hat{j}$$

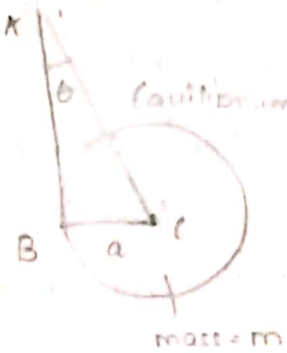
$$F_x = F_2\cos 60^\circ(-\hat{i}) + F_1\cos 30^\circ(\hat{i}) + 4\hat{i}$$

$$= 4/2\,(-\hat{i}) + \frac{4\sqrt{3}}{2}\hat{i} + 4\hat{i}$$

$$= 2 + 2\sqrt{3}\,\hat{i}$$

$$= 2(1+\sqrt{3})\,\hat{i}$$

Scanned with CamScanner




BC = a = radius of sphere.

AC = 2a = length of string

Find ① tension in the string.

② Normal force exerted by wall on sphere.

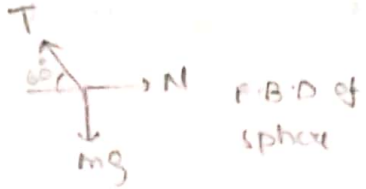



$$\sin\theta = \frac{a}{2a} = \frac{1}{2}$$

$$\theta = 30^\circ$$

$$T = \frac{2mg}{\sqrt{3}}$$

$$N = \frac{mg}{\sqrt{3}}$$



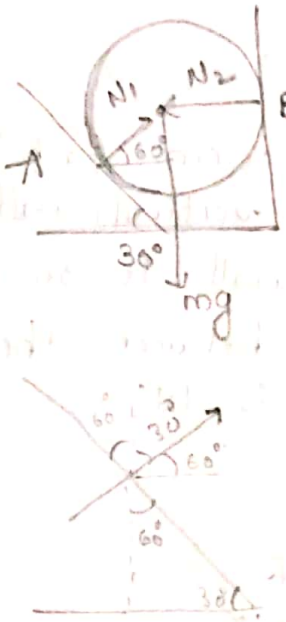

$$T\sin 60^\circ = mg$$

$$T = \frac{mg^2}{\sqrt{3}}$$

$$N = T\cos 60^\circ$$

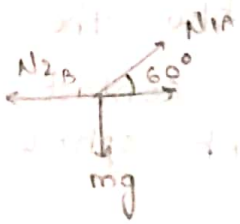
$$= T/2 = \frac{mg}{\sqrt{3}}$$

$m = 50\,kg$




Find the contact force at A & B

$N_2 = \frac{1000}{\sqrt{3}}\,N$ , $N_B = \frac{500}{\sqrt{3}}\,N_1$




$N_A \sin 60 = mg$

$N_A \sin 60 = mg$

$N_A = N_A \frac{mg^2}{\sqrt{3}\sqrt{3}}$

$= \frac{50 \times 2 \times 10}{\sqrt{3}\sqrt{3}}$

$= 1000$




$N_2 = N_1 \cos 60^\circ$

$= \frac{N_1}{2}$ $\frac{N_1}{2}$

$= \frac{1000}{2\sqrt{3}}\,\frac{500}{\sqrt{3}}$

$= \frac{500}{\sqrt{3}}$

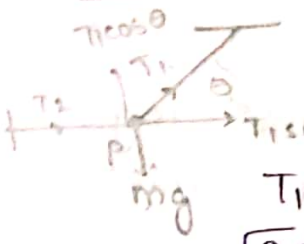
$= \frac{1000}{\sqrt{3}}$

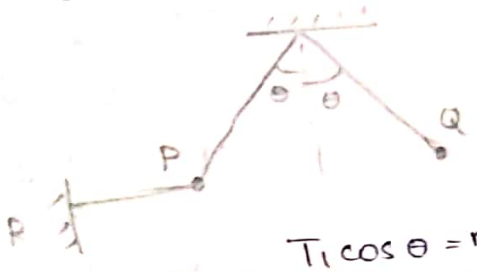
$= \frac{1000}{\sqrt{3}}$

Scanned with CamScanner

A ball of mass 1kg is at rest in position P by means of 2 light strings OP & RP. The string RP is now cut & the ball swings to position Q. If $\theta = 45°$ find the tension in strings in Positions OP when RP was not cut OQ when RP was cut.

Case i:




$T_1 \sin\theta = T_2$

$\sqrt{2}\, mg \times \frac{1}{\sqrt{2}} = T_2 \Rightarrow T_2 = mg.$

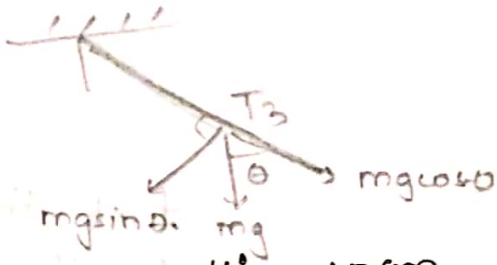
$T_1 \cos\theta = mg$

$T_1 = \sqrt{2}mg.$

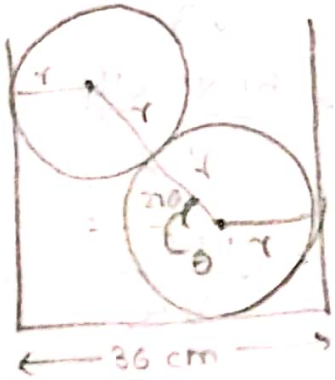
Case ii:

$T_3 = mg\cos\theta$

$T_3 = \frac{mg}{\sqrt{2}}$




→ Two identical spheres of radii 10cm each is placed in between two rigid vertical walls as shown. The space b/w the wall is 36 cm then find the force of contact between the spheres if weight of each sphere is "W".



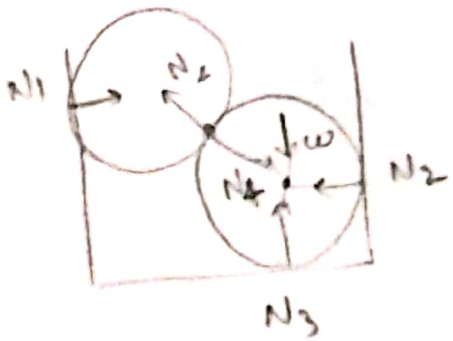

$r + 2r\cos\theta + r = 36$

$2r(1+\cos\theta) = 36$

$1+\cos\theta = \frac{36}{20} \overset{18}{\cancel{}}_{10}$

$\cos\theta = \frac{4}{5}$

$\theta = 37°$



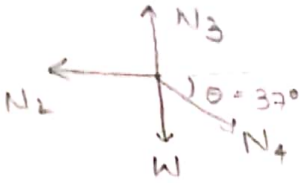

Two spheres are system

$N_1 = N_2 \rightarrow (1)$

& $2W = N_3 \rightarrow (2)$

Scanned with CamScanner




for 1st sphere

$N_4 \cos 37^\circ = N_1$.

$N_4 \sin 37^\circ = W$.

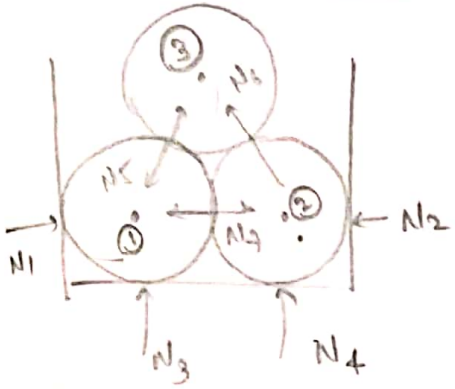
$N_4 \times \frac{3}{5} = W$

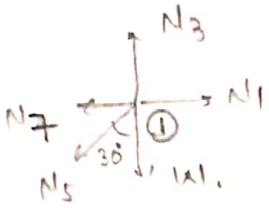
$N_4 = \frac{5W}{3}$

$N_2 = N_4 \cos 37^\circ = N_1$

$N_3 = N_4 \sin 37^\circ + W$.

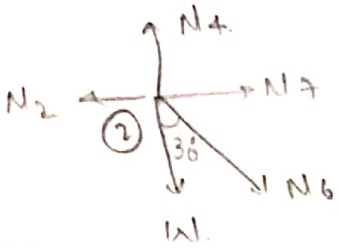
$N_3 = 2W$.



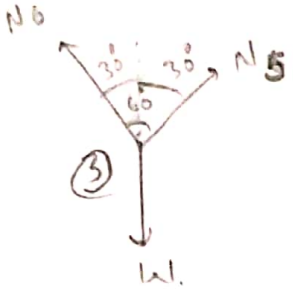




$N_7 + N_5 \sin 30^\circ = N_1 \rightarrow ①$

$N_3 = W + N_5 \cos 30^\circ \rightarrow ②$




$N_4 = N_6 \cos 30^\circ + W$ → ③ | $N_7 = N_2 + N_6 \sin 30^\circ$ → ④.




$(N_5 + N_6) \cos 30^\circ = W$.

$N_5 + N_6 = \frac{2W}{\sqrt{3}.} \rightarrow ⑤$

$N_1 = N_2 \rightarrow ⑥$.

$N_3 + N_4 = 3W \rightarrow ⑦$.

$N_5 = W/\sqrt{3}$.

$N_3 = 3W/2$

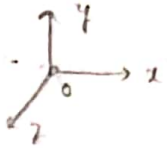
$N_4 = 3W/2$

$N_6 = W/\sqrt{3}$.

Scanned with CamScanner

# Frame of reference

↗ coordinate axes system used to locate the position of body w.r.to. time




① Inertial frame of frame

Any frame of reference which is either at rest or moving with constant velocity is called inertial frame.

→ Earth → Inertial frame of reference.



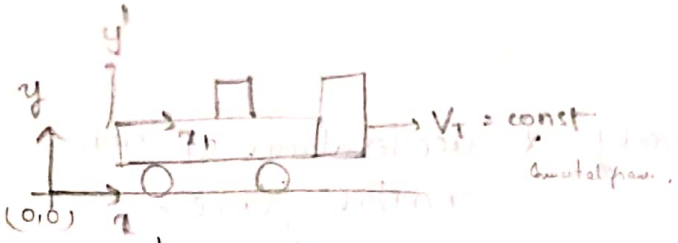

Real forces →
- Gravitational forces
- muscular force
- Spring force, Tension
- Normal reaction, friction

$\Sigma \bar{F}_{real} = m\bar{a}$

↓ Newton's law are valid in inertial frame.

② Non-inertial frame of reference:

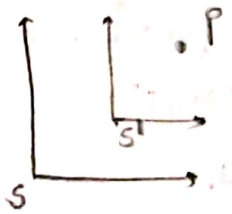
Any frame of reference which are either accelerating or rotating about their own axes w.r.to inertial frame are called non-inertial frames.




In non inertial frame,

$\Sigma \bar{F}_{real} \neq m\bar{a}'$

$\bar{a}'$ = acceleration of body w.r.to truck.




let

$\bar{a}_{p/s'} = \bar{a}$ = acceleration of body w.r.to. non inertial frame.

Scanned with CamScanner

& $\vec{a}_{s's} = \vec{a}_0$ = acc. of non inertial frame w.r.to inertial frame.

we have

$$\vec{a}_{P,S'} = \vec{a}_{P,S} + \vec{a}_{S,S'}$$

$$\vec{a}_{P,S'} = \vec{a}_P - \vec{a}_{S'}$$

$$\vec{a}_{PS} + \vec{a}_{SS'} = \vec{a}_P - \vec{a}_S + \vec{a}_S - \vec{a}_{S'}$$

$$\vec{a} = \vec{a}_{P,S} - \vec{a}_{s's}$$

$$\vec{a} = \vec{a}_{P,S} - \vec{a}_0$$

$$M\vec{a} = M\vec{a}_{PS} - M\vec{a}_0$$

$$M\vec{a} = M\vec{a}_{P,S} + (M\vec{a}_0)$$

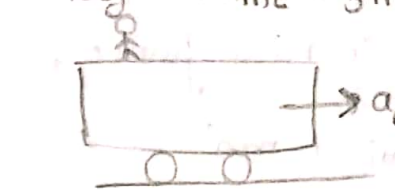
$$M\vec{a} = \Sigma \vec{F}_{real} + \vec{F}_{pseudo\ force}$$

Thus newton's are valid.

## Pseudo force:

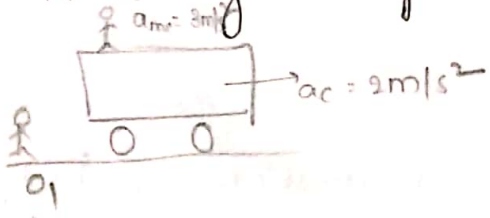
1) It is observed only in "non inertial frame"
2) $F_{pseudo}$ is always "opposite to acceleration" of frame
3) $F_{pseudo}$ = (mass of body) × acceleration of non-inertial frame.

Ex:




Find the force acting on the man?

1) W.r.to ground frame.




now, $\bar{a}_m = \bar{a}_{mc} + \bar{a}_c$

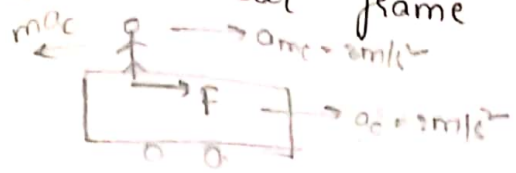
$\bar{a}_m = 3\hat{i} + 2\hat{i} = 5\hat{i}$

$= 5\ m/s^2$

Force $F = ma_m = 70(5) = 350\ N.$

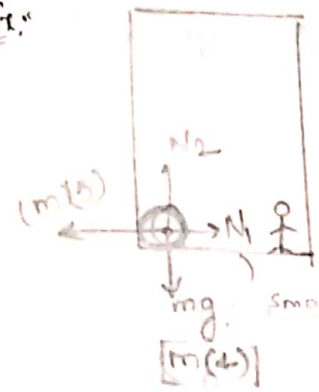
Scanned with CamScanner

2) w.r.to car frame




$$F - ma_c = ma_{mc}$$
$$F - 70(2) = 70(3)$$
$$F = 70 \times 5 = 350\,N$$

Ex:



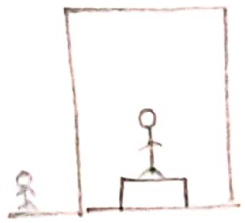

$$\bar{v} = 5t\hat{i} + 4t\hat{j}$$
$$\bar{a} = \frac{d\bar{v}}{dt} = 5\hat{i} + 4\hat{j}$$

$N_1 = 5m$

$N_2 = 4m + mg = 14m$

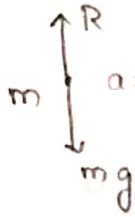
$$N = \sqrt{N_1^2 + N_2^2}$$

## Apparent weight of a man in a lift:



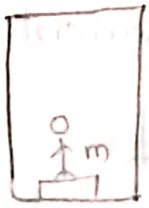
Apparent weight = Normal reaction
= weighing machine reading

1) If lift is at rest | moving with const velocity

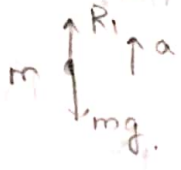



$$\Rightarrow R = mg$$

2) lift accelerates upwards | decelerates downwards.




w.r.to ground.




$$R_1 - mg = ma$$
$$R_1 = ma + mg$$
$$R_1 > mg$$
$$\& \; R_1 = mg\left(1 + \frac{a}{g}\right)$$

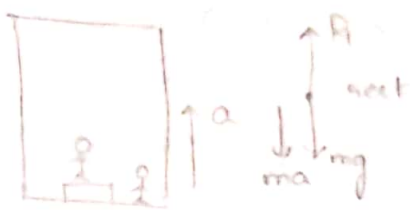



* % error in machine reading

$$\frac{R_1 - mg}{mg} \times 100 = \frac{ma}{mg} \times 100 = \frac{a}{g} \times 100\%$$

Scanned with CamScanner

w.r.to lift frame:

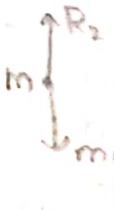



$R_1 = ma + mg.$

3) lift accelerates downwards / decelerates upwards:




w.r.to ground.




$mg - R_2 = mg\,a.$

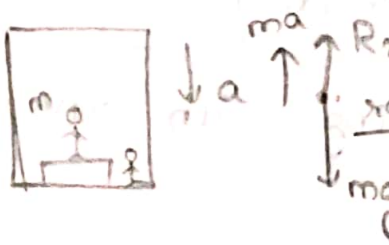
$R_2 = (mg - ma)$

$\& \; R_2 = mg\left(1 - \frac{a}{g}\right)$

% error in machine reading,

$$\frac{R_2 - mg}{mg} \times 100 = \frac{-ma}{mg} \times 100 = \frac{-a}{g} \times 100\%.$$

w.r.to lift frame:




$ma + R_2 = mg$

$R_2 = mg - ma.$

Point:

1) If $a = g$ downwards. then app wt $= 0$
lift is in free fall.

2) If $a < g$ (downwards) then displacement of lift $= \frac{1}{2}at^2$ $[u = 0]$.

displacement of man $= \frac{1}{2}at^2$

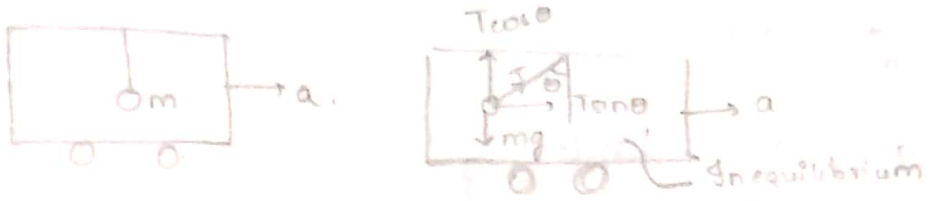
3) If $a > g$ (downwards
then displacement of lift $= \frac{1}{2}at^2$
& displacement of man $= \frac{1}{2}gt^2$

Scanned with CamScanner

After some time head will hit the ceiling of lift.

Important modele:

1).




w.r.to ground.

$T\sin\theta = ma$

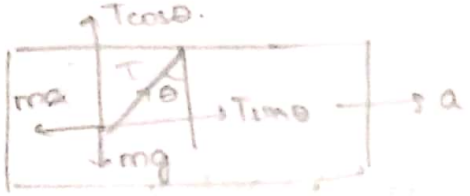
$T\cos\theta = mg.$

& $\tan\theta = a/g.$

$\theta = \tan^{-1}\left(\frac{a}{g}\right)$

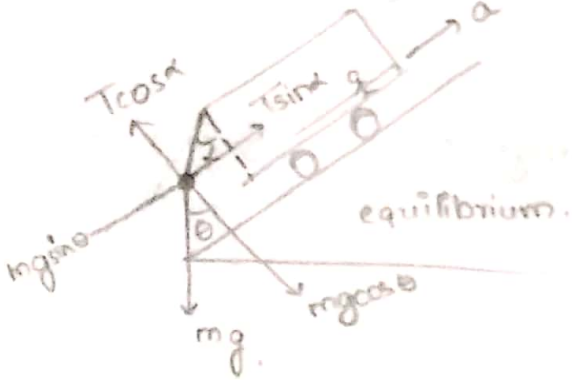
$T = \sqrt{(ma)^2 + (mg)^2}$

$= m\sqrt{a^2+g^2}$




$T\sin\theta = ma$

$T\cos\theta = mg.$



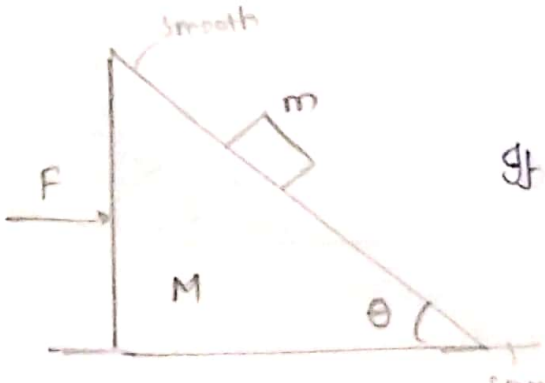

$T\sin\alpha = ma + mg\sin\theta$

$T\cos\alpha = mg\cos\theta.$

$\tan\alpha = \frac{a + g\sin\theta}{g\cos\theta.}$

$\alpha = \tan^{-1}\left(\frac{a+g\sin\theta}{g\cos\theta}\right)$

→




If block is at rest w.r.to wedge

Find ① acceleration of wedge

② Force applied on the wedge

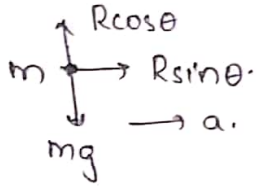
③ Reaction force on block by wedge.

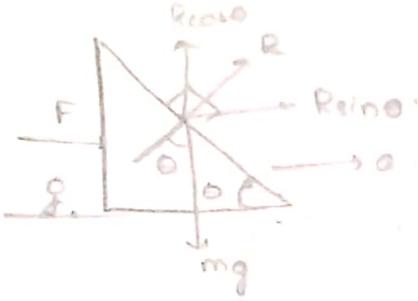
Scanned with CamScanner

I - method

w.r.to ground.

F.B.D of block.




$R\sin\theta = ma$

$R\cos\theta = mg$

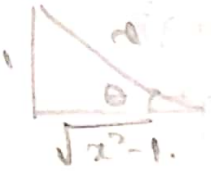
$\tan\theta = \frac{a}{g}$

$a = g\tan\theta$.

If incline is 1 in x.

$\sin\theta = \frac{1}{x}$

$\tan\theta = \frac{1}{\sqrt{x^2-1}}$




$a = \frac{g}{\sqrt{x^2-1}}$.

② $F = (M+m)a = (m+m)g\tan\theta$ //

③ $R\cos\theta = mg$

$R = \frac{mg}{\cos\theta} = mg\sec\theta$ //

II method :

w.r.to wedge

$ma\cos\theta = mg\sin\theta$.

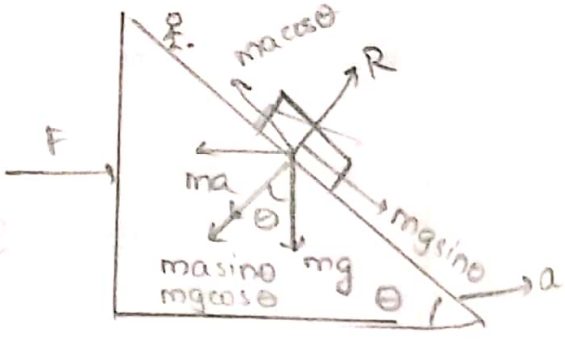
$\tan\theta = \frac{a}{g}$.

& $R = ma\sin\theta + mg\cos\theta$

$a = g\sec\theta$

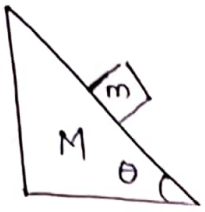
& $R = (M+m)a$

$= (M+m)g\tan\theta$.

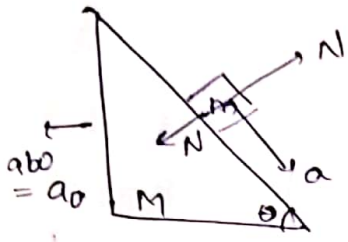



Scanned with CamScanner

(1).




All surfaces are smooth. Find acceleration of block w.r.t to wedge if $a_0$ is acceleration of wedge.

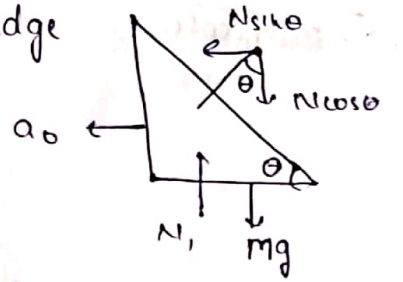



a = acceleration of block w.r.t to wedge. = $a_{bw}$

M - I

From ground frame:

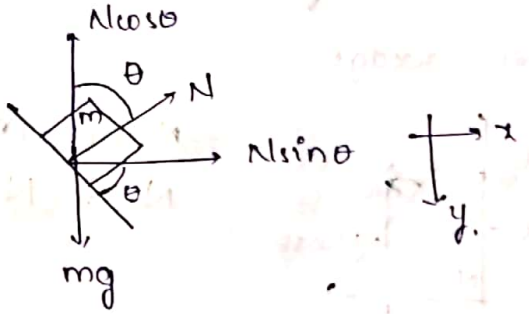
For wedge




$N\sin\theta = Ma_0 \rightarrow (1)$.

$N_1 = N\cos\theta + Mg \rightarrow (2)$

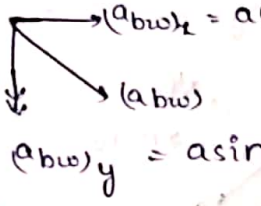
For block




$\bar{a}_b = \bar{a}_{bw} + \bar{a}_w$

$(\bar{a}_b)_x = (\bar{a}_{bw})_x + (\bar{a}_w)_x$

$\bar{a}_{bx} = (a\cos\theta - a_0)\hat{i}$




$(a_b)_x = a\cos\theta - a_0$

& $(\bar{a}_b)_y = (\bar{a}_{bw})_y + (\bar{a}_w)_y$

$= a\sin\theta + 0$

$a_{by} = a\sin\theta$

now, $N\sin\theta = m(a_b)_x$

$N\sin\theta = m(a\cos\theta - a_0) \rightarrow (3)$.

& $mg - N\cos\theta = m(a_b)_y$

$mg - N\cos\theta = ma\sin\theta \rightarrow (4)$.

From eq. (3) & (1) $Ma_0 = m(a\cos\theta - a_0)$

Scanned with CamScanner

$$Ma_0 = ma\cos\theta - ma_0$$

$$(M+m)a_0 = ma\cos\theta$$

$$a = \frac{(M+m)a_0}{m\cos\theta}$$

- net acceleration of block,

$$\bar{a}_b = \bar{a}_{bx} + \bar{a}_{by}$$

$$a_b = \sqrt{a_{bx}^2 + a_{by}^2}$$

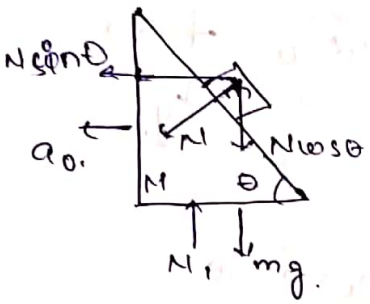
$$a_b = \sqrt{(a\cos\theta - a_0)^2 + (a\sin\theta)^2}$$

$$a_b = \sqrt{a^2 + a_0^2 - 2aa_0\cos\theta}$$

M - II

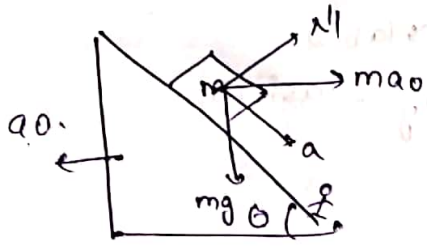
from wedge frame.

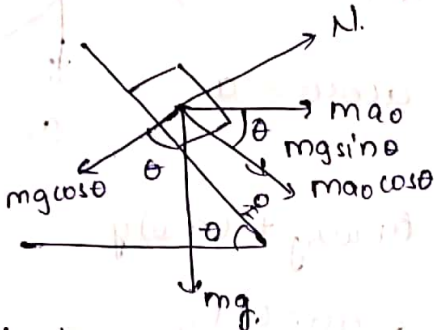
For wedge.




$N\sin\theta = Ma_0$

$N_1 = N\cos\theta + Mg$.




Along the inclined plane

$$ma_0\cos\theta + mg\sin\theta = ma \rightarrow (3)$$

Ier to inclined plane.

$$mg\cos\theta = N + ma_0\sin\theta \rightarrow (4)$$

From eq (4) $mg\cos\theta = \frac{Ma_0}{\sin\theta} + ma_0\sin\theta$ $N\sin\theta = Ma_0$

$$mg\cos\theta = a_0\left(\frac{M + m\sin^2\theta}{\sin\theta}\right)$$

Scanned with CamScanner

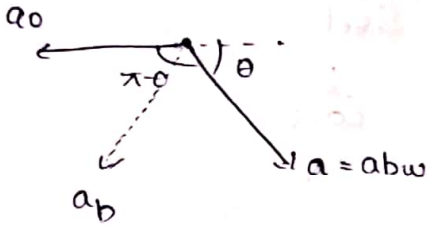
$$g = \frac{a_0(M + m\sin^2\theta)}{m\cos\theta\sin\theta}$$

eq ③ becomes

$$ma_0\cos\theta + m\frac{a_0(M+m\sin^2\theta)}{m\cos\theta\sin\theta} \times \sin\theta = ma$$

$$ma_0\cos^2\theta + Ma_0 + ma_0\sin^2\theta = a\cdot m\cos\theta$$

$$a = \frac{(M+m)a_0}{m\cos\theta}$$

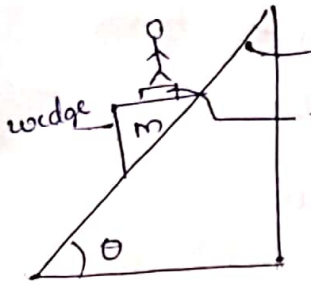



$$\bar{a}_b = \bar{a}_{bw} + \bar{a}_w$$

$$a_b = \sqrt{a_w^2 + a_{bw}^2 + 2a_{bw}a_w\cos(\pi-\theta)}$$

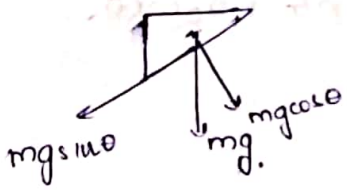
$$a_b = \sqrt{a_0^2 + a^2 - 2aa_0\cos\theta}$$

Eg:




man is at rest w.r.to wedge

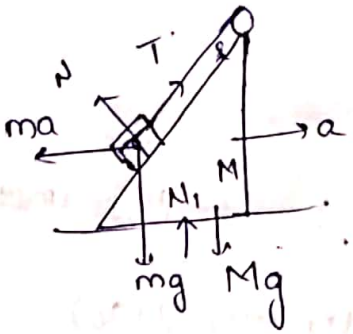
Find weighing machine reading.




∴ acc of block = $g\sin\theta$

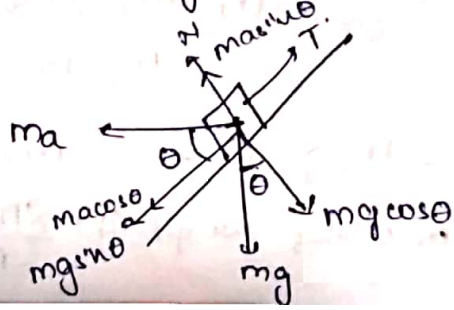
$$mg\sin^2\theta + N = ma$$

$$N = mg(1-\sin^2\theta) = mg\cos^2\theta$$




Find acceleration of wedge for which block just looses contact

Sol. $\boxed{N = 0}$




$$N + ma\sin\theta = mg\cos\theta$$

$$N = 0$$

$$ma\sin\theta = mg\cos\theta$$

$$a = g\cot\theta.$$

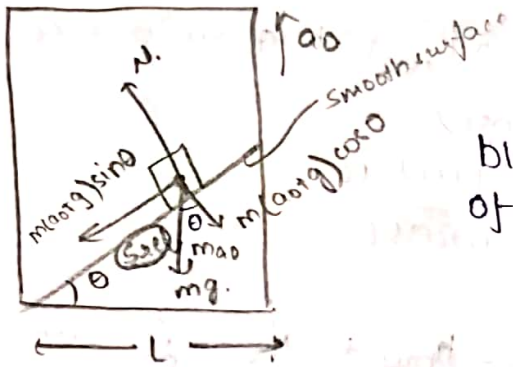
Scanned with CamScanner

$$T = ma\cos\theta + mg\sin\theta$$

$$= mg\cot\theta\cos\theta + mg\sin\theta$$

$$T = mg\,\mathrm{cosec}\,\theta.$$

Ex.




Find the time taken by block to reach bottom of plane.

$$\cos\theta = \frac{L}{S_{rel}}$$

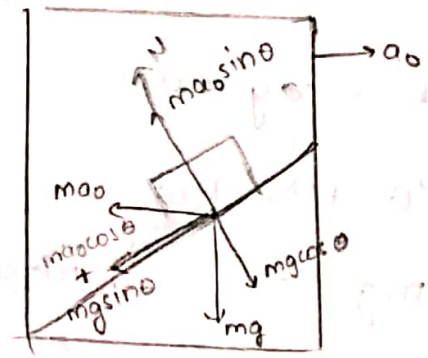
$$S_{rel} = \frac{L}{\cos\theta}$$

$$a_{rel} = (g+a_0)\sin\theta$$

$$S_{rel} = u_{rel}\,t + \frac{1}{2}a_{rel}t^2$$

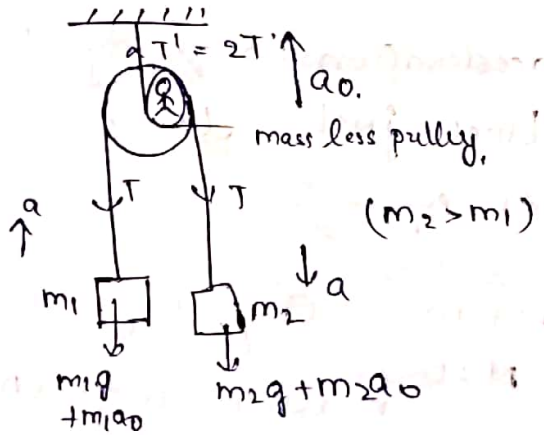
$$\frac{L}{\cos\theta} = \frac{1}{2}(g+a_0)\sin\theta\, t^2$$

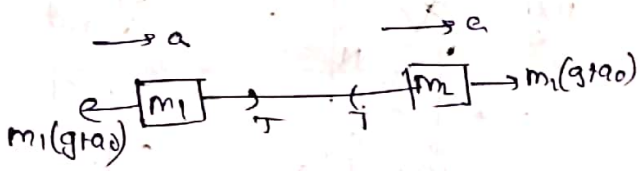
$$t = \sqrt{\frac{2L}{(g+a_0)\sin\theta\cos\theta}}$$




& $a_{rel} = g\sin\theta + a_0\cos\theta$.

Ex:



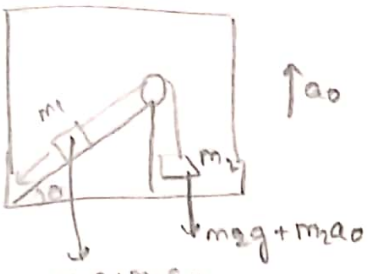



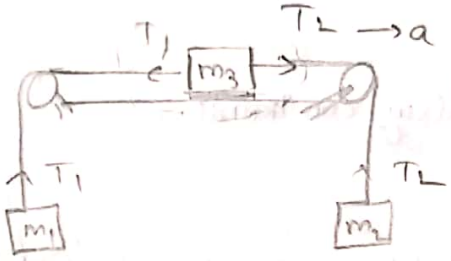

$$a = \frac{(m_2-m_1)(g+a_0)}{m_1+m_2}$$

$$m_2g + a_0 - T = m_2a$$

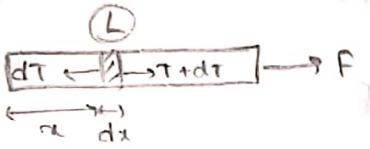
On solving we get $T = \frac{2m_1m_2(g+a_0)}{m_1+m_2}$

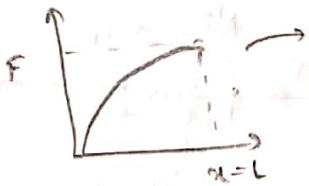
Scanned with CamScanner




$$a = \frac{(m_2 - m_1)(g)}{m_1 + m_2 + m_3}$$




F-x graph is as shown for rod then P of rod will be.

$$dT = (dm)a$$

$$dT = (A\,dx)\rho a$$

$$\frac{dT}{dx} = A\rho a$$

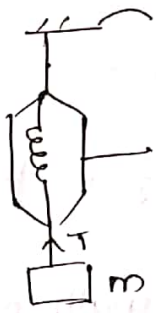
A, a ↳ constant

$$\frac{dT}{dx} \propto \rho$$

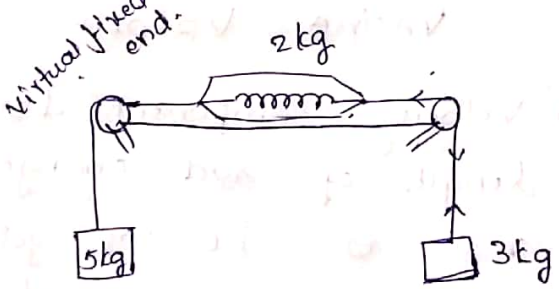
From graph

$$\frac{dT}{dx}(\downarrow) \Longrightarrow \rho(\downarrow)$$

## Spring balance




reading (T) = mg (Newton) or m in kgwt [1 kgwt = 9 N]




$$a = \frac{5g - 3g}{10} = \frac{g}{5} \rightarrow (1)$$

$$T - 3g = 3a$$

$$T = 3(g + a) = 3\left(g + \frac{g}{5}\right)$$

$$= 3 \times \frac{6g}{5} = \frac{18}{5}g$$

$$= 3.6\,g\,N$$

$$= 3.6 \text{ kgwt}$$

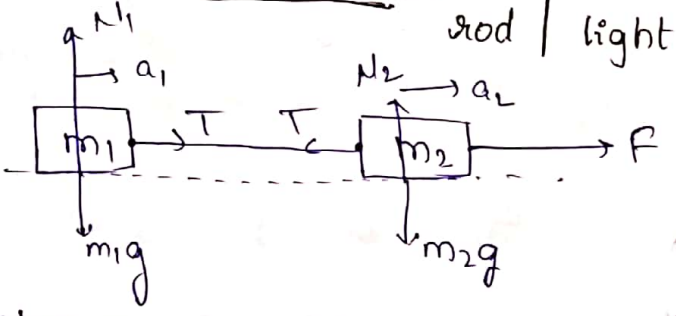
Scanned with CamScanner

# Constraint motion

- ① General constraints ✓
- ② String constraints ✓ ★
- ③ Pulley constraints
- ④ Wedge constraints
- ⑤ Combination, string & pulley constraints.

## General Constraints :

1→ rod / light inextensible string /




unknowns

$N_1, N_2, a_1, a_2, F$

String is inextensible

$V_1 = V_2$ & $a_1 = a_2$

i.e Along the length of string, acceleration, velocity components are equal

$N_1 = m_1 g \rightarrow ①$

$N_2 = m_2 g \rightarrow ②$

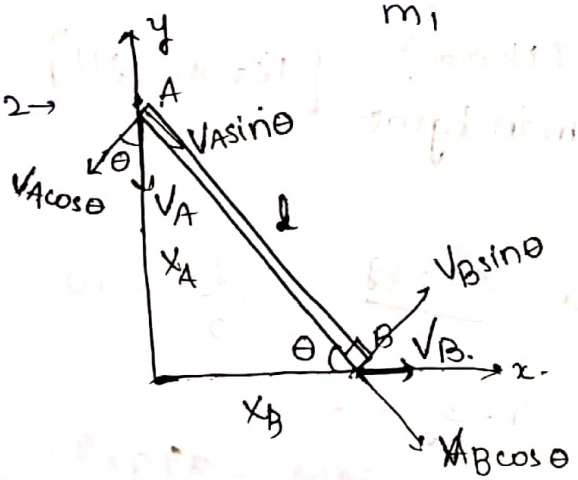
$T = m_1 a_1 \rightarrow ③$

$F - T = m_2 a_2 \rightarrow ④$

ie $a_1 = a_2 \rightarrow ⑤$

$\frac{T}{m_1} = \frac{F-T}{m_2}$ → we get T from this.

2→




★ Along the length of rod,

$V_A \sin\theta = V_B \cos\theta$

★ Velocity components ⊥er to length of rod changes angle θ. i.e we get angular velocity of rod.

## ∫∫ method

$l^2 = X_A^2 + X_B^2 = \text{const}$

Scanned with CamScanner

i.e, we get angular velocity of rod.

II method)

$$l^{\vec{r}} = x_A^{\vec{r}} + x_B^{\vec{r}} \qquad \left[\because l = \text{const} \quad \frac{dl}{dt} = 0\right]$$

$$\cancel{l} \cdot \frac{dl}{dt}^{\to 0} = \cancel{x_A} \cdot \frac{dx_A}{dt} + \cancel{x_B} \cdot \frac{dx_B}{dt}$$

$$0 = x_A(-v_A) + x_B(v_B)$$

$$x_A v_A = x_B v_B$$

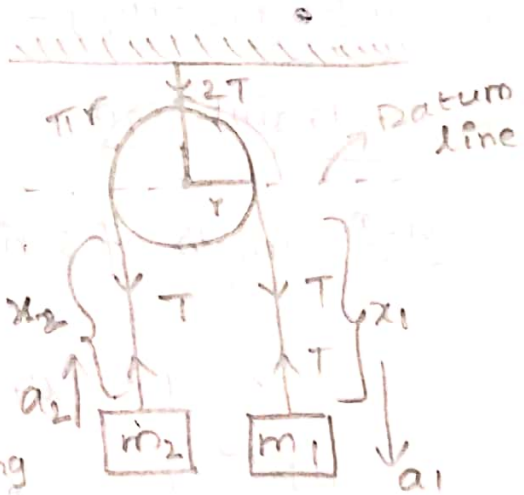
$$v_A = \frac{x_B}{x_A} v_B$$

$$v_A = \tan(90-\theta) v_B$$

$$v_A = \cot\theta \, v_B$$

$$\boxed{v_A \sin\theta = v_B \cos\theta}$$

(2) String constraints




unknowns $T, a_1, a_2$

Total length of string (inextensible)

$$l = x_1 + x_2 + \pi r = \text{const.}$$

$$\frac{dl}{dt}^{\to 0} = \frac{dx_1}{dt} + \frac{dx_2}{dt} + \pi \frac{dr}{dt}^{\to 0}$$

($\frac{dx_1}{dt}$: increasing, $\frac{dx_2}{dt}$: decreasing)

$$-v_2 + v_1 = 0 \implies v_1 = v_2 \implies \boxed{a_1 = a_2}$$

$$m_1 g - T = m_1 a_1 \quad -(1)$$

$$T - m_2 g = m_2 a_2 \quad -(2)$$

$$a_1 = a_2 \quad -(3)$$

$$\frac{m_1 g - T}{m_1} = \frac{T - m_2 g}{m_2}$$

Scanned with CamScanner

II method :

** Virtual work method

Since string is inextensible net work done by tension is zero.

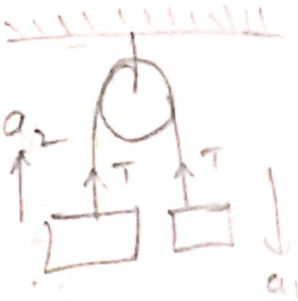
$$\sum \bar{T}\cdot\bar{x} = 0 \Rightarrow \sum T x \cos\theta = 0$$

$$\sum \bar{T}\cdot\bar{v} = 0 \Rightarrow \sum T v \cos\theta = 0$$

$$\sum \bar{T}\cdot\bar{a} = 0 \Rightarrow \sum T a \cos\theta = 0$$

$$\sum \bar{T}\,\bar{a} = 0$$

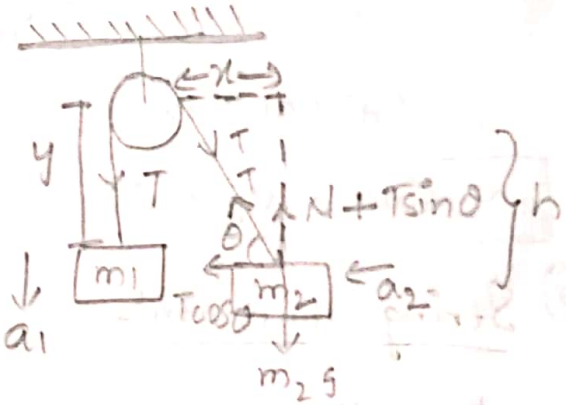
$$-Ta_1 + Ta_2 = 0 \Rightarrow \boxed{a_1 = a_2}$$




$$m_1 g - T = m_1 a_1 \quad -(1)$$

$$T\cos\theta = m_2 a_2 \quad -(2)$$

$$T\sin\theta + N = m_2 g \quad -(3)$$




constraint eqn

M-I length of string

unknowns, $N, T, a_1, a_2$

$$l = y + \sqrt{x^2+h^2} = \text{const} \quad [h = \text{const}]$$

$$\frac{dl}{dt} = \frac{dy}{dt} + \frac{x}{\sqrt{x^2+h^2}}\left(\frac{dx}{dt}\right)$$

$$\left[V_1 = \frac{dy}{dt}, V_2 = \frac{-dx}{dt}\right]$$

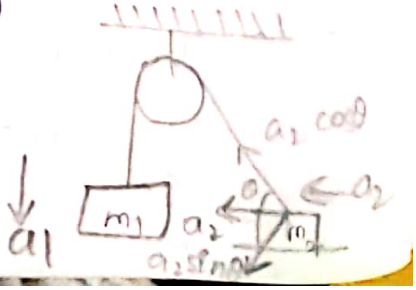
$$0 = V_1 - V_2\cos\theta$$

$$V_1 = V_2\cos\theta \Rightarrow \boxed{a_1 = a_2\cos\theta} \quad -(3)$$

M-II

acceleration component along the string are equal.

$$\boxed{a_2\cos\theta = a_1}$$



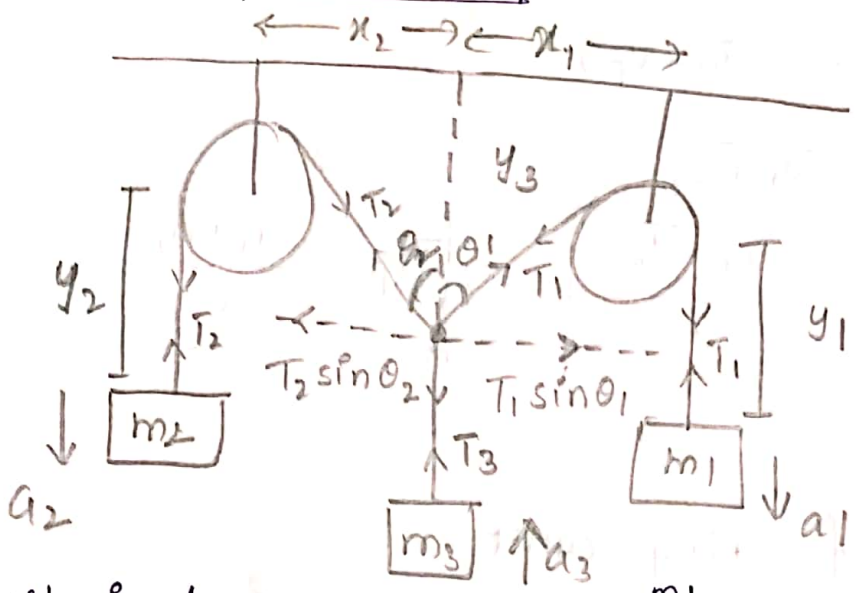

Scanned with CamScanner

II Method

$$\sum \bar{T} \cdot \bar{a} = 0$$

$$-Ta_1 + Ta_2\cos\theta = 0$$

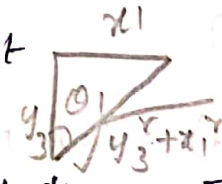
$$\Rightarrow \boxed{a_1 = a_2\cos\theta}$$




constrained eqn

Method - I

$$l_1 = \sqrt{x_1^2 + y_3^2} + y_1 = \text{const}$$




$$\frac{dl_1}{dt} = \frac{2y_3}{2\sqrt{x_1^2+y_3^2}} \times \frac{dy_3}{dt} + \frac{dy_1}{dt}$$

$$V_1 - V_3\cos\theta_1 = 0$$

$$V_1 = V_3\cos\theta_1$$

$$\Rightarrow \boxed{a_1 = a_3\cos\theta_1} \quad -(6)$$

** when $x_1 = x_2$

$\theta_1 = \theta_2$

$\Rightarrow \boxed{T_1 = T_2}$

Method II

$$a_1 = a_3\cos\theta_1$$

$$a_2 = a_3\cos\theta_2$$

$m_1 g - T_1 = m_1 a_1 \quad -(1)$

$m_2 g - T_2 = m_2 a_2 \quad -(2)$

$T_3 - m_3 g = m_3 a_3 \quad -(3)$

$T_1\sin\theta_1 = T_2\sin\theta_2 \quad -(4)$

$T_1\cos\theta_1 + T_2\cos\theta_2 = T_3 \quad -(5)$

and $l_2 = \sqrt{x_2^2 + y_3^2} + y_2$

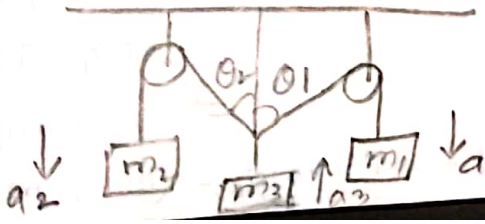
$$\frac{dl_2}{dt} = \frac{dy_2}{dt} + \frac{2y_3}{2\sqrt{x_2^2+y_3^2}} \cdot \frac{dy_3}{dt}$$

$$= 0$$

$$V_2 - V_3\cos\theta_2 = 0$$

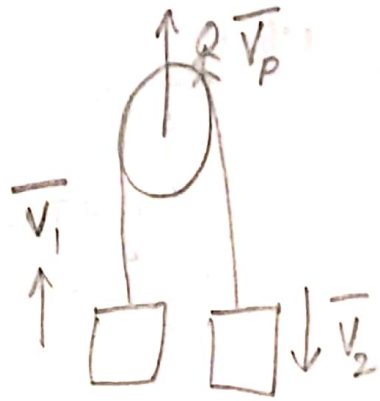
$$V_2 = V_3\cos\theta_2$$

$$\boxed{a_2 = a_3\cos\theta_2} \quad -(7)$$




Scanned with CamScanner

## ③ Pulley constraints




w.r.to pulley

$$\bar{V}_{1P} = -\bar{V}_{2P}$$

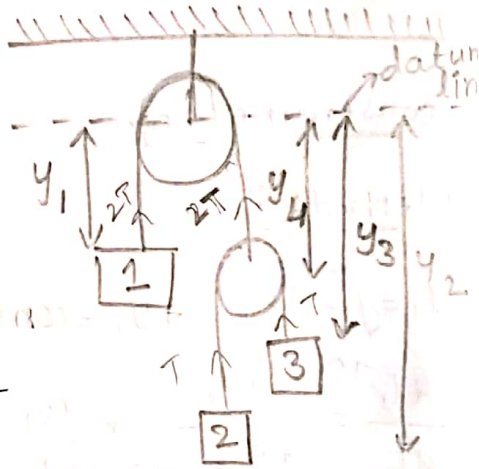
$$\bar{V}_1 - \bar{V}_p = -(\bar{V}_2 - \bar{V}_p)$$

$$2\bar{V}_p = \bar{V}_1 + \bar{V}_2$$

$$\boxed{\bar{V}_p = \frac{\bar{V}_1 + \bar{V}_2}{2}} \Rightarrow \boxed{\bar{a}_p = \frac{\bar{a}_1 + \bar{a}_2}{2}}$$

*we should apply sign conventions*.

length of string = const




$$y_1 + y_4 = \text{const}$$

$$v_1 + v_4 = 0$$

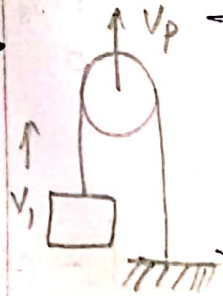
$$v_1 = -v_4 \quad -①$$

$$(y_2 - y_4) + (y_3 - y_4) = \text{const}$$

$$y_2 + y_3 - 2y_4 = \text{const}$$

$$v_2 + v_3 - 2v_4 = 0$$

$$\boxed{v_4 = \frac{v_2 + v_3}{2}}$$




$$\bar{V}_p = \frac{\bar{V}_1 + 0}{2}$$

$$\boxed{2\bar{V}_p = \bar{V}_1}$$

Method - II

Virtual work method

we have,

$$\sum T \cdot \bar{a} = 0$$

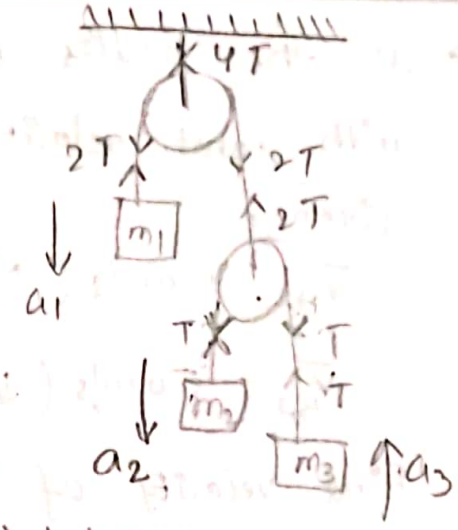
$$-2Ta_1 - Ta_2 + Ta_3 = 0$$

$$\Rightarrow a_3 = 2a_1 + a_2 \quad -①$$

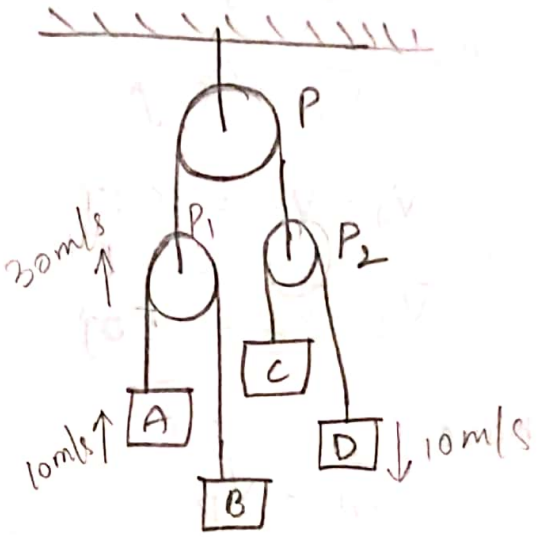
* $m_1 g - 2T = m_1 a_1$ —② , $m_2 g - T = m_2 a_2$ —③

Scanned with CamScanner

$T - m_3 g = m_3 a_3$ —— (4)




→ A lift goes up with $a = 10 m/s^2$, $V_{P_1} = 30 m/s \uparrow$, $V_A = 10 m/s \uparrow$, $V_D = 10 m/s \downarrow$. Find $V_B = ?$, $V_C = ?$




$$V_{P_1 P} = -V_{P_2 P}$$

$$V_{P_1} - V_P = V_P - V_{P_2}$$

$$2V_P = V_{P_1} + V_{P_2}$$

$$V_P = \frac{V_{P_1} + V_{P_2}}{2}$$

$$10\uparrow = \frac{30\uparrow + V_{P_2}}{2}$$

$$20\uparrow = 30\uparrow + V_{P_2}$$

$$\boxed{V_{P_2} = -10\uparrow}$$

(1) $V_{AP_1} = -V_{BP_1}$

$\Rightarrow V_A - V_{P_1} = V_{P_1} - V_B$

$10\uparrow - 30\uparrow = 30\uparrow - V_B$

$\boxed{V_B = 50\uparrow}$

(2) $V_{CP_2} = -V_{DP_2}$

$V_C - V_{P_2} = V_{P_2} - V_D$

$\Rightarrow V_C + V_D = 2V_{P_2}$

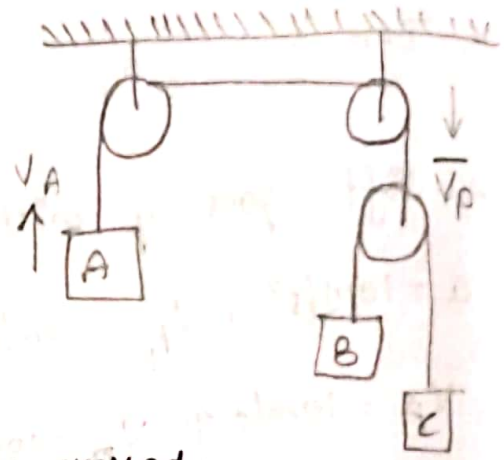
$V_C + 10(-\uparrow)$

$= -20\uparrow$

$\boxed{V_C = -10\uparrow}$

Scanned with CamScanner

→ All the bodies are moving with const velocity,




Given,

$\bar{V}_{AC} = 100 m/s\ (\uparrow)$

$\bar{V}_{BA} = 50 m/s\ (\downarrow)$

Find velocity of 'C' w.r.to ground.

Given, $\bar{V}_{AC} = 100\hat{j}$

$\bar{V}_A - \bar{V}_C = 100\hat{j}$ — ①

$\bar{V}_B - \bar{V}_A = -50\hat{j}$ — ③

$\Rightarrow \bar{V}_B - \bar{V}_C = 50\hat{j}$

we have, $\bar{V}_A = -\bar{V}_P$

$\bar{V}_{BP} = -\bar{V}_{CP} \Rightarrow \bar{V}_B - \bar{V}_P = -\bar{V}_C + \bar{V}_P$

$2\bar{V}_P = \bar{V}_B + \bar{V}_C$

$-2\bar{V}_A = \bar{V}_B + \bar{V}_C$ — ②

① + ② ⇒

$2\bar{V}_B = 50\hat{j} - 2\bar{V}_A$

$\bar{V}_B = 25\hat{j} - \bar{V}_A$

from ③ $25\hat{j} - \bar{V}_A - \bar{V}_A = -50\hat{j}$

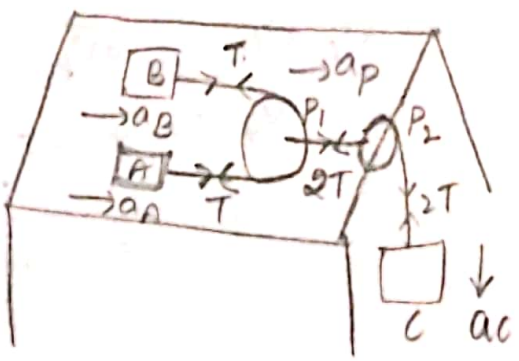
$75\hat{j} = 2\bar{V}_A$

$\boxed{\bar{V}_A = \frac{75}{2}\hat{j}}$

$\frac{75}{2}\hat{j} - \bar{V}_C = 100\hat{j}$

$\boxed{\bar{V}_C = -\frac{125}{2}\hat{j}}$

→




$m_A = 1kg$, $m_B = 2kg$, $m_C = 4k$

Constraint eqn for $P_1$.

Find $a_A, a_B, a_C$?

Scanned with CamScanner

$\overline{a_{P_1}} = \overline{a_C}$

$\overline{a_{BP_1}} = -\overline{a_{AP_1}}$

$2\overline{a_{P_1}} = \overline{a_B} + \overline{a_A}$

$$\boxed{2\overline{a_C} = \overline{a_B} + \overline{a_A}}$$

$$\Rightarrow \boxed{a_C = \frac{1}{2}\left(\frac{T}{m_A} + \frac{T}{m_B}\right)}$$

$m_C g - 2T = m_C a_C$ — ①

$T = m_A a_A$

$T = m_B a_B$

$a_A = \frac{T}{m_A}$, $a_B = \frac{T}{m_B}$

$$m_C g - 2T = m_C \left(\frac{\frac{T}{m_A} + \frac{T}{m_B}}{2}\right)$$

$$m_C g - 2T = \frac{T m_C (m_A + m_B)}{2 m_A m_B}$$

$$40 - 2T = \frac{T \times 4(3)}{2 \times 2}$$

$\Rightarrow 5T = 40 \Rightarrow \boxed{T = 8N}$

$a_A = \frac{8}{1}$, $a_B = 8/2 = 4$

$\boxed{a_A = 8 m/s^2}$ $\boxed{a_B = 4 m/s^2}$

$\Rightarrow a_C = \frac{1}{2}\left[8 + \frac{8}{2}\right]$

$= \frac{1}{2} \times 12 = 6$

$\boxed{a_C = 6 m/s^2}$

→ Find the acceleration of pulley A, B & C find tension in the string

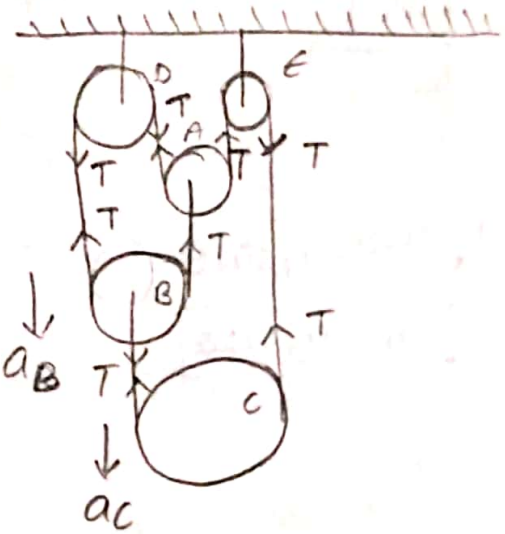
$m_A = m_B = m_C = 1 kg$

$\sum \overline{Ta} = 0$

$-(2T - T)a_A - (2T - T)a_B - 2Ta_C = 0$

$-Ta_A - Ta_B - 2Ta_C = 0$

$$\Rightarrow \boxed{a_A + a_B + 2a_C = 0} \quad — ①$$




$m_C g - 2T = m_C a_C \Rightarrow a_C = \frac{m_C g - 2T}{m_C} = g - 2T$

$m_B g - T = m_B a_B \Rightarrow a_B = \frac{m_B g - T}{m_B} = g - T$

Scanned with CamScanner

$m_A g - T = m_A a_A \Rightarrow a_A = \frac{m_A g - T}{m_A} = g - T$

sub in eqn ①

$2(g - 2T) + g - T + g - T = 0$

$4g = 6T$

~~$3g = 4T$~~

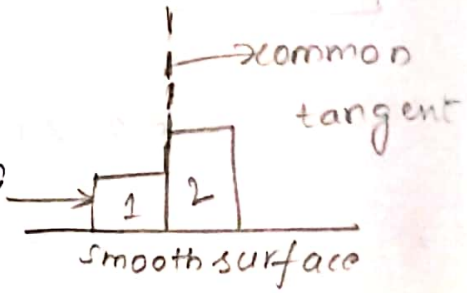
~~$T = 3g/4$~~

$T = 2g/3$

$a_A = g/3 (\downarrow)$ ; $a_B = g/3 (\downarrow)$ ; $a_C = g/3 (\uparrow)$

## Wedge Constraints

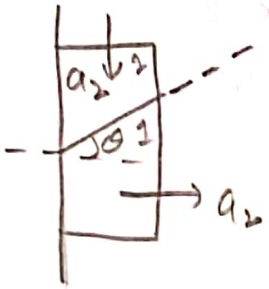
Blocks are moving together,

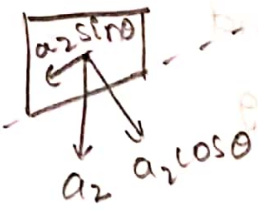


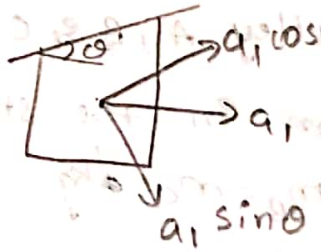

$v_1 = v_2$

$a_1 = a_2$, along normal to common tangent to the surfaces in contact.

Two wedges are moving always in contact find relation b/w $a_1$ & $a_2$.

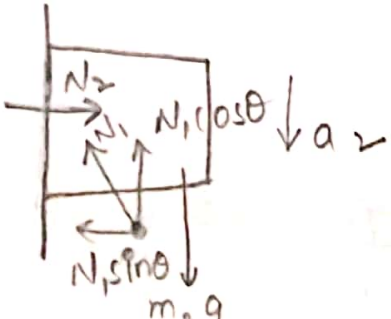


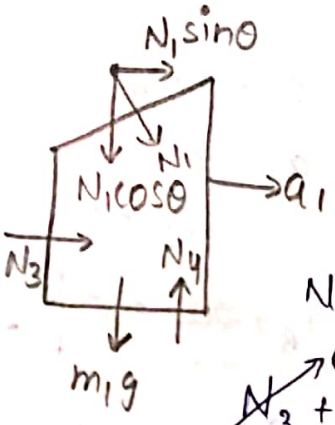




$a_2 \cos\theta = a_1 \sin\theta$

$\boxed{a_1 = a_2 \cot\theta}$



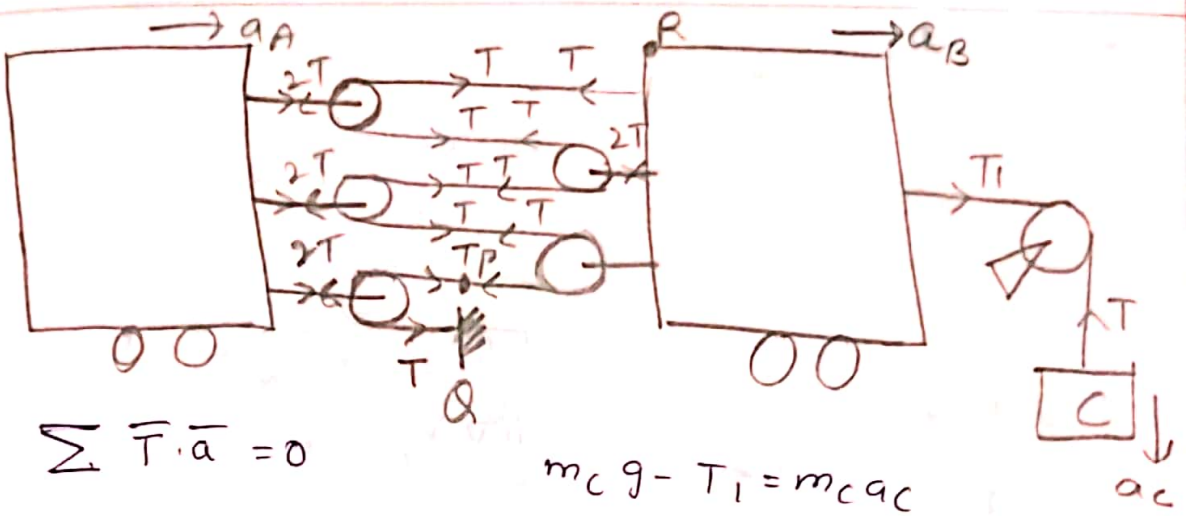

$m_2 g - N_1 \cos\theta = m_2 a_2$ —①

$N_1 \sin\theta = N_2$ —②

$N_1 \cos\theta + m_1 g = N_4$ —③

$N_3 + N_1 \sin\theta = m_1 a_1$ —④

Scanned with CamScanner




$\sum \bar{T} \cdot \bar{a} = 0$

$m_C g - T_1 = m_C a_C$

$6Ta_A - 5Ta_B = 0$

$$\boxed{6a_A = 5a_B}$$

$6V_A = 5V_B \Rightarrow \boxed{6x_A = 5x_B}$

$\bar{V}_{AB} = \bar{V}_A - \bar{V}_B$

$= V_A \uparrow - \frac{6V_A}{5} \uparrow$

$$\boxed{\bar{V}_{AB} = \frac{-V_A}{5} \uparrow}$$

Velocity of part 'P' on the string $V_P = ?$

from 'Q' $\Rightarrow x_P = 2x_A$

$$\boxed{V_P = 2V_A} \ (\longrightarrow)$$

from 'R'

$\bar{V}_P = (4V_A - 5V_B)(\longleftarrow)$

$= 5V_B - 4V_A \ (\longrightarrow)$

$\bar{x}_P = -(4x_A - 5x_B) = 5x_B - 4x_A \ (\longrightarrow)$

If $x_A = 5$ ; $x_P = 2 \times 5 = 10\text{cm} \ (\longrightarrow)$

$x_P = 5x_B - 4x_A$

$= 5 \times 6 - 4 \times 5$

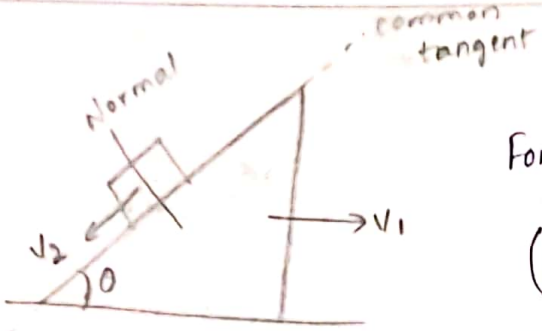
$= 10\text{cm} \ (\longrightarrow)$

$x_B = 6x_A$

$= \frac{6 \times 5}{5}$

$$\boxed{x_B = 6}$$

Scanned with CamScanner




For No slipping,

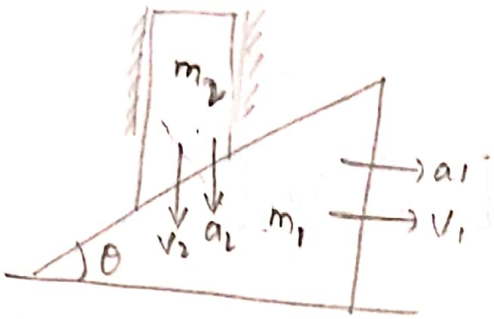
$$(\bar{V}_{12})_N = 0$$

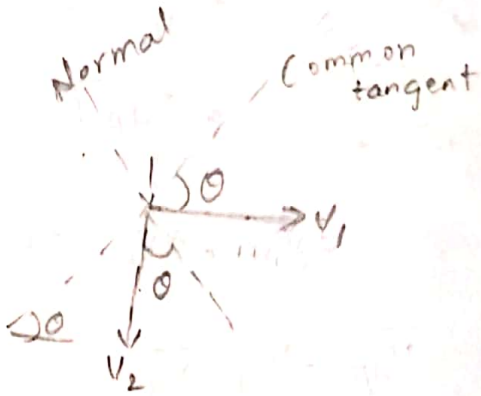
$$(\bar{V}_1)_N - (\bar{V}_2)_N = 0$$

$$\boxed{(\bar{V}_1)_N = (\bar{V}_2)_N}$$

Similarly

$$\Rightarrow \boxed{(\bar{a}_1)_N = (\bar{a}_2)_N}$$




For no slipping,

$$V_{1N} = V_{2N}$$

$$\boxed{V_1 \sin\theta = V_2 \cos\theta}$$

Similarly,

$$\boxed{a_1 \sin\theta = a_2 \cos\theta} \rightarrow (1)$$

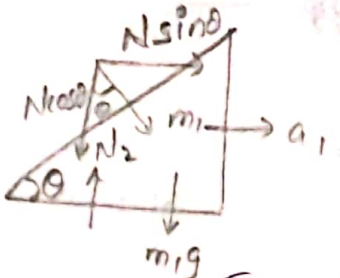
For rod:-




$$N_1 = N\sin\theta \quad -(2)$$

$$m_2 g - N\cos\theta = m_2 a_2 \quad -(3)$$

For wedge:-




$$N\sin\theta = m_1 a_1 \quad -(4)$$

$$m_1 g + N\cos\theta = N_2 \quad -(5)$$

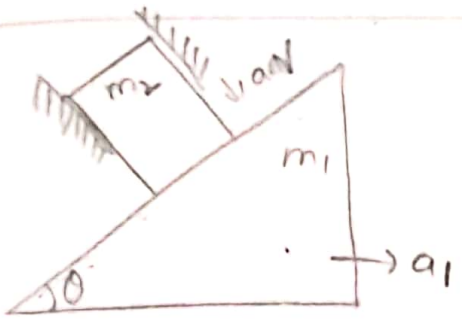
unknowns $\rightarrow a_1, a_2, N_1, N_2, N$.

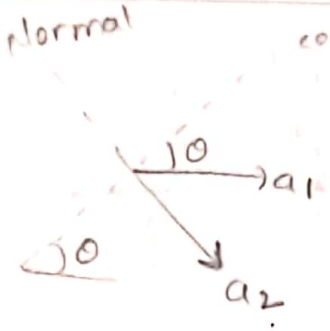
* From eq (1), (3), (5) $\Rightarrow \left(\frac{N\sin\theta}{m_1}\right)\sin\theta = \frac{(m_2 g - N\cos\theta)}{m_2}\cos\theta$

solve for $N$.

Scanned with CamScanner

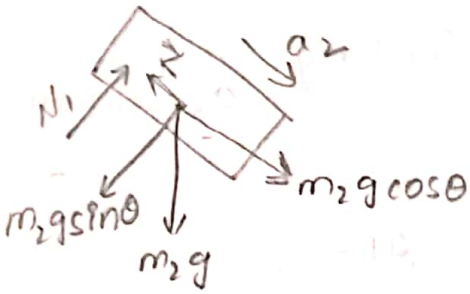
→



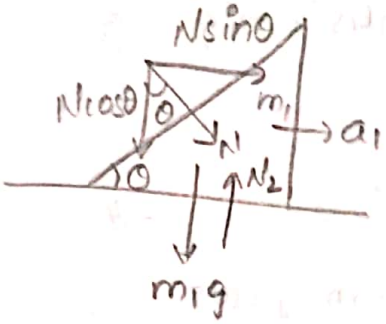

$a_{1N} = a_1 \sin\theta$ , $a_{2N} = a_2$ ⇒ $a_{1N} = a_{2N}$

$\therefore \boxed{a_1 \sin\theta = a_2}$ — ①




$m_2 g \cos\theta - N = m_2 a_2$ — ②

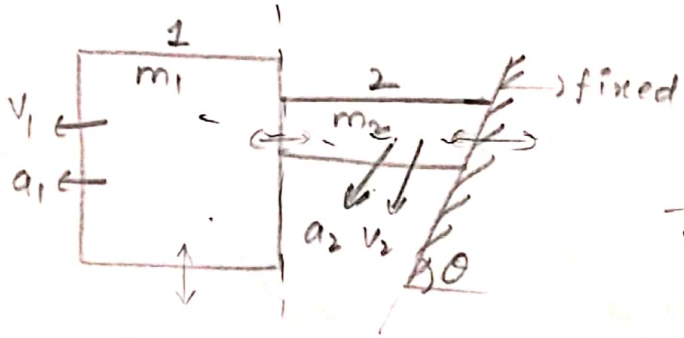
$m_2 g \sin\theta = N_1$ — ③

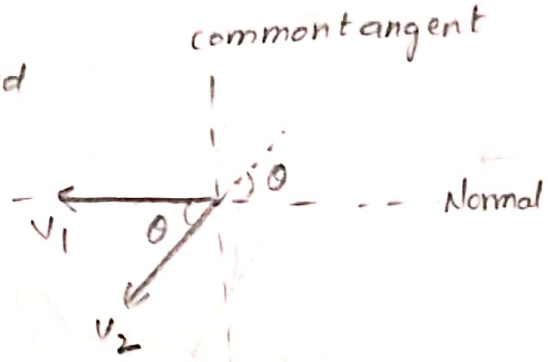



$N\cos\theta + m_1 g = N_2$ — ④

$N \sin\theta = m_1 a_1$ — ⑤

→

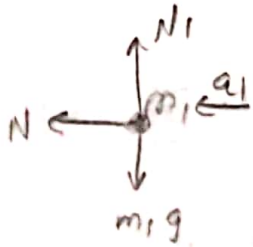



$(V_1)_N = V_1$ , $(V_2)_N = V_2 \cos\theta$ ⇒ $(V_1)_N = (V_2)_N$

$\boxed{V_1 = V_2 \cos\theta}$ similarly, $\boxed{a_1 = a_2 \cos\theta}$ — ①

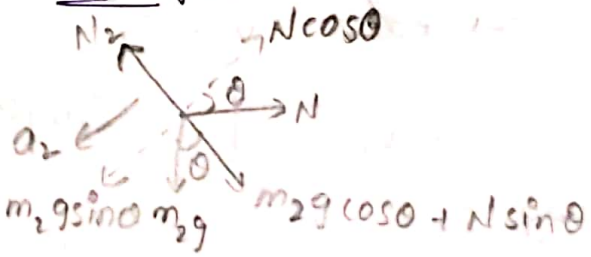
For body 1:-




$N = m_1 a_1$ — ②

$N_1 = m_1 g$ — ③

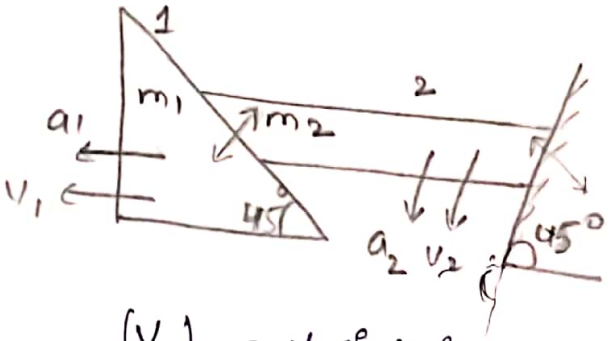
For body 2:-

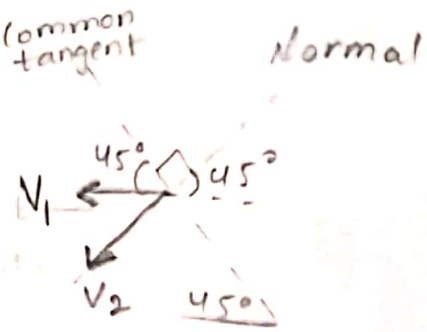



$m_2 g \sin\theta - N\cos\theta = m_2 a_2$ — ④

$N_2 = m_2 g \cos\theta + N \sin\theta$ — ⑤

Scanned with CamScanner

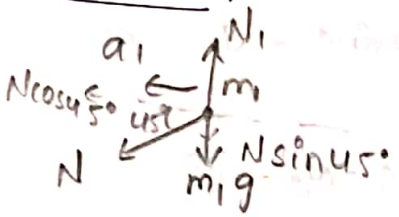



$(V_1)_N = V_1 \sin 45°$ , $(V_2)_N = V_2 \Rightarrow (V_1)_N = (V_2)_N$

$\boxed{V_2 = V_1 \sin 45°}$ Similarly, $\boxed{a_2 = a_1 \sin 45°}$ — (1)

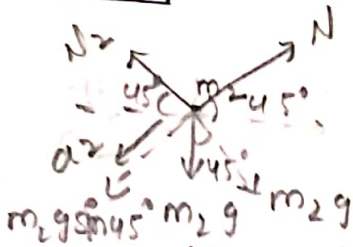
For body 1:-




$N\cos 45° = m_1 a_1$ — (2)

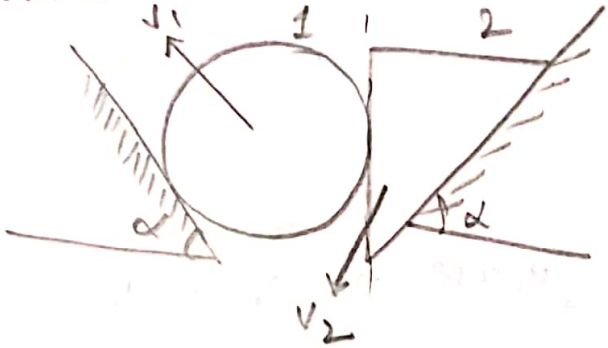
$N_1 = N\sin 45° + m_1 g$ — (3)

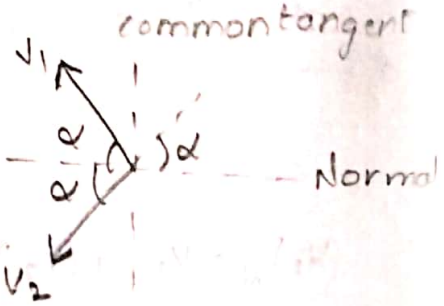
For body 2:-




$m_2 g \sin 45° - N = m_2 a_2$ — (4)

$N_2 = m_2 g \cos 45°$ — (5)



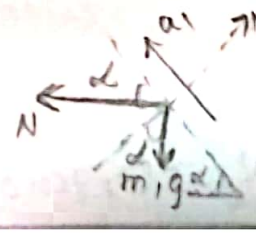




$(V_1)_N = V_1 \cos\alpha$ , $(V_2)_N = V_2 \cos\alpha \Rightarrow (V_1)_N = (V_2)_N$

$V_1 \cos\alpha = V_2 \cos\alpha$

$\boxed{V_1 = V_2}$ Similarly, $\boxed{a_1 = a_2}$ — (1)

For body 1:-

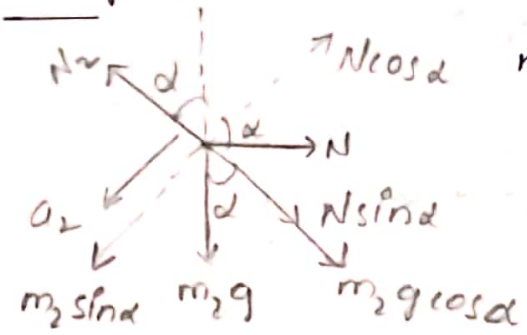



$N\cos\alpha - m_1 g \sin\alpha = m_1 a_1$ — (2)

$N\sin\alpha + m_1 g \cos\alpha = N_1$ — (3)

Scanned with CamScanner

For body 2 :-

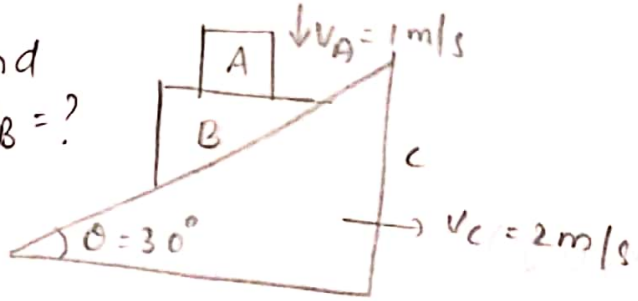



$m_2 g\sin\alpha - N\cos\alpha = m_2 a_2$ — (4)

$N_2 = N\sin\alpha + m_2 g\cos\alpha$ — (5)

→

Find $V_B = ?$

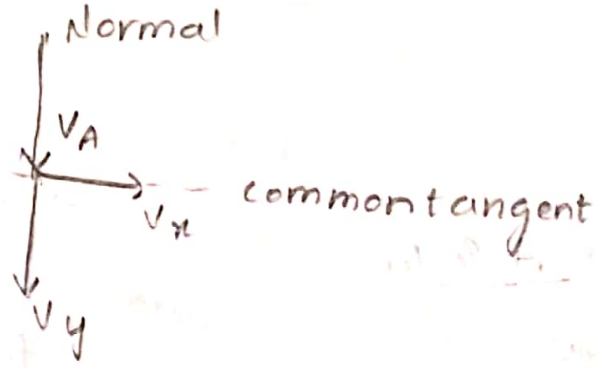



Let $V_B = V_x\hat{i} + V_y(-\hat{j})$

For A & B :-

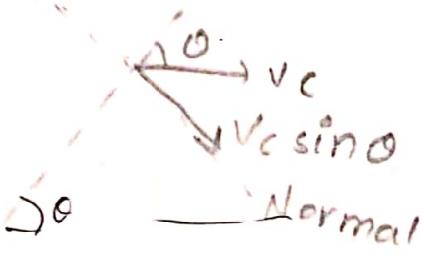
$V_A = V_y$

$\Rightarrow V_y = 1$ m/s



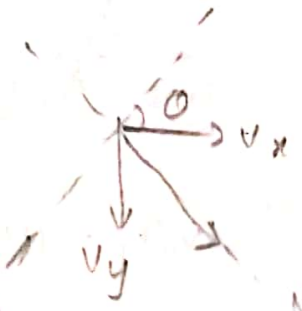

For B & C :-

for C :- e.x




$(V_C)_N = V_C \sin\theta$.

for B :-




$(V_B)_N = V_x \sin\theta + V_y \cos\theta$

$\Rightarrow (V_C)_N = (V_B)_N$

$V_C \sin 30° = V_x \sin 30° + V_y \cos 30°$

$2 \times \frac{1}{2} = V_x \times \frac{1}{2} + 1 \times \frac{\sqrt{3}}{2}$

$2 - \sqrt{3} = V_x$

$\Rightarrow V_B = (2-\sqrt{3})\hat{i} \cdot \hat{j}$

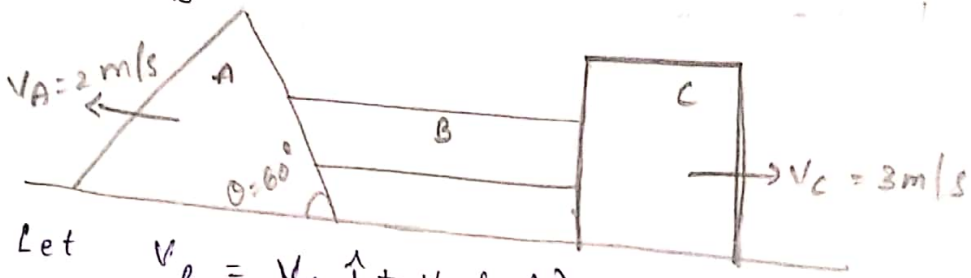
Scanned with CamScanner

$$|V_B| = \sqrt{V_x^2 + V_y^2} = \sqrt{(2-\sqrt{3})^2 + 1}$$

$$V_B = \sqrt{4+3-4\sqrt{3}+1}$$

$$= \sqrt{8-4\sqrt{3}} \Rightarrow \boxed{V_B = 2\sqrt{2-\sqrt{3}}\ m/s}$$

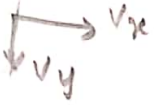
→ Find $V_B$.



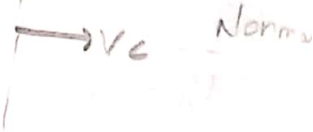

Let $V_B = V_x \hat{i} + V_y(-\hat{j})$

For B & C:-

for B:-



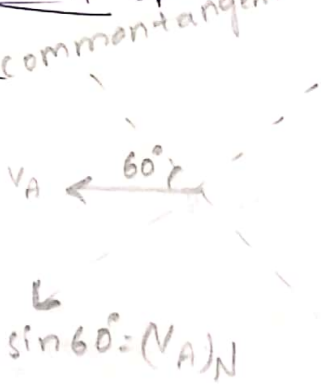

for C:-



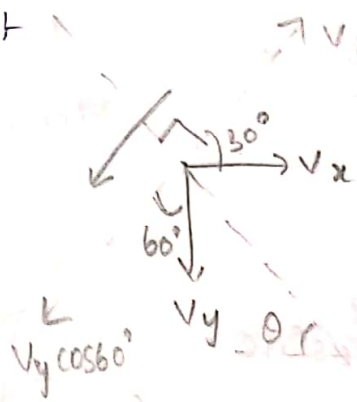

$\Rightarrow V_x = V_C = 3$ —(1)

For A & B:-

for A:-




for B:-




$$(V_A)_N = (V_B)_N$$

$$V_A \sin 60° = V_y \cos 60° - V_x \cos 30°$$

$$\cancel{2} \times \frac{\sqrt{3}}{\cancel{2}} = V_y\left(\frac{1}{2}\right) - 3\left(\frac{\sqrt{3}}{2}\right)$$

$$\sqrt{3}\left(1+\frac{3}{2}\right) = V_y\left(\frac{1}{2}\right)$$

$$\boxed{5\sqrt{3} = V_y} \Rightarrow \boxed{V_B = 3\hat{i} - 5\sqrt{3}\hat{j}}$$

Scanned with CamScanner

$$|V_B| = \sqrt{V_x^2 + V_y^2} = \sqrt{9+75} = \sqrt{84} = 2\sqrt{21}$$

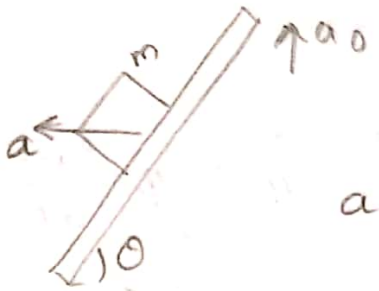
$$\boxed{V_B = 2\sqrt{21}\ m/s}$$

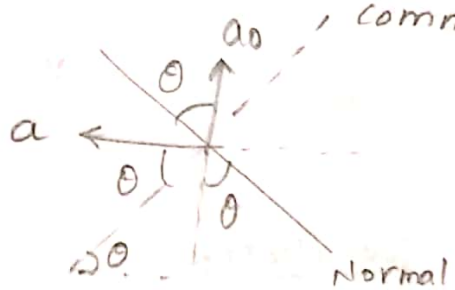
→




Same as above problem.

→ Find the value of $a_0$ so that block's vertical component of acceleration is zero.

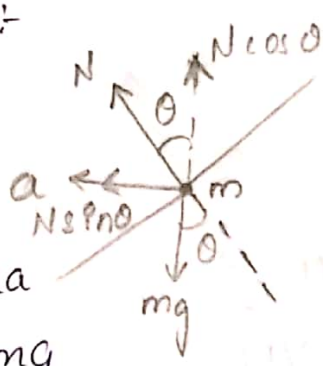



$$a\sin\theta = a_0\cos\theta \quad -①$$

$$\tan\theta = \frac{a_0}{a}$$

For m:-




$$N\sin\theta = ma$$

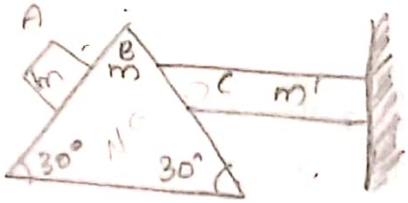
$$N\cos\theta = mg$$

$$\tan\theta = \frac{a}{g}$$

$$\tan^2\theta = \frac{a_0}{g}$$

$$\boxed{a_0 = g\tan^2\theta}$$

→

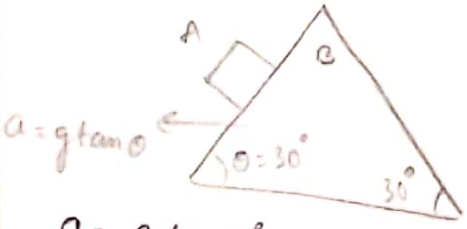



Find the value of m' so that blocks A & B move together.



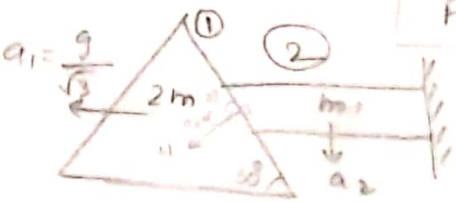

$a = g\tan\theta$

$F = (m_1+m_2)g\tan\theta$

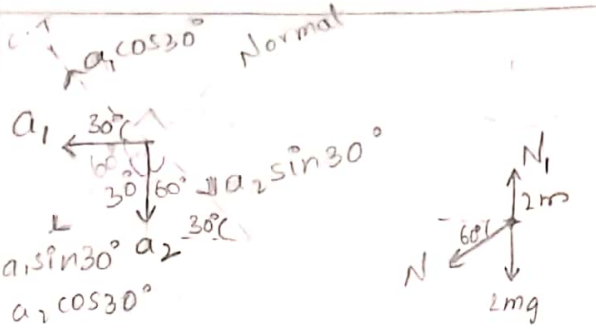


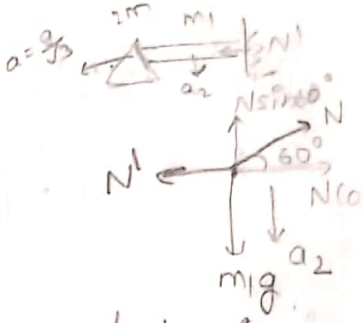

$$a = g\tan 30°$$

$$\boxed{a = g/\sqrt{3}}$$




Scanned with CamScanner




$a_1 \sin 30° = a_2 \cos 30°$

$\frac{g}{\sqrt{3}} \times \frac{1}{2} = a_2 \times \frac{\sqrt{3}}{2}$

$\boxed{a_2 = g/3}$

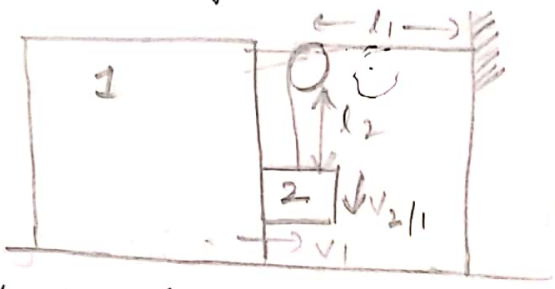
$N' = N \cos 60°$

$m'g - N \sin 60° = m' a_2$

$m'(g - g/3) = N \sin 60°$

$m' = \frac{\sqrt{3} N \times 3}{4}$

## String – Wedge Constraints:-




$V_1 = -\frac{dl_1}{dt}$, $V_{2/1} = +\frac{dl_2}{dt}$

*Contact exists blw blocks:-

length $l = l_1 + l_2 = \text{const}$

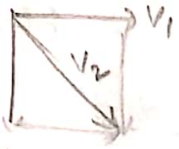
$\frac{dl}{dt} = 0 \Rightarrow \frac{dl_1}{dt} + \frac{dl_2}{dt} = 0$

$-V_1 + V_{2/1} = 0$

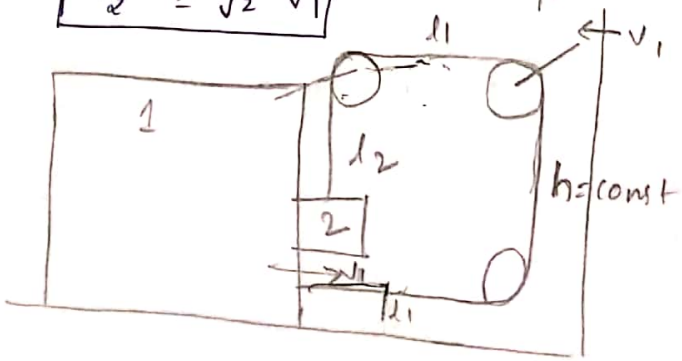
$\boxed{V_1 = V_{2/1}}$

$\therefore \; \overline{V_2} = \overline{V_{2/1}} + \overline{V_1}$

$\overline{V_2} = \sqrt{V_{2/1}^2 + V_1^2}$




$\boxed{V_2 = \sqrt{2} V_1}$




$l = 2l_1 + h + l_2 = \text{const}$

Scanned with CamScanner

$$\frac{dl}{dt} = 2\frac{dl_1}{dt} + \frac{dh}{dt} + \frac{dl_2}{dt}$$

$$-2v_1 + 0 + v_{2/1} = 0$$

$$v_{2/1} = 2v_1 \longrightarrow (1)$$

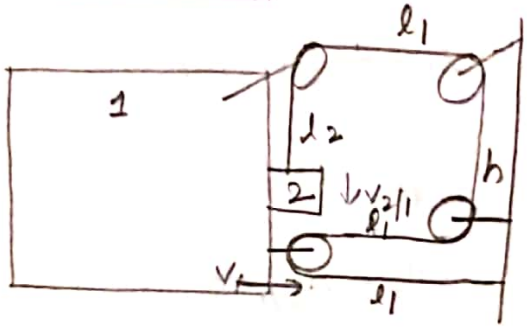
net velocity,

$$\overline{v_2} = \overline{v_{2/1}} + \overline{v_1}$$

$$|v_2| = \sqrt{v_1^2 + v_{2/1}^2}$$

$$= \sqrt{v_1^2 + (2v_1)^2}$$

$$\boxed{|v_2| = \sqrt{5}\, v_1}$$




$$l = 2l_1 + h + l_1 + l_2 = 3l_1 + l_2 + h$$

$$\frac{dl}{dt} = 3\frac{dl_1}{dt} + \frac{dl_2}{dt} + \frac{dh}{dt}$$

$$\frac{3dl_1}{dt} + \frac{dl_2}{dt} = 0$$

$$-3v_1 + v_{2/1} = 0$$

$$\boxed{v_{2/1} = 3v_1}$$

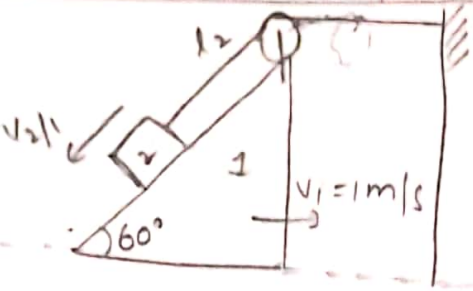
net velocity,

$$\overline{v_2} = \sqrt{v_{2/1}^2 + v_1^2}$$

$$= \sqrt{9v_1^2 + v_1^2}$$

$$\boxed{|v_2| = \sqrt{10}\, v_1}$$

Scanned with CamScanner

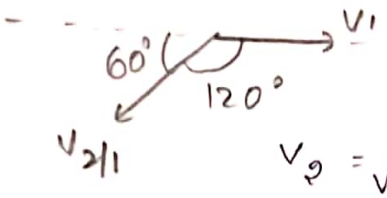



$l = l_1 + l_2 = \text{const}$

$$\frac{dl}{dt} = \frac{dl_1}{dt} + \frac{dl_2}{dt} = 0$$

$$-V_1 + V_{2/1} = 0$$

$$\boxed{V_{2/1} = V_1} = 1\,m/s$$

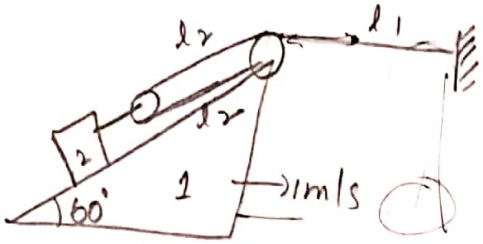



$$V_2 = \sqrt{V_1^2 + V_{2/1}^2 + 2V_1 V_{2/1}\cos\theta}$$

$$= \sqrt{1 + 1 + 2(1)\cos 120°}$$

$$= \sqrt{2 + 2\left(-\frac{1}{2}\right)}$$

$$\boxed{V_2 = 1\,m/s}$$



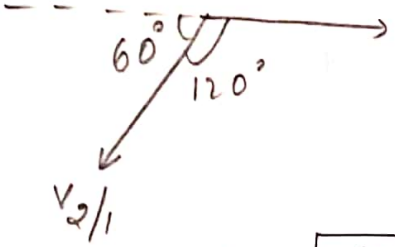

$l_1 + 2l_2 = \text{const}$

$$\frac{dl_1}{dt} + 2\frac{dl_2}{dt} = 0$$

$$-V_1 + 2V_{2/1} = 0$$

$$V_{2/1} = \frac{V_1}{2} = \frac{1}{2}$$

$$\boxed{V_{2/1} = 1/2\,m/s}$$

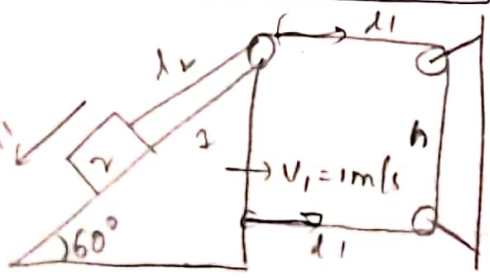



$$V_2 = \sqrt{V_1^2 + V_{2/1}^2 + 2V_1 V_{2/1}\cos\theta}$$

$$= \sqrt{V_1^2 + \frac{V_1^2}{4} + 2\frac{V_1^2}{2}\left(-\frac{1}{2}\right)}$$

$$= \frac{V_1}{2}$$

$$\boxed{V_2 = 1/2\,m/s}$$



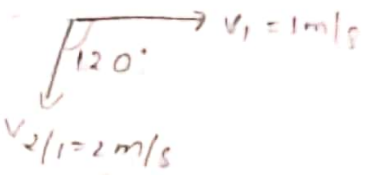

$l = 2l_1 + h + l_2 = \text{const}$

$$\frac{dl}{dt} = 0 = 2\frac{dl_1}{dt} + \frac{dh}{dt}^{0} + \frac{dl_2}{dt}$$

$$-2V_1 + V_{2/1} = 0 \Rightarrow V_{2/1} = 2V_1 = 2\,m/s$$

Scanned with CamScanner

now, $\overline{V_2} = \overline{V_{2/1}} + \overline{V_1}$

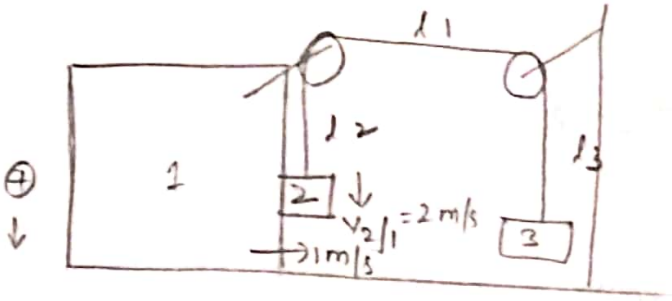



$$\overline{V_2} = \sqrt{V_1^2 + V_{2/1}^2 + 2V_1V_{2/1}\cos\theta}$$

$$= \sqrt{V_1^2 + 4V_1^2 + 2V_1^2\left(-\frac{1}{2}\right)}$$

$$V_2 = 2V_1 = 2\,m/s$$

$\boxed{V_2 = 2\,m/s}$




$l_T = l_1 + l_2 + l_3 = \text{const}$

$\frac{dl}{dt} = \frac{dl_1}{dt} + \frac{dl_2}{dt} + \frac{dl_3}{dt} = 0$

$-(1) + 2 + V_3 = 0$

$V_3 = -1$

$\boxed{V_3 = 1\,m/s}\ (\uparrow)$

net velocity,

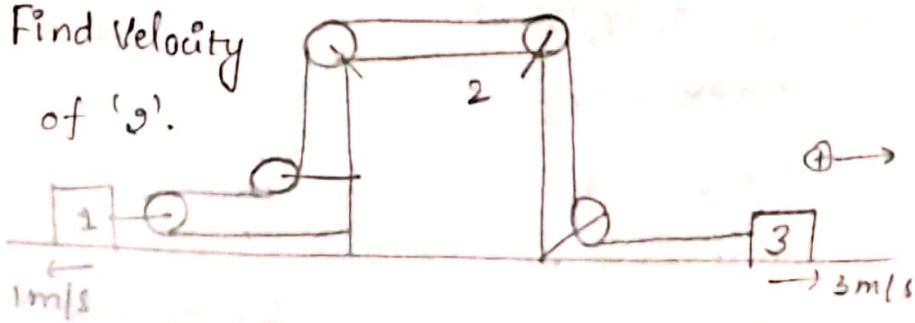
$\overline{V_2} = \overline{V_{2/1}} + \overline{V_1}$

$\overline{V_2} = \sqrt{V_{2/1}^2 + V_1^2}$

$= \sqrt{4+1}$

$\boxed{V_2 = \sqrt{5}\,m/s}$

Find Velocity of '2'.




Scanned with CamScanner

we solve this problem w.r.to '2'.

$$l_{3/2} + 2l_{1/2} = \text{const}$$

$$\frac{dl_{3/2}}{dt} + 2\frac{dl_{1/2}}{dt} = 0$$

$$(V_3 - V_2) - 2(V_1 - V_2) = 0$$

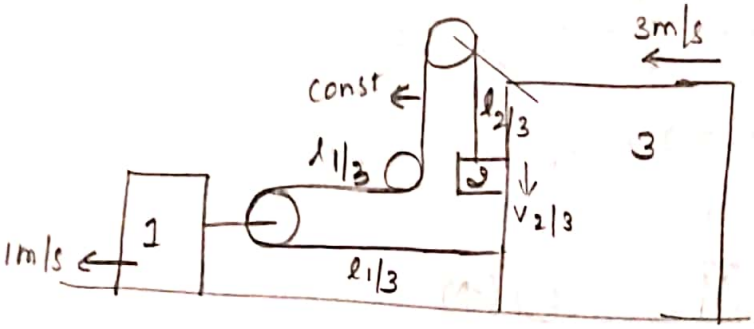
$$(3 - V_2) - 2(-1 - V_2) = 0$$

$$3 - V_2 = -2V_2 - 2$$

$$5 = -V_2$$

$$V_2 = -5\ m/s$$

$$\boxed{V_2 = 5\ m/s} \quad (\leftarrow)$$




we solve this problem w.r.to '3'.

$$2l_{1/3} + l_{2/3} = \text{const}$$

$$2\frac{dl_{1/3}}{dt} + \frac{dl_{2/3}}{dt} = 0$$

$$2(V_{1/3}) + V_{2/3} = 0$$

$$2(V_1 - V_3) + V_{2/3} = 0$$

$$2(1-3) + V_{2/3} = 0$$

$$\boxed{V_{2/3} = 4\ m/s}$$

net velocity, $\overline{V_2} = \overline{V_{2/3}} + \overline{V_3}$

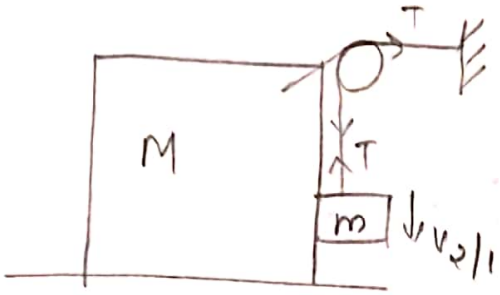
Scanned with CamScanner

$$v_2 = \sqrt{4^2 + 3^2}$$

$$= \sqrt{5^2}$$

$$\boxed{v_2 = 5\,m/s}$$

→

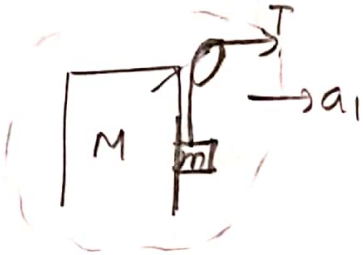



If the system is released from rest. Find accelerations at this moment.

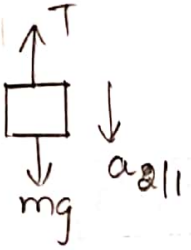
$$v_1 = v_{2/1}$$

$$a_1 = a_{2/1} \quad \text{—①}$$

$$\therefore \boxed{a_1 = \frac{T}{M+m}} \rightarrow ②$$




& for block,




$$mg - T = m a_{2/1}$$

$$mg - T = m a_1$$

$$mg - T = \frac{mT}{M+m}$$

$$mg = T\left(\frac{2m+M}{M+m}\right)$$

$$\Rightarrow \boxed{T = \frac{m(M+m)g}{M+2m}}$$

$$\therefore a_1 = \frac{m(M+m)g}{(M+2m)(M+m)}$$

$$\boxed{a_1 = \frac{mg}{M+2m} = a_{2/1}}$$

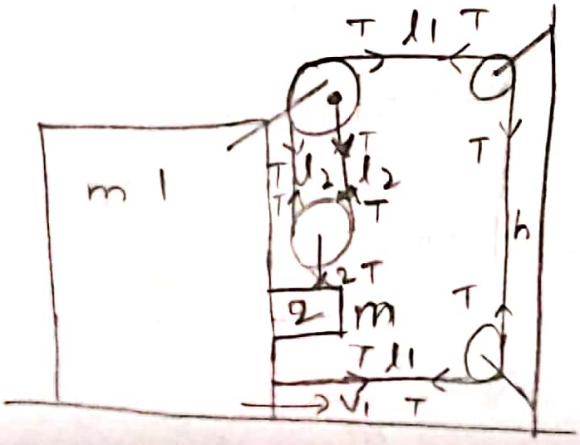
net acceleration of $a_2$

$$\overline{a_2} = \overline{a_{2/1}} + \overline{a_1}$$

$$a_2 = \sqrt{2}\,a_1$$

$$\boxed{a_2 = \frac{\sqrt{2}mg}{M+2m}}$$

→




Find acceleration of blocks just after release.

$$v_1 = -\frac{dl_1}{dt}, \quad v_{2/1} = \frac{dl_2}{dt}$$

Scanned with CamScanner

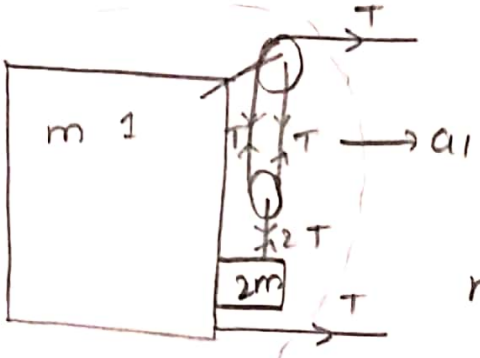
$$l_1 + h + l_1 + 2l_2 = \text{const}$$

$$\frac{d}{dt}(2l_1) + \frac{d}{dt}(+2l_2) = 0$$

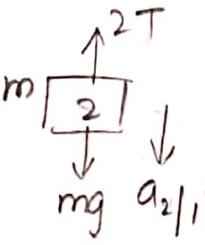
$$\frac{dl_1}{dt} + \frac{dl_2}{dt} = 0$$

$$-V_1 + V_{2/1} = 0$$

$$\boxed{V_1 = V_{2/1}} \Rightarrow \boxed{a_1 = a_{2/1}}$$




$$a_1 = \frac{2T}{m+m} = \frac{T}{m} \rightarrow (2)$$




$$mg - 2T = m a_{2/1}$$

$$\frac{T}{m} = \frac{mg - 2T}{m}$$

$$\Rightarrow 3T = mg \Rightarrow \boxed{T = mg/3}$$

$$a_1 = \frac{T}{m} = \frac{mg}{3m}$$

$$\boxed{a_1 = g/3}$$

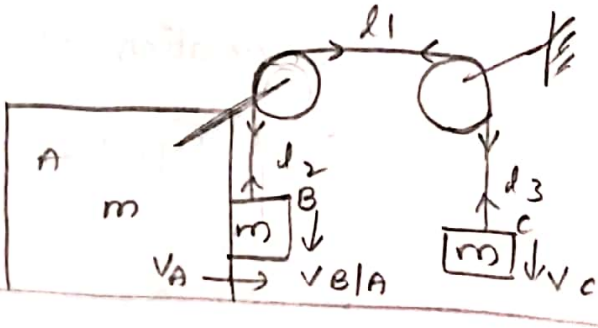
$$\boxed{a_{2/1} = g/3}$$

net acceleration:

$$\vec{a_2} = \vec{a}_{2/1} + \vec{a_1}$$

$$a_2 = \sqrt{2}\, a_1$$

$$\boxed{a_2 = \sqrt{2}\, g/3}$$

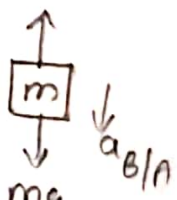



Find $a_A$, $a_B$ & $a_C$ if system is released from rest.

$$-V_A + V_{B/A} + V_C = 0$$

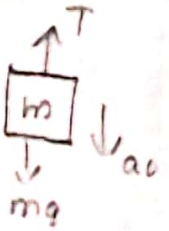
$$-a_A + a_{B/A} + a_C = 0$$

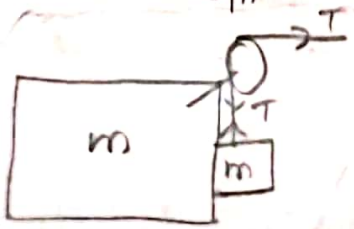
$$a_A = \frac{T}{2m}$$




$$a_{B/A} = \frac{mg - T}{m}$$

$$a_C = \frac{mg - T}{m}$$




Scanned with CamScanner

$$-\frac{T}{2m} + \frac{mg-T}{m} + \frac{mg-T}{m} = 0$$

$$2mg - 2T = T/2$$

$$mg = \frac{T}{4} + T$$

$$\boxed{\frac{4mg}{5} = T}$$

$$\therefore a_A = \frac{T}{2m} = \frac{2g}{5}$$

$$a_{B/A} = g - \frac{4g}{5} = g/5$$

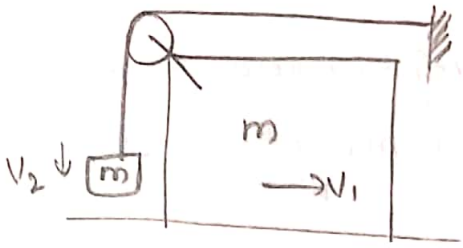
$$a_C = g - 4g/5 = g/5$$

now,

$$\vec{a_B} = \vec{a_{B/A}} + \vec{a_A}$$

$$= \frac{g}{5}\sqrt{5}$$

$$\boxed{a_B = g/\sqrt{5}}$$




Find acceleration of block just after release.

* Contact does not exist b/w blocks.

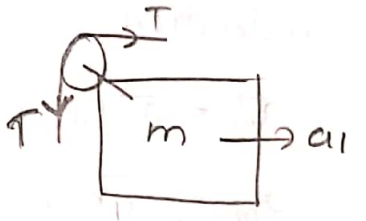
$$l_1 + l_2 = \text{const}$$

$$\frac{dl_1}{dt} + \frac{dl_2}{dt} = 0$$

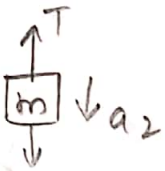
$$-v_1 + v_2 = 0$$

$$\boxed{v_1 = v_2}$$

$$\Rightarrow \boxed{a_1 = a_2} \rightarrow (1)$$




$$T = ma_1 \quad (2)$$



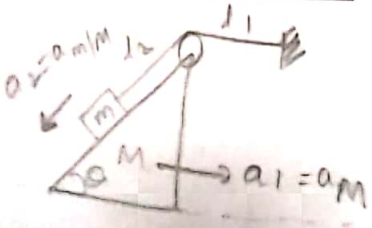

$$mg - T = ma_2 \quad (3)$$

$$\therefore \frac{T}{m} = g - \frac{T}{m}$$

$$\frac{2T}{m} = g \Rightarrow \boxed{T = \frac{mg}{2}}$$

$$\therefore \frac{mg}{2} = ma_1 \Rightarrow \boxed{a_1 = g/2}$$

$$\Rightarrow \boxed{a_1 = a_2 = g/2}$$




Find the acceleration of m & M at initial state.

Scanned with CamScanner

$l = l_1 + l_2 = \text{const}$

$\frac{dl_1}{dt} + \frac{dl_2}{dt} = 0$

$-v_M + v_{m/M} = 0$

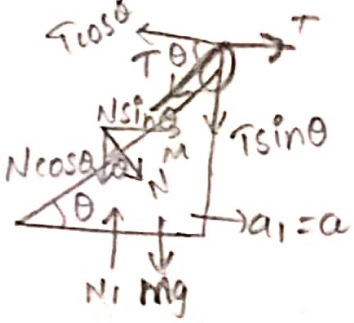
$v_M = v_{m/M} \Rightarrow \boxed{a_{m/M} = a_M}$

$\boxed{a_1 = a_2 = a}$ (say) —⑤

unknows:

$a_1, a_2, N, N', T$.

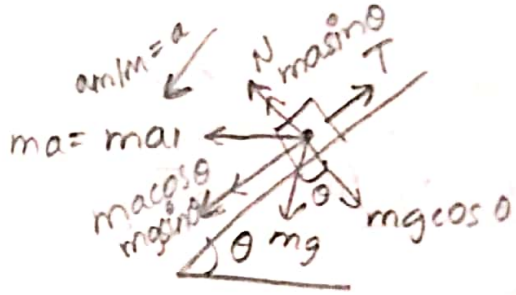
For wedge:




$N\sin\theta + T - T\cos\theta = Ma$ — ①

$N\cos\theta + T\sin\theta + Mg = N_1$ —②

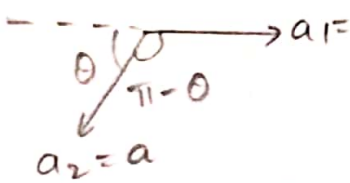
For block on wedge:




$mg\sin\theta + ma\cos\theta - T = ma$ —③

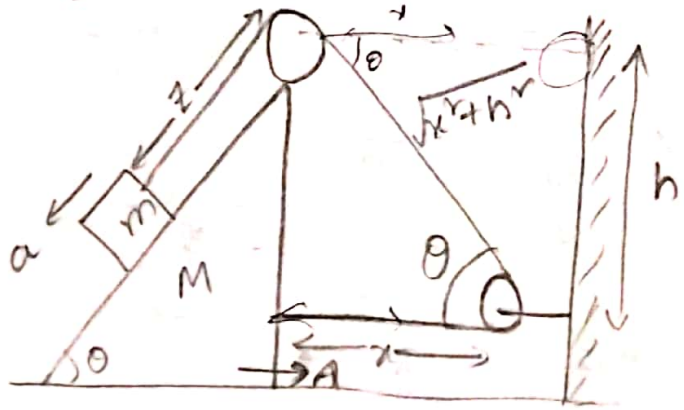
$N + ma\sin\theta = mg\cos\theta$ —④

Finally, net acceleration:




$\vec{a_m} = \vec{a}_{m/M} + \vec{a}_M$

$a_m = \sqrt{a_1^2 + a_2^2 + 2a_1a_2\cos(\pi - \theta)}$




A = acceleration of M

a = acceleration of m on

length of string,

$l = x + \sqrt{x^2+h^2} + z = \text{const}$

$\frac{dl}{dt} = 0 \Rightarrow \frac{dx}{dt} + \frac{2x}{2\sqrt{x^2+h^2}} \cdot \frac{dx}{dt} + \frac{dz}{dt} = 0$

$\frac{dx}{dt}\left(1 + \frac{x}{\sqrt{x^2+h^2}}\right) + \frac{dz}{dt} = 0$

Scanned with CamScanner

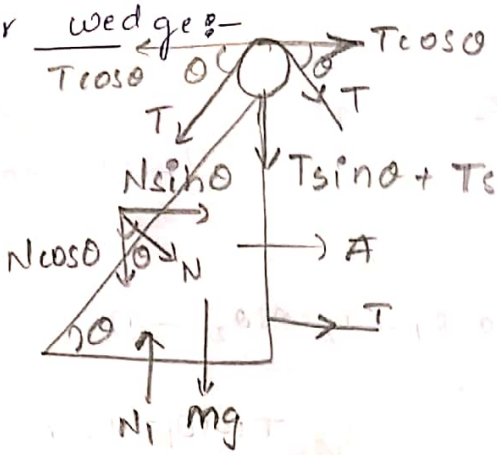
$$-V_m(1+\cos\theta) + V_{m/M} = 0.$$

$$V_m(1+\cos\theta) = V_{m/M}$$

$$a_M(1+\cos\theta) = a_{m/M}$$

$$a = A(1+\cos\theta) \quad \text{—— ①}$$

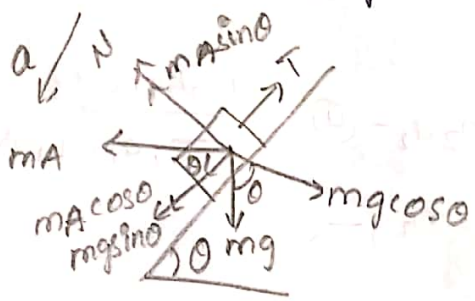
For wedge :-




$$T + N\sin\theta = MA \quad \text{—— ②}$$

$$T\sin\theta + N\cos\theta + Mg = N_1 \quad \text{—— ③}$$

For block on wedge :-



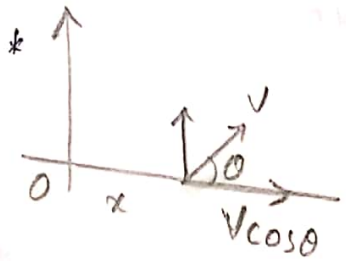

$$mg\sin\theta + mA\cos\theta - T = ma \quad \text{—— ④}$$

$$mA\sin\theta + N = mg\cos\theta \quad \text{—— ⑤}$$

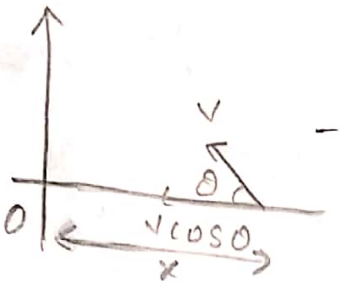
unknowns : $a, A, T, N, N_1$.

Strings at an angle 'θ' :-

Concept :-




$$\frac{dx}{dt} = V\cos\theta$$

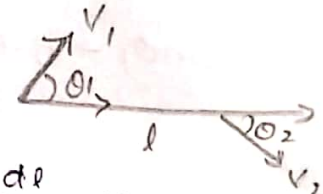



$$-\frac{dx}{dt} = V\cos\theta.$$

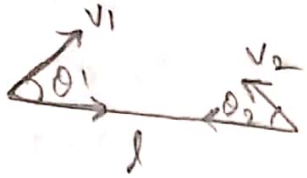
For string :




$$\frac{dl}{dt} = V\cos\theta$$



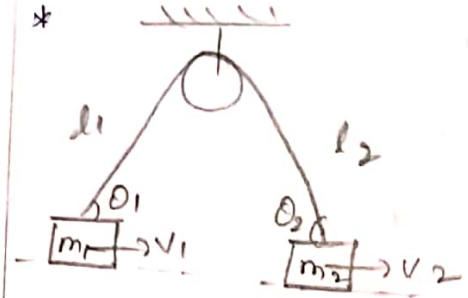

$$\frac{dl}{dt} = V_2\cos\theta_2 - V_1\cos\theta_1$$




$$-\frac{dl}{dt} = V_1\cos\theta_1 + V_2\cos\theta_2$$

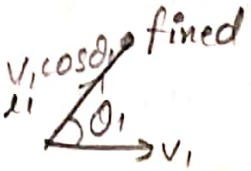
Scanned with CamScanner

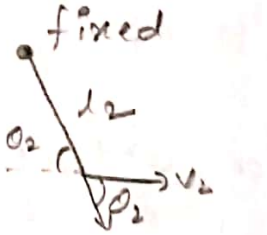
*




$l = l_1 + l_2 = \text{const}$

$$\frac{dl_1}{dt} + \frac{dl_2}{dt} = 0 \longrightarrow (1)$$



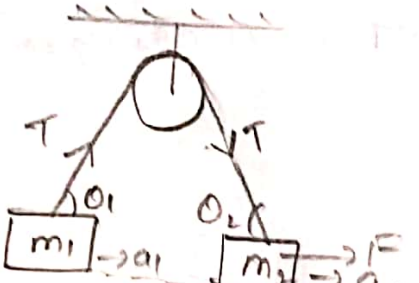



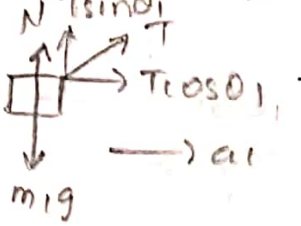

$\Rightarrow \boxed{v_1\cos\theta_1 = v_2\cos\theta_2}$

$\frac{dl_1}{dt} = -v_1\cos\theta_1$

$dl_2 = v_2\cos\theta_2$

*

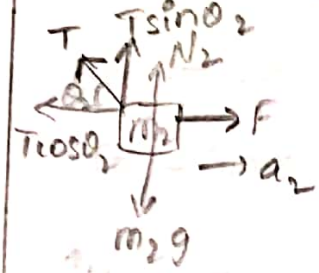



$a_1\cos\theta_1 = a_2\cos\theta_2 \longrightarrow (1)$

$T\cos\theta_1 = m_1 a_1 \longrightarrow (2)$

$T\sin\theta_1 + N_1 = m_1 g \longrightarrow (3)$

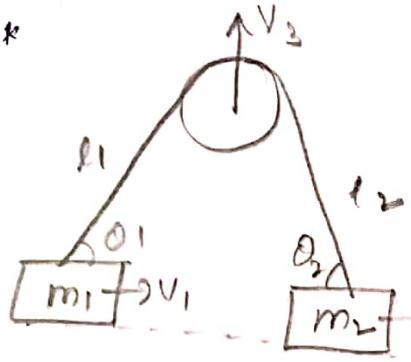



$F - T\cos\theta_2 = m_2 a_2 \longrightarrow (4)$

$T\sin\theta_2 + N_2 = m_2 g \longrightarrow (5)$

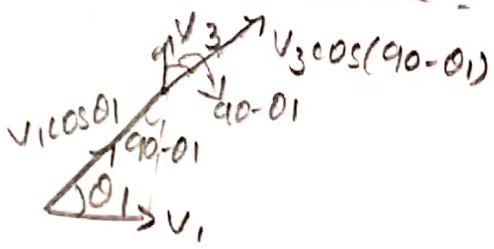
Unknows:
$T, a_1, a_2, N_1, N_2$

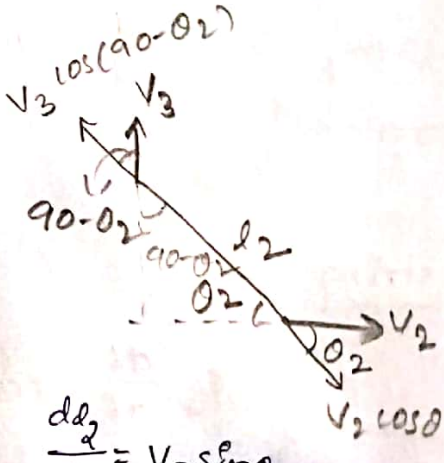
*




$l = l_1 + l_2 = \text{const}$

$$\frac{dl_1}{dt} + \frac{dl_2}{dt} = 0$$




$\frac{dl_1}{dt} = v_3\sin\theta_1 - v_1\cos\theta_1 \longrightarrow (1)$

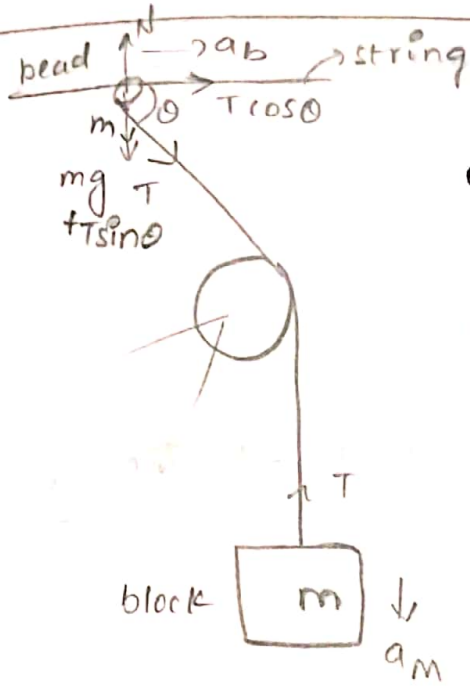
$\frac{dl_2}{dt} = v_3\sin\theta_2 + v_2\cos\theta_2 \longrightarrow (2)$

$(1) + (2) \Rightarrow$

$$v_3\sin\theta_1 - v_1\cos\theta_1 + v_3\sin\theta_2 + v_2\cos\theta_2 = 0$$

$$v_3(\sin\theta_1 + \sin\theta_2) - (v_1\cos\theta_1 - v_2\cos\theta_2) = 0.$$

Scanned with CamScanner




when the system is released,

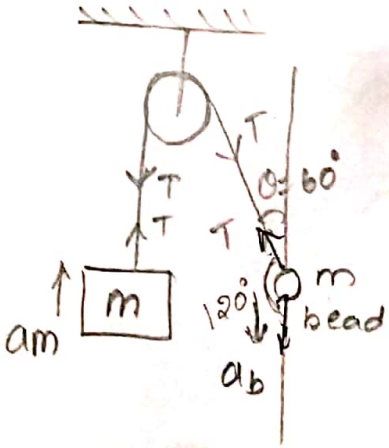
* $a_b\cos\theta = a_M$ — ①

For bead, $T\cos\theta = ma_b$ — ②

$N = T\sin\theta + mg$ — ③

For block :-

$Mg - T = Ma_M$ — ④




Find acceleration of block & bead.

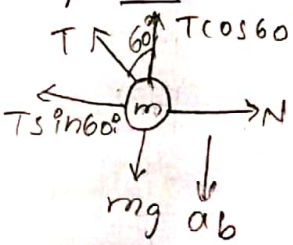
$T = \frac{6mg}{5}$, $a_b = \frac{2g}{5}$ (↓)

$a_m = \frac{g}{5}$ (↑)

$a_m = a_b\cos 180° \Rightarrow a_m = -\frac{1}{2}a_b$

$\Rightarrow \boxed{a_m = \frac{1}{2}a_b}$ — ④ So $a_m$ ↑ , $a_b$ ↓

for bead :-



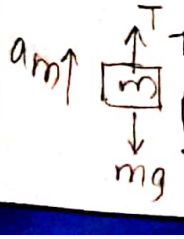

$T\cos 60° - mg = -ma_b$

$\frac{T}{2} = m(g - a_b)$

$\boxed{T = 2m(g - a_b)}$ — ①

$N = T\sin 60°$ — ②

for block :-




$T - mg = ma_m$

$\boxed{T = m(g + a_m)}$ — ⑤

① = ② $\Rightarrow 2m(g - a_b) = m(g + a_m)$

$2g - g = 2a_b + a_m$

$g = 2a_b + \frac{1}{2}a_b$

$g = \frac{5a_b}{2} \Rightarrow a_b = g \times 2/5$

$\boxed{a_b = 2g/5}$ — ⑤

put ⑤ in ① $\Rightarrow T = 2m(g - 2g/5)$

$T = 2m(3g/5)$

$\boxed{T = 6mg/5}$

put ⑤ in ④

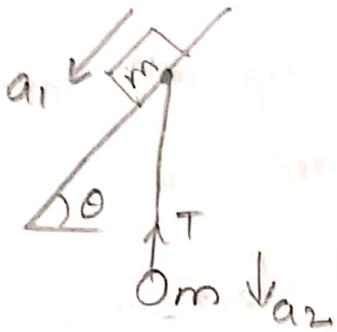
$a_m = \frac{1}{2}(2g/5)$

$\boxed{a_m = g/5}$

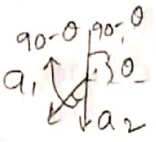
$N = \frac{6mg}{5} \times \frac{\sqrt{3}}{2} \Rightarrow \boxed{N = \frac{3\sqrt{3}mg}{5}}$

Scanned with CamScanner

Find $a_1, a_2, T$ at initial state.



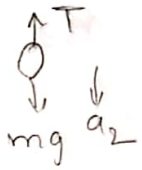

$$T = \frac{mg\cos^2\theta}{1+\sin^2\theta}, \quad a_1 = \frac{2g\sin\theta}{1+\sin^2\theta}$$




$$a_2 = \frac{2g\sin^2\theta}{1+\sin^2\theta}$$

$$a_1\cos(90-\theta) = a_2 \Rightarrow \boxed{a_1\sin\theta = a_2} \quad -(1)$$

for bead :-

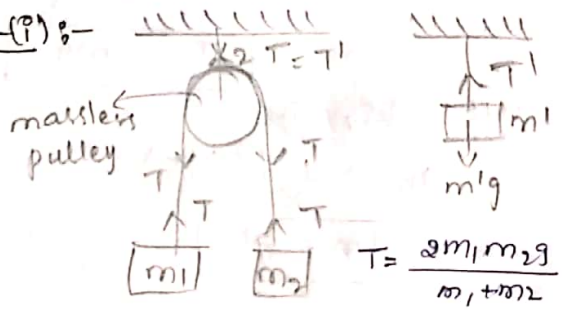



$$T - mg = -ma_2$$

$$\boxed{T = m(g-a_2)}$$

Equivalent mass Method :-

case-(i) :-




$$T' = m'g$$

$$m'g = \frac{4m_1m_2g}{m_1+m_2}$$

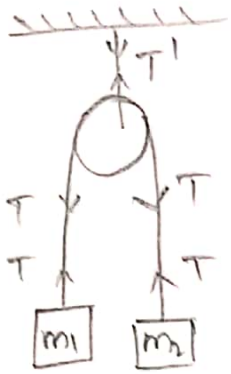
$$\boxed{m' = \frac{4m_1m_2}{m_1+m_2}} \rightarrow \text{equivalent mass}$$

$$T = \frac{2m_1m_2g}{m_1+m_2}$$

$$T' = \frac{4m_1m_2g}{m_1+m_2}$$

Scanned with CamScanner

case(iii):-

* 


$$T = \frac{2m_1m_2}{m_1+m_2}(g+a)$$

& $T' = 2T = \frac{4m_1m_2}{m_1+m_2}(g+a)$

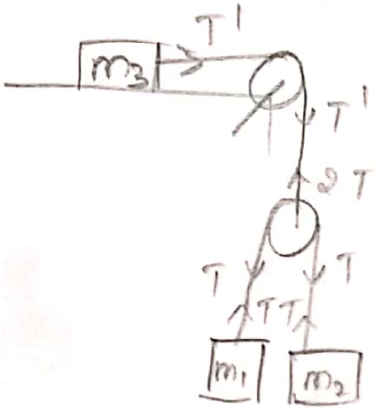



$T' = m'(g+a)$

$$m'(g+a) = \frac{4m_1m_2}{m_1+m_2}(g+a)$$

$$\boxed{m' = \frac{4m_1m_2}{m_1+m_2}}$$

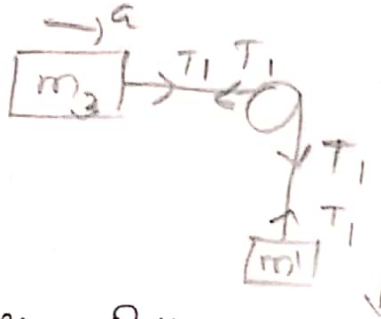
*




$$T = \frac{2m_1m_2g}{m_1+m_2}, \quad T' = \frac{4m_1m_2g}{m_1+m_2}$$

$$m'g = \frac{4m_1m_2g}{m_1+m_2}$$

$$\boxed{m' = \frac{4m_1m_2}{m_1+m_2}}$$

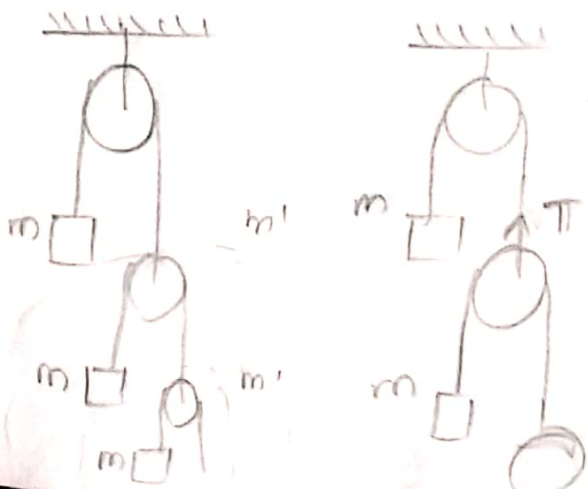



$$\boxed{a = \frac{m'g}{m'+m_3}}$$

$$\boxed{T_1 = m_3 a}$$

Infinite Pulley problem:-

All pulleys and strings are massless. Then the acceleration of top mass is ____.




$$T' = \frac{4mm'g}{m+m'}$$

$$m'g = \frac{4mm'g}{m+m'}$$

$$\boxed{m' = 3m}$$

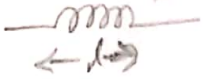
$$a = \frac{3m-m}{3m+m}(g)$$

$$= \frac{2m}{4m}g$$

$$\boxed{a = g/2}$$

Scanned with CamScanner

## Springs :-



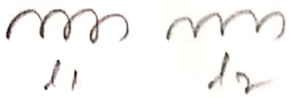

* usually, 'k' depends on
  1. length of spring
  2. Nature of material of spring.
  3. radius of wire used to make the spring.

* we have, $\boxed{k \propto \frac{1}{l}}$ ie, $\boxed{kl = \text{const}}$

## Cutting of springs :-




*cut into two parts* $kl_1 = \text{const}$, $kl_2 = \text{const}$



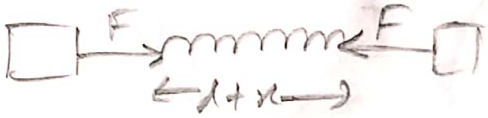

$\boxed{l = l_1 + l_2}$ $\boxed{kl = k_1 l_1 = k_2 l_2 = \text{const}}$

$\therefore k_1 = \frac{kl}{l_1} = \frac{k(l_1+l_2)}{l_1}$ & $k_2 = \frac{k(l_1+l_2)}{l_2}$

* If $l_1 = l_2 = l/2$ then $k_1 = 2k = k_2$.

* when spring is at it's "natural length", then spring force = 0.




*elongation = x

spring force $\propto$ elongation / compression.

$F \propto -x$

$\boxed{F = -kx}$,

* $\boxed{k = \frac{F}{x}}$ # force per unit elongation / compression.

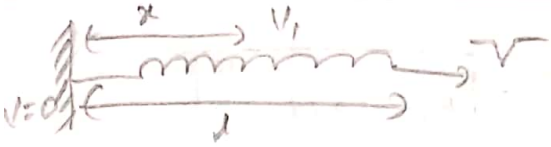
Units:

S.I – N/m

C.G.S – dyne/cm




Scanned with CamScanner




$$\frac{V-0}{l} = \frac{v_1-0}{x}$$

$$\boxed{v_1 = \frac{Vx}{l}}$$




$$kx = m_1 a_1$$

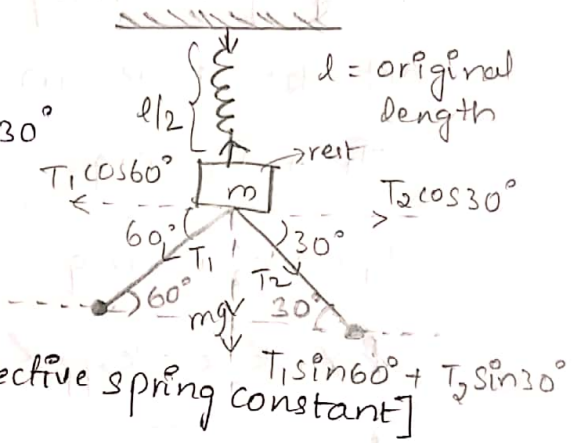
$$F - kx = m_2 a_{x_2}$$

$$\boxed{F = m_1 a_1 + m_2 a_2}$$

$T_1 \cos 60^\circ = T_2 \cos 30^\circ$ —①

$k(l/2) = mg + T_1 \sin 60^\circ + T_2 \sin 30^\circ$ —②

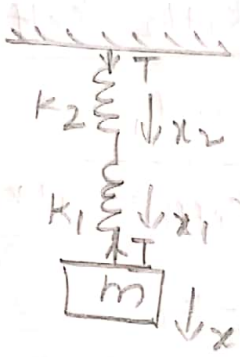
Solve for $T_1$ & $T_2$




## Combination of Springs [effective spring constant]

### ① Series Combination

* Tension or spring force is same in all springs.
* elongations in springs may or may not be equal.

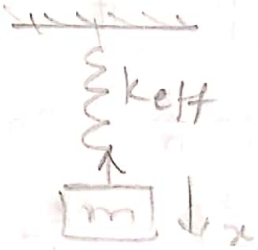



$T = mg$ ; $T = k_1 x_1$ ; $T = k_2 x_2$

$T = k_{eff}\, x = mg$

$x = x_1 + x_2$

$$\frac{T}{k_{eff}} = \frac{T}{k_1} + \frac{T}{k_2} \Rightarrow \boxed{\frac{1}{k_{eff}} = \frac{1}{k_1} + \frac{1}{k_2}}$$




In general, $\boxed{\frac{1}{k_{eff}} = \frac{1}{k_1} + \frac{1}{k_2} + \cdots + \frac{1}{k_n}}$

If $k_1 = k_2 = k_3 = \cdots = k_n = k$ (n times)

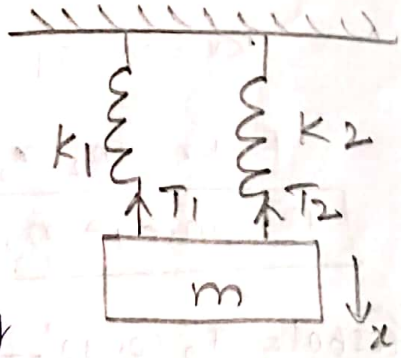
Scanned with CamScanner

$$\frac{1}{k_{eff}} = \underbrace{\frac{1}{k} + \frac{1}{k} + \cdots + \frac{1}{k}}_{n \text{ times}} \Rightarrow \boxed{k_s = k_{eff} = \frac{k}{n}}$$

$\Rightarrow$ $k_{eff}$ decreases.

## (2) Parallel combination

* Here elongation is same in all springs. $\boxed{T_1 = k_1 x \quad T_2 = k_2 x}$



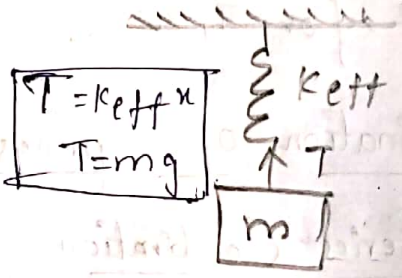

* Spring force may or may not be same in all springs in equilibrium,

$$T_1 + T_2 = mg$$

$$k_1 x + k_2 x = mg$$

$$k_1 x + k_2 x = k_{eff} x$$

$$\boxed{k_1 + k_2 = k_{eff}}$$




In general,

$$\boxed{k_{eff} = k_1 + k_2 + \cdots + k_n}$$

If $k_1 = k_2 = \cdots = k_n = k$

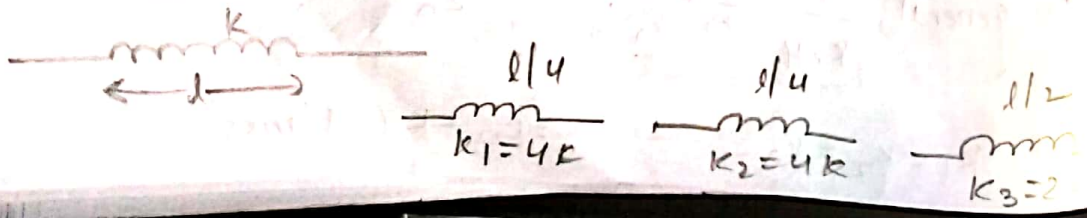
$$\boxed{k_p = k_{eff} = nk} \Rightarrow k_{eff} \text{ increases}$$

* $k_s = \frac{k}{n}$ ; $k_p = nk$

$$\frac{k_p}{k_s} = \frac{nk}{k/n} = n^2 \longrightarrow \text{for identical springs.}$$

$$\boxed{k_p = n^2 k_s}$$

→




Scanned with CamScanner

$k_1$ & $k_3$ are in series.

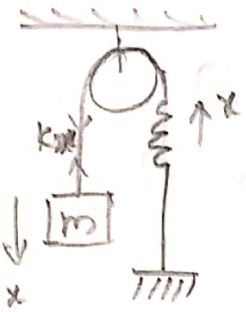
$$k_s = \frac{k_1 * k_3}{k_1 + k_3} = \frac{4k \times 2k}{4k + 2k} = \frac{4k}{3}.$$

$k_1$ & $k_3$ are in parallel

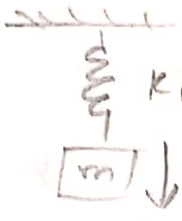
$$k_p = 4k + 2k = 6k.$$

→ Find effective spring constants:-

*



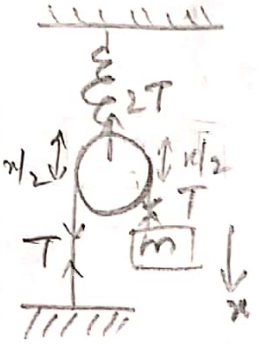

$kx = mg$ —— ①




$k_{eff}\, x = mg$ —— ②

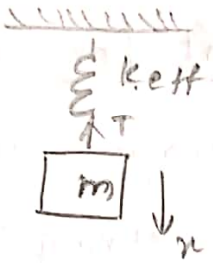
$kx = k_{eff}x \Rightarrow \boxed{k = k_{eff}}$

*




$2T = k(x/2)$

$T = \frac{kx}{4}$ ←①

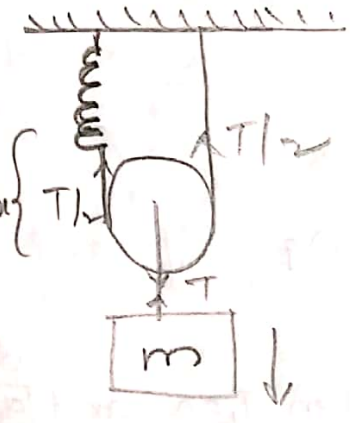



$T = k_{eff}\, x$ —— ③

$k_{eff}x = \frac{kx}{4}$

$\boxed{k_{eff} = k/4}$

*




$T/2 = 2kx$

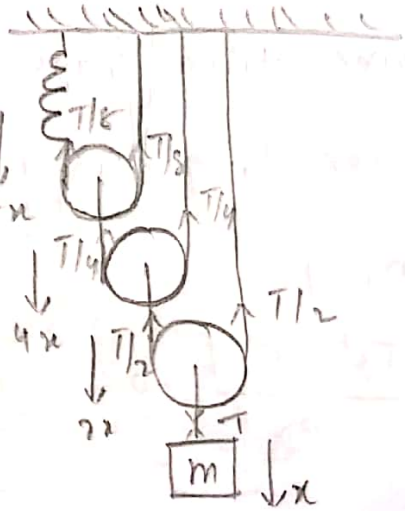
$T = 4kx$

$\boxed{k_{eff} = 4k}$




$T = k_{eff}\, x$

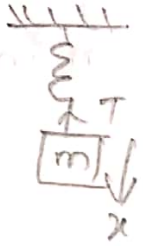
*




$\frac{T}{8} = 8kx$

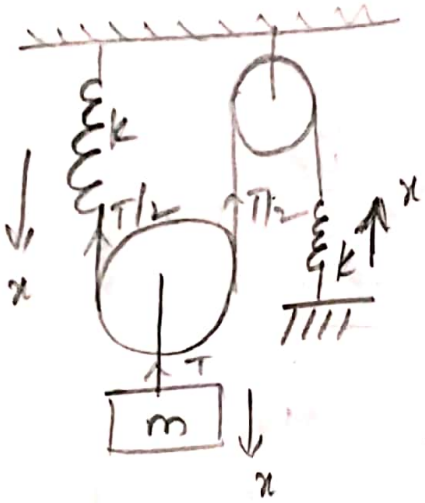
$T = 64kx$

$\boxed{k_{eff} = 64k}$




$T = k_{eff}\, x$

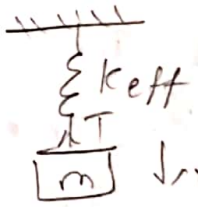
Scanned with CamScanner



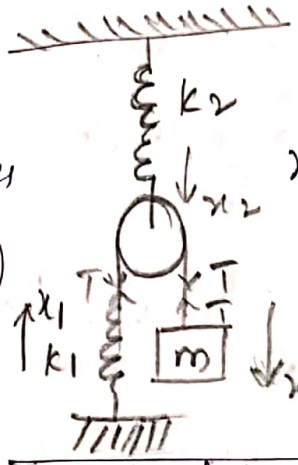

$T/2 = kx$

$T = 2kx$

$\boxed{k_{eff} = 2k}$



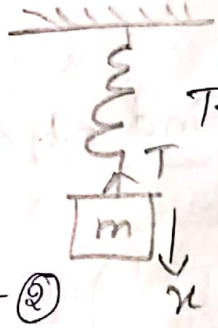

$T = k_{eff}\, x$




$x = 2x_2 + x_1$ — ①

$T = k_1 x_1$ — ②

$2T = k_2 x_2$ — ③




$T = k_{eff}\, x$ — ④

Now, eqⁿ ① becomes

$$\frac{T}{k_{eff}} = \frac{T}{k_1} + 2\left(\frac{2T}{k_2}\right)$$

$$\frac{1}{k_{eff}} = \frac{1}{k_1} + \frac{4}{k_2} \Rightarrow \boxed{k_{eff} = \frac{k_1 k_2}{4k_1 + k_2}}$$

## Breaking Support for springs




$\boxed{T = mg}$

when string snaps (cuts), Tension in that string is zero.

$$a_m = \frac{mg}{m} \Rightarrow \boxed{a_m = g}\ (\downarrow)$$

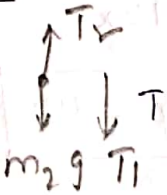



for $m_1$:




$T_1 = m_1 g$

for $m_2$:




(↓)

① when only lower string snaps then $T_1 = 0$ and $\boxed{a_1 = g}$

and tension in upper string changes immediately.

$$T_2' = m_2 g$$

$$a_2 = \frac{T_2' - m_2 g}{m_2} = 0$$

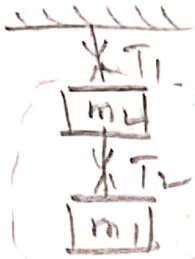



Scanned with CamScanner

(9) when only upper string snaps then $T_2 = 0$ and * Both the blocks are moving with common acceleration.

$$a_1 = a_2 = \frac{(m_1+m_2)g}{m_1+m_2}$$

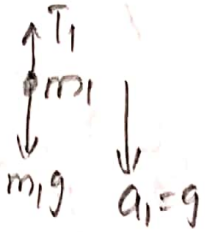
$$\boxed{a_1 = a_2 = g} \quad (\downarrow)$$




now for $m_1$ :

$$m_1 g - T_1 = m_1 a_1$$

$$\boxed{T_1 = 0}$$




for springs




$$T_2 = T_1 + m_2 g$$

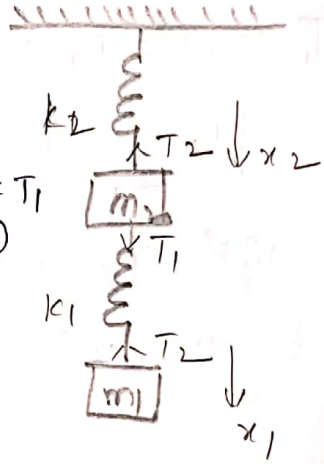
$$k_2 x_2 = k_1 x_1 + m_2 g$$

$$k_2 x_2 = m_1 g + m_2 g$$

$$T_2 = k_2 x_2 = (m_1+m_2)g \quad \text{— (2)}$$




$$k_1 x_1 = m_1 g = T_1 \quad \text{— (1)}$$




a) when only lower spring is cut $\boxed{T_1 = 0}$

But tension in "other spring" will not change "immediately" because of inertia.

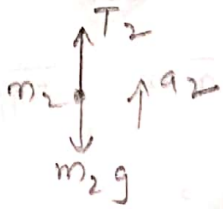
* $\boxed{a_1 = g}$ $(\downarrow)$

* for $m_2$ :

$$a_2 = \frac{T_2 - m_2 g}{m_2}$$

$$a_2 = \frac{(m_1+m_2)g - m_2 g}{m_2}$$

$$\boxed{a_2 = \frac{m_1}{m_2} g}$$




b) when only upper spring is cut $\boxed{T_2 = 0}$ but $T_1$ doesnot change immediately. $\boxed{T_1 = m_1 g}$

Scanned with CamScanner

for $m_2$:

$m_2 a_2 = T_1 + m_1 g$

$m_2 a_2 = (m_1 + m_2) g$

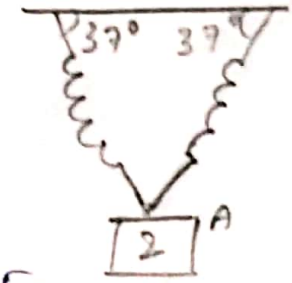
$\boxed{a_2 = \left(\frac{m_1+m_2}{m_2}\right) g}$ (1)

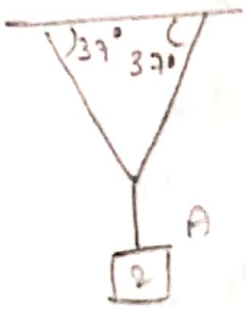
for $m_1$:

$a_1 = \frac{m_1 g - T_1}{m_1}$

$a_1 = \frac{m_1 g - m_1 g}{m_1}$

$\boxed{a_1 = 0}$




when string & spring snaps. find the ratio of immediate accelerating of A & B.

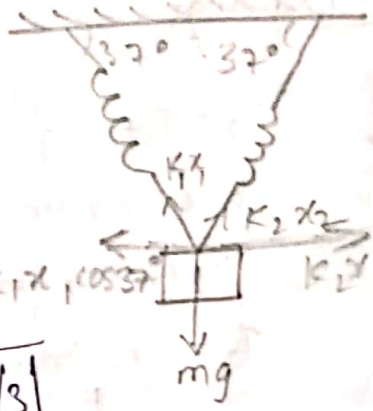
For spring:

$T_1 \cos 37° = T_2 \cos 37°$

$\boxed{T_1 = T_2}$

$T_1 \sin 37° + T_2 \sin 37° = mg$

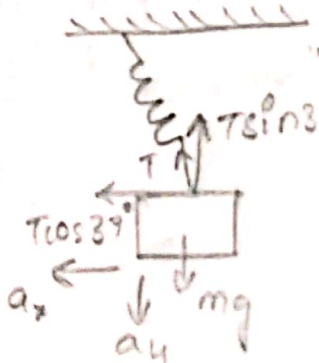
$T_1 \times \frac{3}{5} = 2g \Rightarrow \boxed{T_1 = 5g/3}$




$2a_x = T\cos 37°$

$a_x = \frac{5g}{3} \times \frac{4}{5} \times \frac{1}{2}$

$\boxed{a_x = 2g/3}$




$2g - T\sin 37° = 2a_y$

$2g - \frac{5g}{3} \times \frac{3}{5} = 2a_y$

$2a_y = g$

$\boxed{a_y = g/2}$

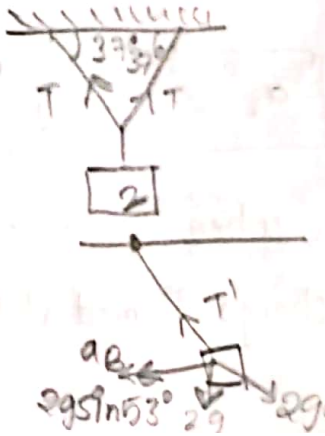
$a_A = \sqrt{a_x^2 + a_y^2} = \sqrt{(2g/3)^2 + (g/2)^2} = g\sqrt{\frac{4}{9} + \frac{1}{4}} = 5/6\, g.$

For string:

$T' = 2g\cos 53° = 6g/5$

$2T\sin 37° = 2g$

$\boxed{T = 5g/3}$

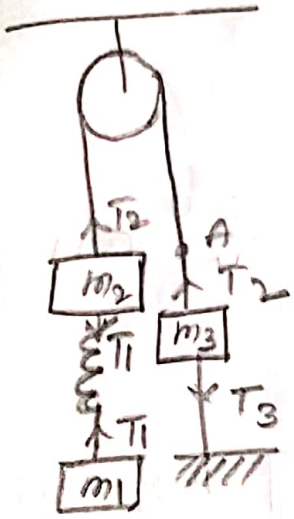



$2a_B = 2g\sin 53°$

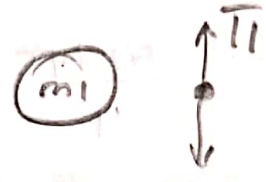
$\boxed{a_B = 4g/5}$

$\Rightarrow \frac{a_A}{a_B} = \frac{\frac{5g}{6}}{\frac{4g}{5}} = \frac{25}{24}$

Scanned with CamScanner




If the string is cut at A find immediate accelerations of $m_1, m_2, a_3$.




$\boxed{T_1 = m_1 g}$ —①

$T_2 = T_1 + m_2 g$

$\Rightarrow T_2 = (m_1 + m_2) g$ —②

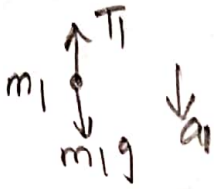
$T_2 = T_3 + m_3 g$ —③

If string cut at 'A'.

$\boxed{T_2 = 0}$ , $\boxed{T_3 = 0}$

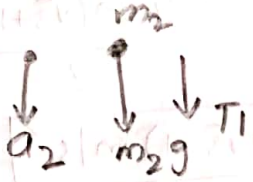
$m_3 a_3 = m_3 g \Rightarrow \boxed{a_3 = g}$.

$T_1$ will not change immediately.




$\Rightarrow m a_1 = m_1 g - T_1$

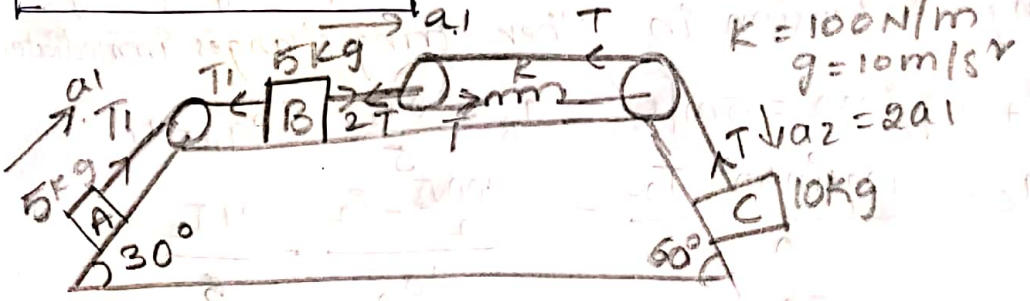
$\boxed{a_1 = 0}$




$m_2 a_2 = T_1 + m_2 g$

$a_2 = \frac{(m_1 + m_2) g}{m_2}$

$\boxed{a_2 = \left(1 + \frac{m_1}{m_2}\right) g}$




① Find accelerations of A, B & C and extension in spring.

② If string is snapped at 'P'. Find $a_A$, $a_B$ & $a_C$ (immediate)

③ If string is snapped at 'p'. Find $a_A$, $a_B$ & $a_C$ put in this case spring is replaced with string.

④ $T_1 - 5g\sin 30° = 5a_1 \Rightarrow T_1 = 5a_1 + \frac{5}{2} \times 10^5$

$2T - T_1 = 5a_1 \Rightarrow 2T - 5a_1 = 5a_1 \Rightarrow 10a_1 = 2T - 5/2$

$2T = 10a_1 + 5/2$

$g \times 10 \sin 60 - T = 10(2a_1)$

$5\sqrt{3} - 5a_1 - \frac{5}{4} = 20a_1 \Rightarrow 25a_1 = 5\sqrt{3} - \frac{5}{4} \Rightarrow 5a_1 = \sqrt{3} - \frac{1}{4}$

Scanned with CamScanner

$$\boxed{a_1 = \frac{4\sqrt{3}-1}{20}\ m/s^2}$$

$$2T = \frac{4\sqrt{3}-1}{2} + \frac{5}{2} \Rightarrow T = \frac{4\sqrt{3}-1+5}{4} = \frac{4\sqrt{3}+4}{4} = \sqrt{3}+1$$

$$T_1 = \frac{4\sqrt{3}-1}{4} + \frac{5}{4} \Rightarrow \boxed{T_1 = \frac{4\sqrt{3}+9}{4}}$$

(2) If string snaps at 'P'. $\boxed{T_1 = 0}$ but 'T' will not change immediately.

$$a_A = \frac{5g\sin 30^\circ}{5} \Rightarrow \boxed{a_A = g/2} = 5\ m/s^2 (\swarrow)$$

$$a_B = \frac{2T}{5} = \frac{2(10(\sqrt{3}+1))}{5} = 4(\sqrt{3}+1)\ m/s^2$$

$$\boxed{a_C = (4\sqrt{3}-1)\ m/s^2}$$

(3) If string snaps at 'P' after replacing spring with string & tension in other string changes immediately.

$$a_A = g\sin 30^\circ = 5\ m/s^2 \qquad a_B = \frac{2T_2}{5} \qquad 2a_B = a_C$$

$$a_C = \frac{10g\sin 60^\circ - T_2}{10} = \frac{\frac{100\sqrt{3}}{2} - T_2}{10} = \frac{4T_2}{5}$$

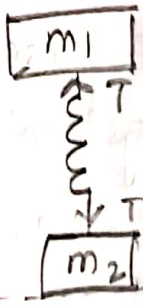
$$\frac{100\sqrt{3}}{2} = 9T_2$$

$$\boxed{T_2 = \frac{50\sqrt{3}}{9}}$$

$$a_B = \frac{20\sqrt{3}}{9}$$

$$a_C = \frac{40\sqrt{3}}{9}$$

→ when support is suddenly removed. Find immediate acceleration of $m_1$ & $m_2$.




Initially,

$T = m_1 g$ — ①

$m_2 g + T = N$ — ②

after release,

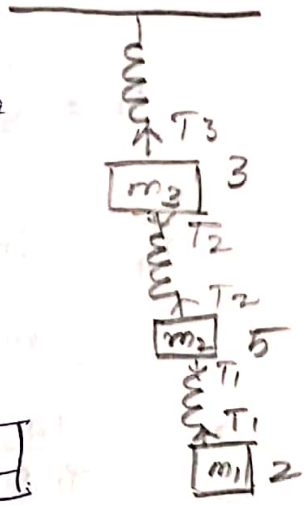
$m_2 g - T = m_2 a_2$

$m_2 g - m_1 g = m_2 a_2$

$$\boxed{a_2 = \frac{(m_2 - m_1)g}{m_2}}$$

Scanned with CamScanner

→ when middle spring, find immediate accelerations of $m_1, m_2, m_3$.




$T_1 = m_1 g = 2g$, $T_2 = (m_1 + m_2)g = 7g$

$T_3 = T_2 + 3g = 10g$

when spring acts $\boxed{T_2 = 0}$

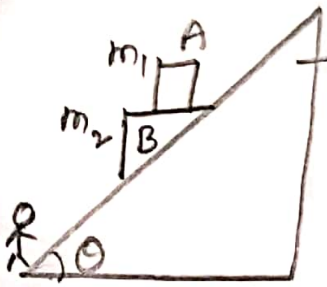
$(m_1)$ $m_1 g - T_1 = m_1 a_1 \Rightarrow \boxed{a_1 = 0}$

$(m_2)$ $T_1 + m_2 g = m_2 a_2 \Rightarrow \boxed{7g/5 = a_2}$

$(m_3)$ $m_3 g - T_3 = m_3 a_3$

$3g - 10g = 3a_3 \Rightarrow \boxed{a_3 = -\frac{7g}{3}}$

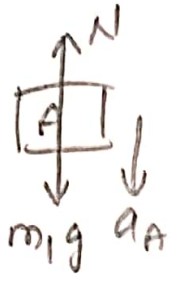
→



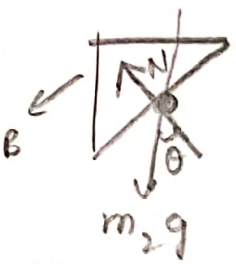

all surfaces are smooth.




$a_A = a_B \sin\theta$ — (9)




$m_1 g - N = m_1 a_A$ — (1)




$(m_2 g + N)\sin\theta = m_2 a_B$ — (2)

$N_1 = (N + m_2 g)\cos\theta$ — (8)

$$\frac{m_1 g - N}{m_1} = \frac{(m_2 g + N)\sin^2\theta}{m_2}$$

$$m_1 m_2 g - N m_2 = m_1 m_2 g \sin^2\theta + m N \sin^2\theta$$

$$m_1 m_2 g \cos^2\theta = N m_1 \sin^2\theta + N m_2$$

$$m_1 m_2 g \cos^2\theta = N (m_1 \sin^2\theta + m_2)$$

$$\boxed{N = \frac{m_1 m_2 g \cos^2\theta}{m_1 \sin^2\theta + m_2}}$$

$$m_1 g - \frac{m_1 m_2 g \cos^2\theta}{m_1 \sin^2\theta + m_2} = m_1 a_A$$

$$\frac{g m_1 \sin^2\theta + m_2 g - g m_2 \cos^2\theta}{m_1 \sin^2\theta + m_2} = a_A$$

Scanned with CamScanner

$$a_A = \frac{m_1 g (\sin^2\theta) + (1-\cos^2\theta) m_2 g}{m_1 \sin^2\theta + m_2}$$

$$a_A = \frac{(m_2 g + m_1 g) \sin^2\theta}{m_1 \sin^2\theta + m_2}$$

$$\boxed{a_A = \frac{(m_1 + m_2) g \sin^2\theta}{m_1 \sin^2\theta + m_2}}$$

$$a_B = \frac{a_A}{\sin\theta} \Rightarrow \boxed{a_B = \frac{(m_1 + m_2) g \sin\theta}{m_1 \sin^2\theta + m_2}}$$

Scanned with CamScanner